Problem 7-1

The column is fixed to the floor and is subjected to the loads shown. Determine the internal normal force, shear force, and moment at points *A* and *B*.

Units Used:

$kN = 10^3 N$

Given:

$F_1 = 6$ kN

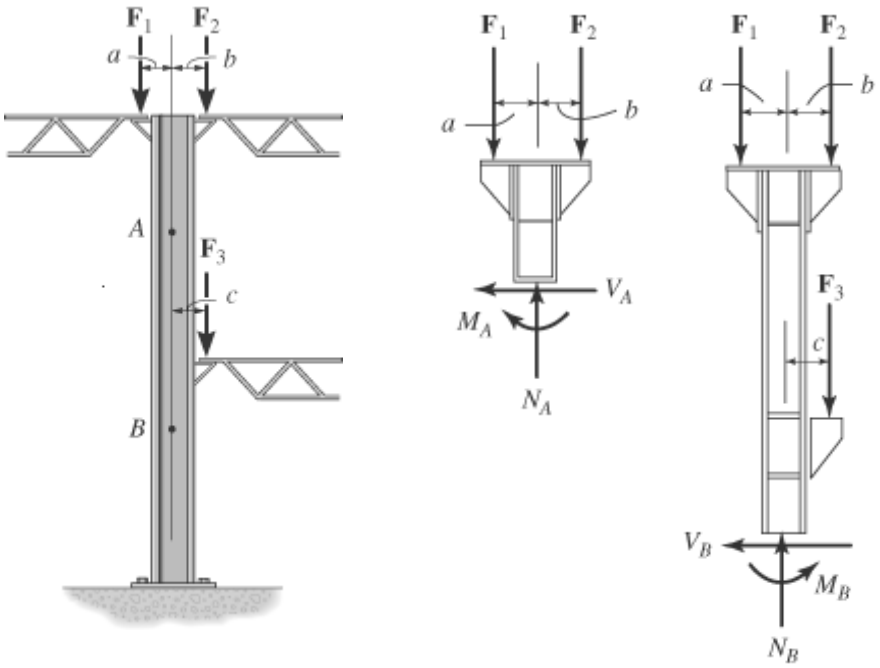
$F_2 = 6$ kN

$F_3 = 8$ kN

$a = 150$ mm

$b = 150$ mm

$c = 150$ mm



Solution:

Free body Diagram: The support reaction need not be computed in this case.

Internal Forces: Applying equations of equillibrium to the top segment sectioned through point A, we have

$\xrightarrow{+} \Sigma F_x = 0;$ $V_A = 0$ $V_A = 0$

$+\uparrow \Sigma F_y = 0;$ $N_A - F_1 - F_2 = 0$ $N_A = F_1 + F_2$ $N_A = 12.0\,\text{kN}$

$\circlearrowleft + \Sigma M_A = 0;$ $F_1 a - F_2 b - M_A = 0$ $M_A = F_1 a - F_2 b$ $M_A = 0\,\text{kN}\cdot\text{m}$

Applying equations of equillibrium to the top segment sectioned through point *B*, we have

$\xrightarrow{+} \Sigma F_x = 0;$ $V_B = 0$ $V_B = 0$

$+\uparrow \Sigma F_y = 0;$ $N_B - F_1 - F_2 - F_3 = 0$ $N_B = F_1 + F_2 + F_3$ $N_B = 20.0\,\text{kN}$

$+\Sigma M_B = 0;$ $F_1 a - F_2 b - F_3 c + M_B = 0$ $M_B = -F_1 a + F_2 b + F_3 c$ $M_B = 1.20\,\text{kN}\cdot\text{m}$

Problem 7-2

The axial forces act on the shaft as shown. Determine the internal normal forces at points *A* and *B*.

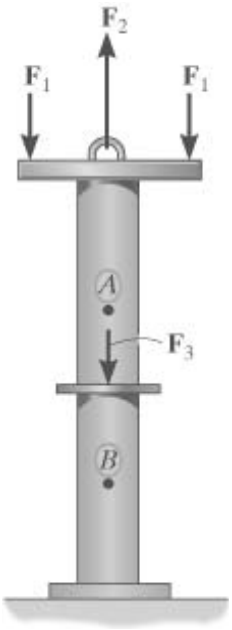
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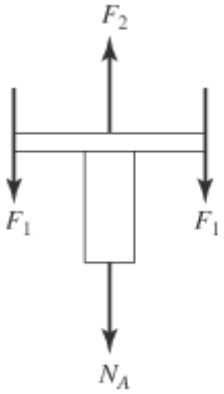
Given:

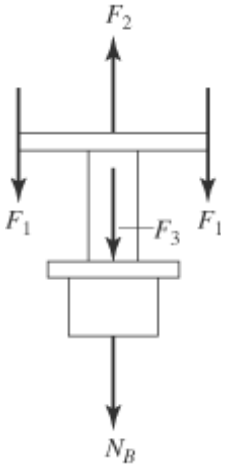
$F_1 = 20$ lb

$F_2 = 50$ lb

$F_3 = 10$ lb







Solution:

Section *A*:

$\Sigma F_z = 0; \quad F_2 - 2F_1 - N_A = 0$

$N_A = F_2 - 2\,F_1$

$N_A = 10.00$ lb

Section *B*:

$\Sigma F_z = 0; \quad F_2 - 2F_1 - N_A + N_B = 0$

$N_B = -F_2 + 2F_1 + N_A$

$N_B = 0.00$ lb

Problem 7-3

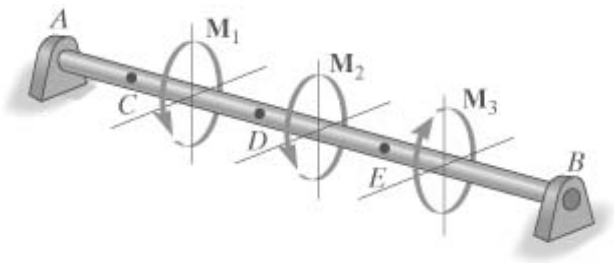
The shaft is supported by smooth bearings at *A* and *B* and subjected to the torques shown. Determine the internal torque at points *C*, *D*, and *E*.

Given:

$M_1 = 400$ N·m

$M_2 = 150$ N·m

$M_3 = 550$ N·m



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Solution:

Section *C*:

$\Sigma M_x = 0; \quad T_C = 0$



Section *D*:

$\Sigma M_x = 0; \quad T_D - M_1 = 0$

$T_D = M_1$

$T_D = 400.00\,\text{N}\cdot\text{m}$

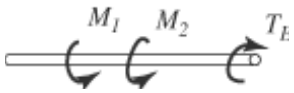


Section *E*:

$\Sigma M_x = 0; \quad M_1 + M_2 - T_E = 0$

$T_E = M_1 + M_2$

$T_E = 550.00\,\text{N}\cdot\text{m}$



Problem 7-4

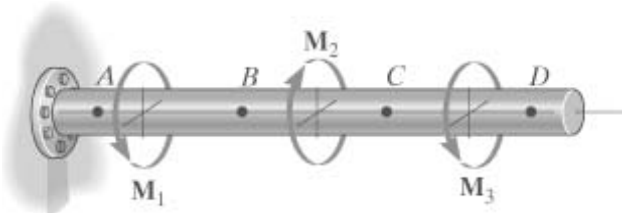
Three torques act on the shaft. Determine the internal torque at points *A*, *B*, *C*, and *D*.

Given:

$M_1 = 300\ \text{N}\cdot\text{m}$

$M_2 = 400\ \text{N}\cdot\text{m}$

$M_3 = 200\ \text{N}\cdot\text{m}$



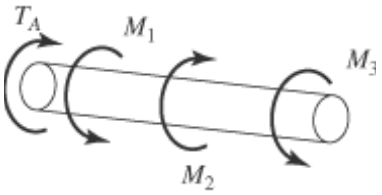
Solution:

Section *A*:

$\Sigma M_x = 0; \quad -T_A + M_1 - M_2 + M_3 = 0$

$T_A = M_1 - M_2 + M_3$

$T_A = 100.00\,\text{N}\cdot\text{m}$



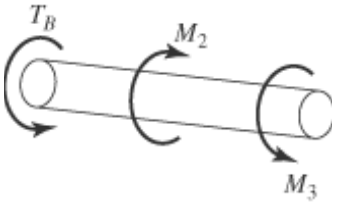
Section *B*:

$\Sigma M_x = 0; \quad T_B + M_3 - M_2 = 0$

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$$T_B = -M_3 + M_2$$

$$T_B = 200.00\,\text{N}\cdot\text{m}$$

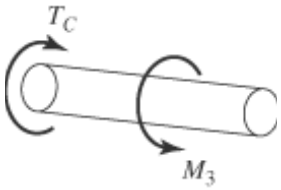


Section C:

$$\Sigma M_x = 0; \qquad -T_C + M_3 = 0$$

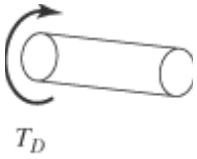
$$T_C = M_3$$

$$T_C = 200.00\,\text{N}\cdot\text{m}$$



Section D:

$$\Sigma M_x = 0; \qquad T_D = 0$$



Problem 7-5

The shaft is supported by a journal bearing at A and a thrust bearing at B. Determine the normal force, shear force, and moment at a section passing through (a) point C, which is just to the right of the bearing at A, and (b) point D, which is just to the left of the force F_2.

Units Used:

$$\text{kip} = 10^3\ \text{lb}$$

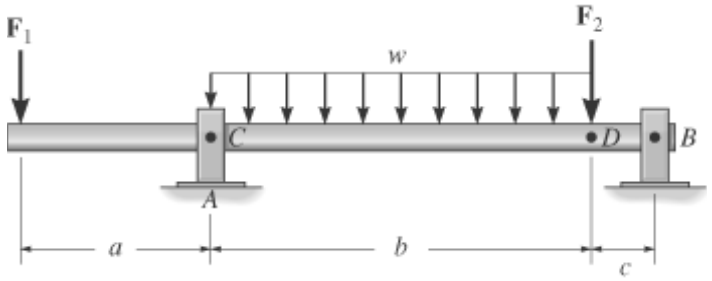
Given:

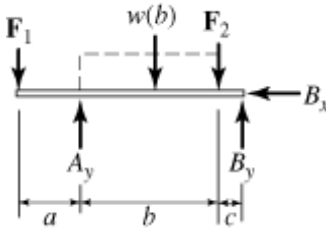
$$F_1 = 2.5\ \text{kip}$$

$$F_2 = 3\ \text{kip}$$

$$w = 75\ \frac{\text{lb}}{\text{ft}} \qquad b = 12\ \text{ft}$$

$$a = 6\ \text{ft} \qquad c = 2\ \text{ft}$$





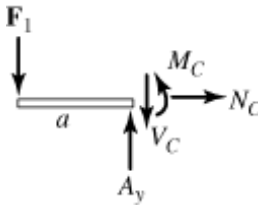
Solution:

$$\circlearrowleft + \quad \Sigma M_B = 0;$$

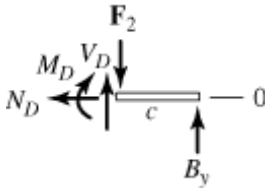
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$$-A_y(b+c) + F_1(a+b+c) + wb\left(c+\frac{b}{2}\right) + F_2 c = 0$$

$$A_y = \frac{F_1(a+b+c) + wb\left(c+\frac{b}{2}\right) + F_2 c}{b+c}$$



$A_y = 4514\,\text{lb}$



$\rightarrow \Sigma F_x = 0; \qquad B_x = 0\ \text{lb}$

$+\uparrow \Sigma F_y = 0; \qquad A_y - F_1 - wb - F_2 + B_y = 0$

$B_y = -A_y + F_1 + wb + F_2 \qquad B_y = 1886\,\text{lb}$

$\circlearrowleft + \Sigma M_C = 0; \qquad F_1 a + M_c = 0 \qquad M_C = -F_1 a \qquad M_C = -15.0\,\text{kip}\cdot\text{ft}$

$\rightarrow \Sigma F_x = 0; \qquad N_C = 0\ \text{lb} \qquad N_C = 0.00\,\text{lb}$

$+\uparrow \Sigma F_y = 0; \qquad -F_1 + A_y - V_C = 0 \qquad V_C = A_y - F_1 \qquad V_C = 2.01\,\text{kip}$

$\circlearrowleft + \Sigma M_D = 0; \qquad -M_D + B_y c = 0 \qquad M_D = B_y c \qquad M_D = 3.77\,\text{kip}\cdot\text{ft}$

$\rightarrow \Sigma F_x = 0; \qquad N_D = 0\ \text{lb} \qquad N_D = 0.00\,\text{lb}$

$+\uparrow \Sigma F_y = 0; \qquad V_D - F_2 + B_y = 0 \qquad V_D = F_2 - B_y \qquad V_D = 1.11\,\text{kip}$

Problem 7-6

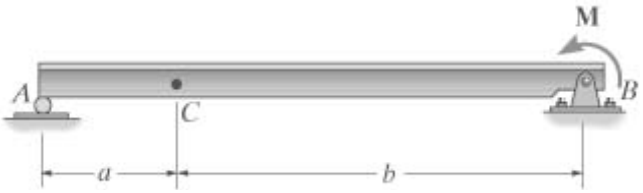
Determine the internal normal force, shear force, and moment at point *C*.

Given:

$M = 400\ \text{lb}\cdot\text{ft}$

$a = 4\ \text{ft}$

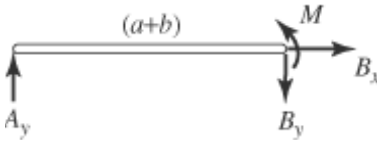
$b = 12\ \text{ft}$



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Solution:

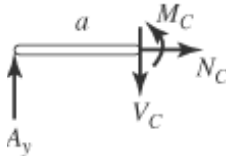
Beam:



$\Sigma M_B = 0; \quad M - A_y(a + b) = 0$

$A_y = \dfrac{M}{a + b} \qquad A_y = 25.00\ \text{lb}$

Segment *AC*:



$\Sigma F_x = 0; \quad N_C = 0$

$\Sigma F_y = 0; \quad A_y - V_C = 0$

$V_C = A_y$

$V_C = 25.00\ \text{lb}$

$\Sigma M_C = 0; \quad -A_y a + M_C = 0$

$M_C = A_y a$

$M_C = 100.00\ \text{lb}\cdot\text{ft}$

Problem 7-7

Determine the internal normal force, shear force, and moment at point *C*.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 30\ \text{kN}$

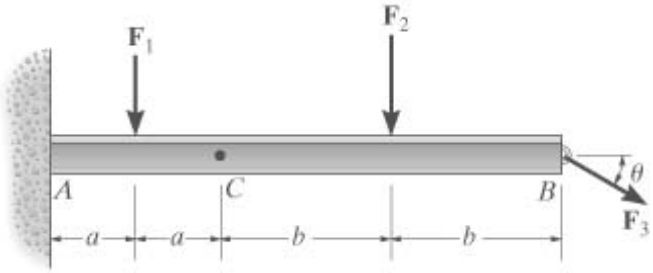
$F_2 = 50\ \text{kN}$

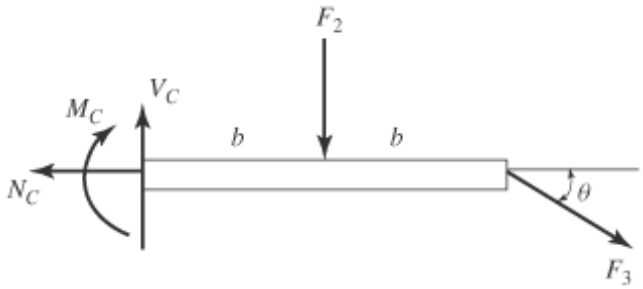
$F_3 = 25\ \text{kN}$

$a = 1.5\ \text{m}$

$b = 3\ \text{m}$

$\theta = 30\ \text{deg}$





Solution:

$\Sigma F_x = 0; \quad -N_C + F_3\cos(\theta) = 0$

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$$N_C = F_3 \cos(\theta)$$

$$N_C = 21.7\,\text{kN}$$

$\Sigma F_y = 0;$ $\quad V_C - F_2 - F_3 \sin(\theta) = 0$

$$V_C = F_2 + F_3 \sin(\theta)$$

$$V_C = 62.50\,\text{kN}$$

$\Sigma M_C = 0;$ $\quad -M_C - F_2 b - F_3 \sin(\theta)\, 2b = 0$

$$M_C = -F_2 b - F_3 \sin(\theta)\, 2b$$

$$M_C = -225.00\,\text{kN}\cdot\text{m}$$

Problem 7-8

Determine the normal force, shear force, and moment at a section passing through point *C*. Assume the support at *A* can be approximated by a pin and *B* as a roller.

Units used:

$$\text{kip} = 10^3\ \text{lb}$$

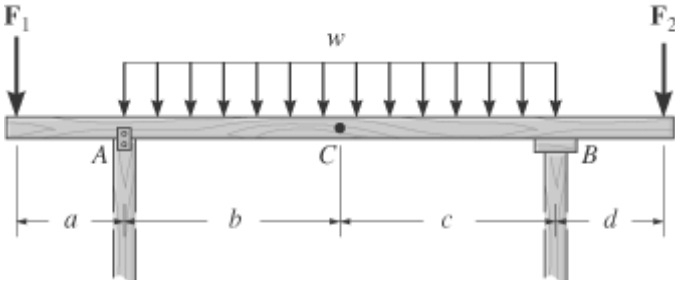
Given:

$F_1 = 10$ kip $\quad a = 6$ ft

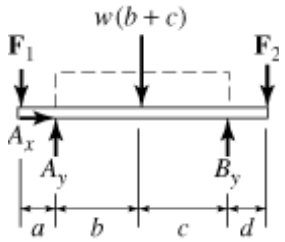
$F_2 = 8$ kip $\quad b = 12$ ft

$w = 0.8 \dfrac{\text{kip}}{\text{ft}}$ $\quad c = 12$ ft

$d = 6$ ft



Solution:



$\curvearrowleft + \Sigma M_A = 0;$

$$-w(b+c)\left(\frac{b+c}{2}\right) - F_2(b+c+d) + B_y(b+c) + F_1 a = 0$$

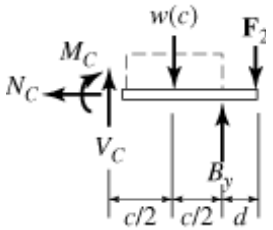
$$B_y = \frac{w\,\dfrac{(b+c)^2}{2} + F_2(b+c+d) - F_1 a}{b+c} \qquad B_y = 17.1\,\text{kip}$$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad N_C = 0$

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$+\uparrow \Sigma F_y = 0; \qquad V_C - wc + B_y - F_2 = 0$

$V_C = wc - B_y + F_2 \qquad V_C = 0.5\,\text{kip}$



$\circlearrowleft + \Sigma M_C = 0; \qquad -M_C - wc\left(\frac{c}{2}\right) + B_y c - F_2(c + d) = 0$

$M_C = -w\left(\frac{c^2}{2}\right) + B_y c - F_2(c + d) \qquad M_C = 3.6\,\text{kip}\cdot\text{ft}$

Problem 7-9

The beam *AB* will fail if the maximum internal moment at *D* reaches M_{max} or the normal force in member *BC* becomes P_{max}. Determine the largest load *w* it can support.

Given:

$M_{max} = 800\ \text{N}\cdot\text{m}$

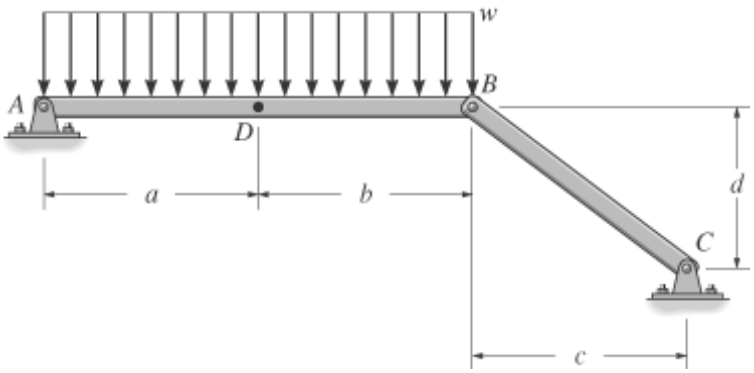
$P_{max} = 1500\ \text{N}$

$a = 4\ \text{m}$

$b = 4\ \text{m}$

$c = 4\ \text{m}$

$d = 3\ \text{m}$



Solution:

$w\,(a + b)\left(\frac{a + b}{2}\right) - A_y(a + b) = 0$

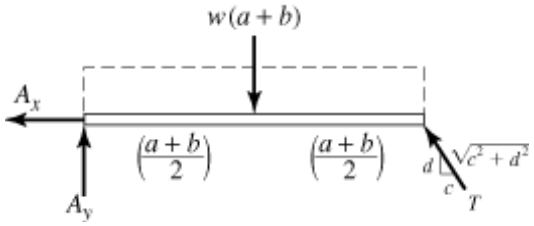
$A_y = \frac{w(a + b)}{2}$

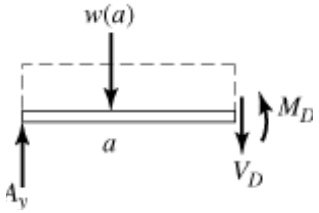
$w\,a\left(\frac{a}{2}\right) - A_y a + M_D = 0$

$M_D = w\left(\frac{a\,b}{2}\right)$

$A_y - w(a + b) + \left(\frac{d}{\sqrt{c^2 + d^2}}\right)T = 0$

$T = \frac{w(a + b)\sqrt{c^2 + d^2}}{2d}$





Assume the maximum moment has been reached

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$M_D = M_{max}$ $\quad w_1 = \dfrac{2M_D}{a\,b}$ $\quad w_1 = 100\,\dfrac{\text{N}}{\text{m}}$

Assume that the maximum normal force in *BC* has been reached

$T = P_{max}$ $\quad w_2 = \dfrac{T2d}{(a+b)\sqrt{c^2+d^2}}$ $\quad w_2 = 225\,\dfrac{\text{N}}{\text{m}}$

Now choose the critical load $\quad w = \min(w_1, w_2)$ $\quad w = 100\,\dfrac{\text{N}}{\text{m}}$

Problem 7-10

Determine the shear force and moment acting at a section passing through point *C* in the beam.

Units Used:

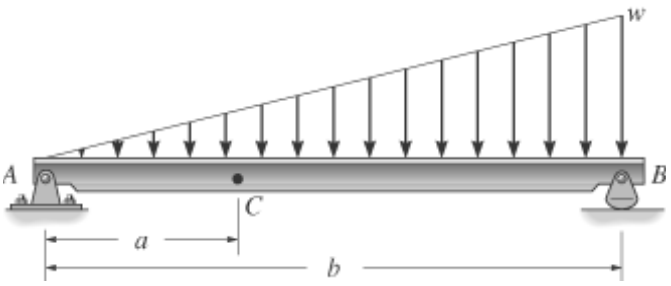
$\text{kip} = 10^3\ \text{lb}$

Given:

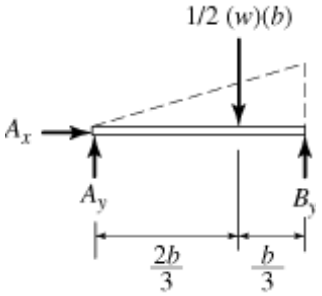
$w = 3\ \dfrac{\text{kip}}{\text{ft}}$

$a = 6\ \text{ft}$

$b = 18\ \text{ft}$



Solution:



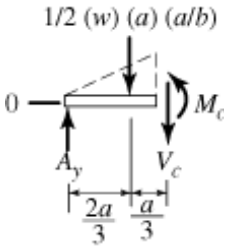
$+\circlearrowleft \Sigma M_B = 0;$ $\quad -A_y b + \dfrac{1}{2} w b \left(\dfrac{b}{3}\right) = 0$

$A_y = \dfrac{1}{6} w b$ $\quad A_y = 9\,\text{kip}$

$+\circlearrowleft \Sigma M_C = 0;$ $\quad -A_y a + \dfrac{1}{2}\left(w\,\dfrac{a}{b}\right) a \left(\dfrac{a}{3}\right) + M_C = 0$

$M_C = A_y a - \dfrac{w a^3}{6b}$

$M_C = 48\,\text{kip}\cdot\text{ft}$



$+\uparrow \Sigma F_y = 0;$ $\quad A_y - \dfrac{1}{2}\left(w\,\dfrac{a}{b}\right) a - V_C = 0$

$V_C = A_y - \dfrac{w a^2}{2b}$

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$V_C = 6\,\text{kip}$

Problem 7-11

Determine the internal normal force, shear force, and moment at points *E* and *D* of the compound beam.

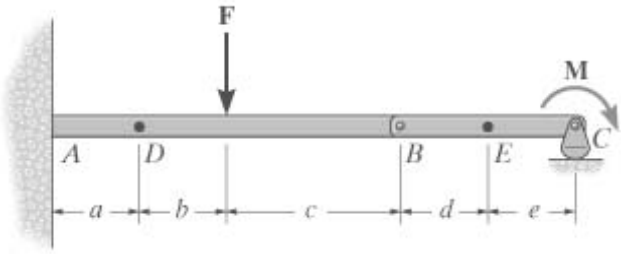
Given:

$M = 200\ \text{N·m}$ $c = 4\ \text{m}$

$F = 800\ \text{N}$ $d = 2\ \text{m}$

$a = 2\ \text{m}$ $e = 2\ \text{m}$

$b = 2\ \text{m}$

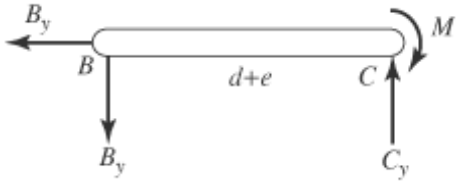


Solution:

Segment *BC* :

$-M + C_y(d + e) = 0 \qquad C_y = \dfrac{M}{d + e}$

$-B_y + C_y = 0 \qquad B_y = C_y$

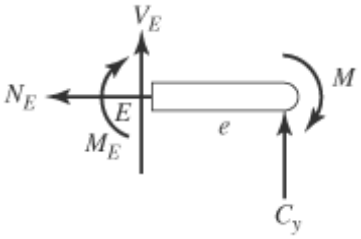


Segment *EC* :

$-N_E = 0 \qquad N_E = 0\text{N} \qquad N_E = 0.00$

$V_E + C_y = 0 \qquad V_E = -C_y \qquad V_E = -50.00\ \text{N}$

$-M_E - M + C_y e = 0 \qquad M_E = C_y e - M \qquad M_E = -100.00\,\text{N·m}$



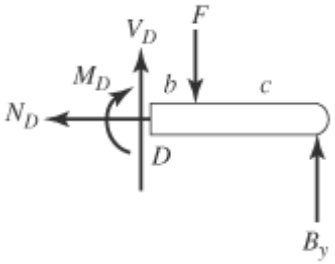
Segment *DB* :

$-N_D = 0 \qquad N_D = 0\text{N} \qquad N_D = 0.00$

$V_D - F + B_y = 0 \qquad V_D = F - B_y \qquad V_D = 750.00\ \text{N}$

$-M_D - F\,b + B_y(b + c) = 0$

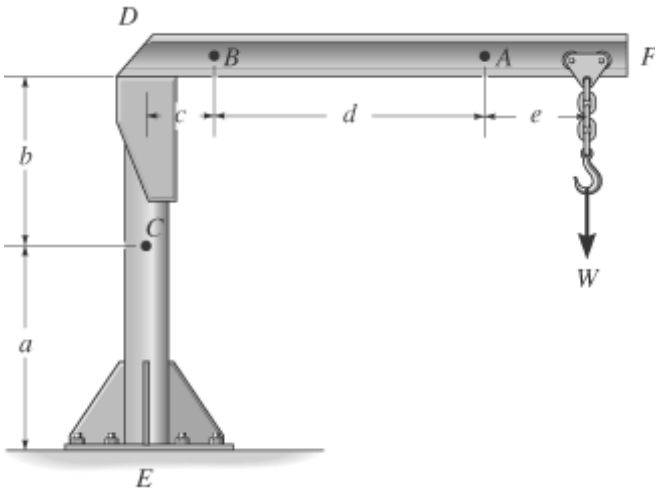
$M_D = -F\,b + B_y(b + c) \qquad M_D = -1300\,\text{N·m}$



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Problem 7-12

The boom *DF* of the jib crane and the column *DE* have a uniform weight density γ. If the hoist and load have weight *W*, determine the normal force, shear force, and moment in the crane at sections passing through points *A*, *B*, and *C*. Treat the boom tip, beyond the hoist, as weightless.



Given:

$W = 300$ lb

$\gamma = 50 \dfrac{\text{lb}}{\text{ft}}$

$a = 7$ ft

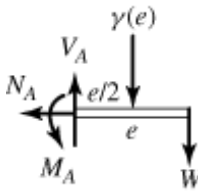
$b = 5$ ft

$c = 2$ ft

$d = 8$ ft

$e = 3$ ft

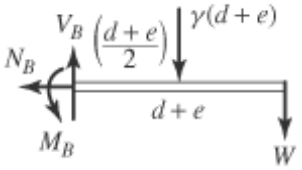
Solution:



$\xrightarrow{+} \Sigma F_x = 0; \quad -N_A = 0 \quad N_A = 0 \text{lb} \quad N_A = 0.00\,\text{lb}$

$+\uparrow \Sigma F_y = 0; \quad V_A - W - \gamma e = 0$

$V_A = W + \gamma\, e \quad V_A = 450\,\text{lb}$



$\circlearrowleft + \Sigma M_A = 0; \quad M_A - \gamma\, e\left(\dfrac{e}{2}\right) - We = 0$

$M_A = \gamma\left(\dfrac{e^2}{2}\right) + We \quad M_A = 1125.00\,\text{lb}\cdot\text{ft}$

$\xrightarrow{+} \Sigma F_x = 0; \quad -N_B = 0 \quad N_B = 0\text{lb} \quad N_B = 0.00\,\text{lb}$

$+\uparrow \Sigma F_y = 0; \quad V_B - \gamma(d+e) - W = 0 \quad V_B = \gamma(d+e) + W \quad V_B = 850\,\text{lb}$

$\circlearrowleft + \Sigma M_B = 0; \quad M_B - \gamma(d+e)\left(\dfrac{d+e}{2}\right) - W(d+e) = 0$

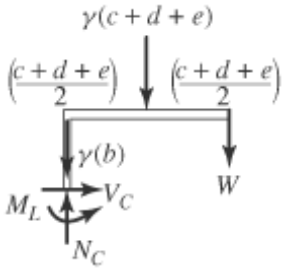
$M_B = \dfrac{1}{2}\,\gamma\,(d+e)^2 + W(d+e) \quad M_B = 6325.00\,\text{lb}\cdot\text{ft}$

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$\xrightarrow{+}\ \Sigma F_x = 0;\quad V_C = 0 \quad V_C = 0\text{lb} \quad V_C = 0.00\,\text{lb}$

$+\uparrow\ \Sigma F_y = 0;\quad N_C - (c + d + e)\,\gamma - W - \gamma(b) = 0$

$N_C = \gamma(c + d + e + b) + W \quad N_C = 1200.00\,\text{lb}$



$\circlearrowleft\ \Sigma M_C = 0;\quad M_C - (c + d + e)\gamma\left(\frac{c + d + e}{2}\right) - W(c + d + e) = 0$

$M_C = (c + d + e)\gamma\left(\frac{c + d + e}{2}\right) + W(c + d + e) \quad M_C = 8125.00\,\text{lb}\cdot\text{ft}$

Problem 7-13

Determine the internal normal force, shear force, and moment at point *C*.

Units Used:

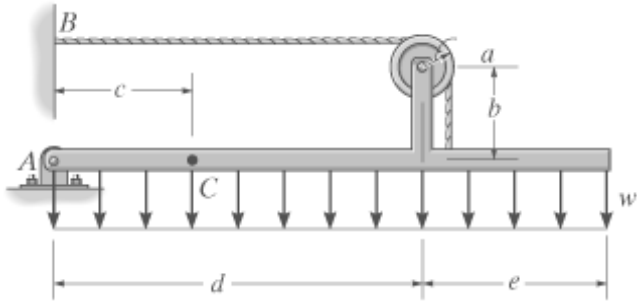
$\text{kip} = 10^3\ \text{lb}$

Given:

$a = 0.5\ \text{ft} \quad d = 8\text{ft}$

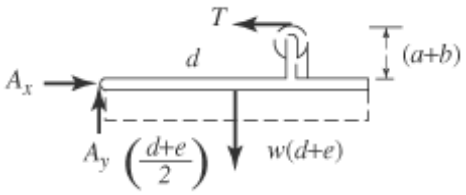
$b = 2\ \text{ft} \quad e = 4\ \text{ft}$

$c = 3\ \text{ft} \quad w = 150\ \frac{\text{lb}}{\text{ft}}$



Solution:

Entire beam:

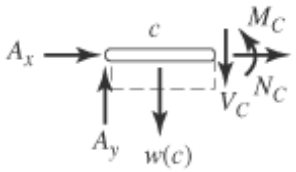


$\Sigma M_A = 0;\quad -w(d + e)\left(\frac{d + e}{2}\right) + T(a + b) = 0$

$T = \frac{w(d + e)^2}{2(a + b)} \quad T = 4.32\,\text{kip}$

$\Sigma F_x = 0;\quad A_x - T = 0$

$A_x = T \quad A_x = 4.32\,\text{kip}$



$\Sigma F_y = 0;\quad A_y - w(d + e) = 0$

$A_y = w(d + e) \quad A_y = 1.80\,\text{kip}$

Segment *AC*:

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$\Sigma F_x = 0;\quad A_x + N_C = 0 \qquad N_C = -A_x \qquad N_C = -4.32\,\text{kip}$

$\Sigma F_y = 0;\quad A_y - w\,c - V_C = 0 \qquad V_C = A_y - w\,c \qquad V_C = 1.35\,\text{kip}$

$\Sigma M_C = 0;\quad -A_y\,c + w\,c\left(\frac{c}{2}\right) + M_C = 0 \qquad M_C = A_y\,c - w\left(\frac{c^2}{2}\right) \qquad M_C = 4.72\,\text{kip}\cdot\text{ft}$

Problem 7-14

Determine the normal force, shear force, and moment at a section passing through point *D* of the two-member frame.

Units Used:

$\text{kN} = 10^3\ \text{N}$

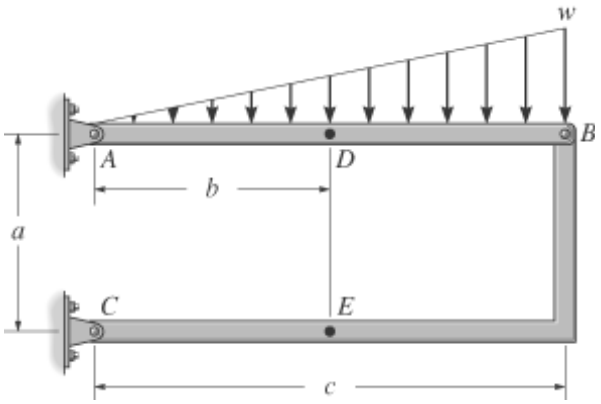
Given:

$w = 400\ \frac{\text{N}}{\text{m}}$

$a = 2.5\ \text{m}$

$b = 3\ \text{m}$

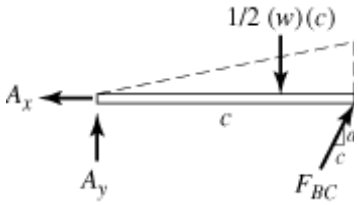
$c = 6\ \text{m}$



Solution:

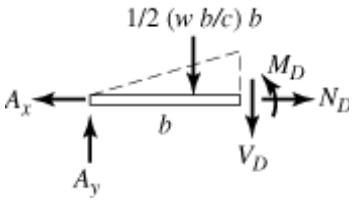
$\circlearrowleft + \Sigma M_A = 0;\quad \frac{-1}{2}wc\left(\frac{2}{3}c\right) + F_{BC}\left(\frac{a}{\sqrt{a^2+c^2}}\right)c = 0$

$F_{BC} = \frac{1}{3}\,wc\left(\frac{\sqrt{a^2+c^2}}{a}\right) \qquad F_{BC} = 2080\ \text{N}$



$\xrightarrow{+} \Sigma F_x = 0;\quad \left(\frac{c}{\sqrt{a^2+c^2}}\right)F_{BC} - A_x = 0$

$A_x = \left(\frac{c}{\sqrt{a^2+c^2}}\right)F_{BC} \qquad A_x = 1920\ \text{N}$



$+\uparrow \Sigma F_y = 0;\quad A_y - \frac{1}{2}wc + \left(\frac{a}{\sqrt{a^2+c^2}}\right)F_{BC} = 0$

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$$A_y = \frac{1}{2} w c - \left(\frac{a}{\sqrt{a^2 + c^2}} \right) F_{BC} \qquad A_y = 400 \text{ N}$$

$$\xrightarrow{+} \Sigma F_x = 0; \quad N_D - A_x = 0 \qquad N_D = A_x \qquad N_D = 1.920 \text{ kN}$$

$$+\uparrow \Sigma F_y = 0; \quad A_y - \frac{1}{2}\left(w \frac{b}{c} \right) b - V_D = 0$$

$$V_D = A_y - \frac{1}{2} w \left(\frac{b^2}{c} \right) \qquad V_D = 100 \text{ N}$$

$$\circlearrowleft + \Sigma F_y = 0; \quad -A_y b + \frac{1}{2}\left(w \frac{b}{c} \right) b \left(\frac{b}{3} \right) + M_D = 0$$

$$M_D = A_y b - \frac{1}{6} w \left(\frac{b^3}{c} \right) \qquad M_D = 900 \text{ N m}$$

Problem 7-15

The beam has weight density γ. Determine the internal normal force, shear force, and moment at point C.

Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

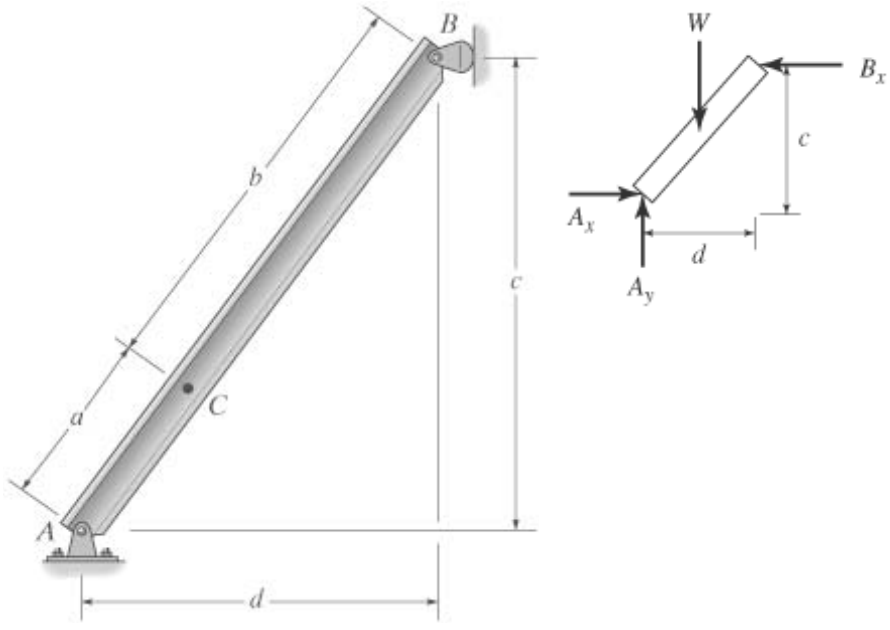
$$\gamma = 280 \frac{\text{lb}}{\text{ft}}$$

$$a = 3 \text{ ft}$$

$$b = 7 \text{ ft}$$

$$c = 8 \text{ ft}$$

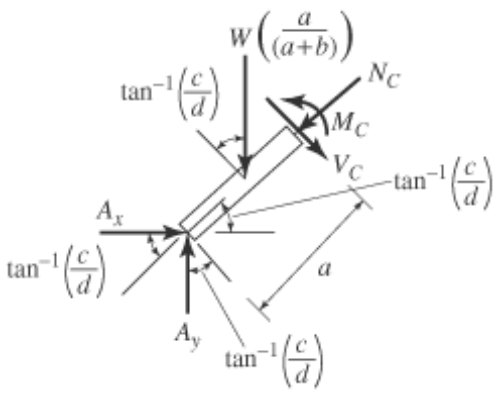
$$d = 6 \text{ ft}$$



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Solution:

$\theta = \text{atan}\left(\frac{c}{d}\right)$ $W = \gamma\,(a+b)$



Guesses

$A_x = 1\ \text{lb}$ $A_y = 1\ \text{lb}$ $B_x = 1\ \text{lb}$

$N_C = 1\ \text{lb}$ $V_C = 1\ \text{lb}$ $M_C = 1\ \text{lb ft}$

Given

Entire beam:

$A_x - B_x = 0$ $A_y - W = 0$ $B_x c - W\left(\frac{d}{2}\right) = 0$

Bottom Section

$A_x - N_C\cos(\theta) + V_C\sin(\theta) = 0$

$A_y - W\left(\frac{a}{a+b}\right) - N_C\sin(\theta) - V_C\cos(\theta) = 0$

$M_C - V_C a - W\left(\frac{a}{a+b}\right)\left(\frac{a}{2}\right)\cos(\theta) = 0$

$\begin{pmatrix} A_x \\ A_y \\ B_x \\ N_C \\ V_C \\ M_C \end{pmatrix} = \text{Find}(A_x, A_y, B_x, N_C, V_C, M_C)$ $\begin{pmatrix} A_x \\ A_y \\ B_x \end{pmatrix} = \begin{pmatrix} 1.05 \\ 2.80 \\ 1.05 \end{pmatrix}\text{kip}$ $\begin{pmatrix} N_C \\ V_C \end{pmatrix} = \begin{pmatrix} 2.20 \\ 0.34 \end{pmatrix}\text{kip}$

$M_C = 1.76\,\text{kip}\cdot\text{ft}$

Problem 7-16

Determine the internal normal force, shear force, and moment at points *C* and *D* of the beam.

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Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

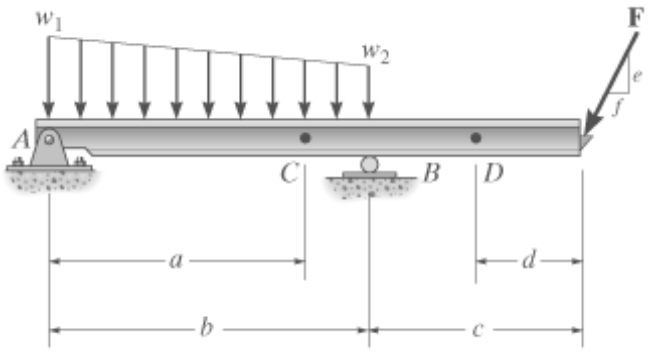
$w_1 = 60 \dfrac{\text{lb}}{\text{ft}}$ $a = 12$ ft

$b = 15$ ft

$w_2 = 40 \dfrac{\text{lb}}{\text{ft}}$ $c = 10$ ft

$F = 690$ lb $d = 5$ ft

$e = 12$ $f = 5$



Solution: $\theta = \text{atan}\left(\dfrac{e}{f}\right)$

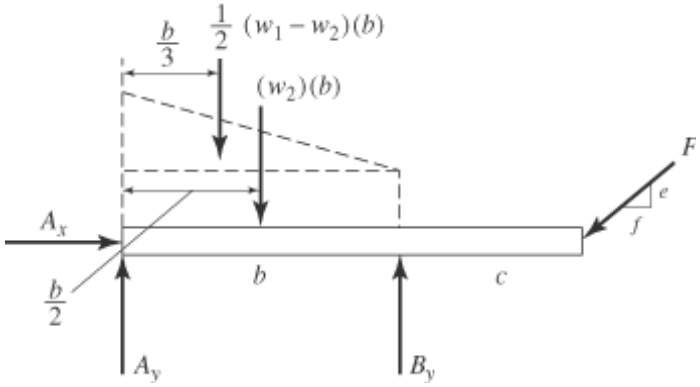
Guesses

$B_y = 1$ lb $N_C = 1$ lb

$V_C = 1$ lb $M_C = 1$ lb·ft

$N_D = 1$ lb $V_D = 1$ lb

$M_D = 1$ lb·ft



Given

$$B_y b - F\sin(\theta)(b + c) - w_2 b\left(\frac{b}{2}\right) \ldots = 0$$

$$+ \frac{-1}{2}(w_1 - w_2)b\left(\frac{b}{3}\right)$$

$$-N_C - F\cos(\theta) = 0$$

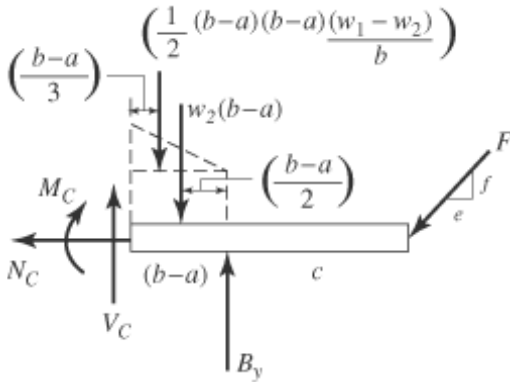
$$V_C - \frac{1}{2}(w_1 - w_2)\left(\frac{b - a}{b}\right)(b - a) \ldots = 0$$

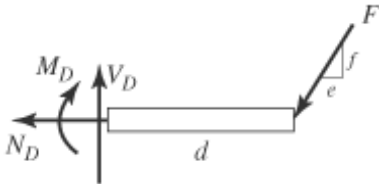
$$+ B_y - w_2(b - a) - F\sin(\theta)$$

$$-N_D - F\cos(\theta) = 0$$

$$V_D - F\sin(\theta) = 0$$

$$-M_D - F\sin(\theta)d = 0$$





$$-M_C - w_2\,(b - a)\left(\frac{b - a}{2}\right) - \frac{1}{2}(w_1 - w_2)\left(\frac{b - a}{b}\right)(b - a)\left(\frac{b - a}{3}\right) \ldots = 0$$

$$+ B_y\,(b - a) - F\sin(\theta)(c + b - a)$$

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$$\begin{pmatrix} B_y \\ N_C \\ V_C \\ M_C \\ N_D \\ V_D \\ M_D \end{pmatrix} = \text{Find}\left(B_y, N_C, V_C, M_C, N_D, V_D, M_D\right)$$

$$\begin{pmatrix} N_C \\ V_C \\ N_D \\ V_D \end{pmatrix} = \begin{pmatrix} -265 \\ -649 \\ -265 \\ 637 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} M_C \\ M_D \end{pmatrix} = \begin{pmatrix} -4.23 \\ -3.18 \end{pmatrix} \text{kip}\cdot\text{ft}$$

Problem 7-17

Determine the normal force, shear force, and moment acting at a section passing through point *C*.

Units Used: $\text{kip} = 10^3\ \text{lb}$

Given:

$F_1 = 800\ \text{lb}$

$F_2 = 700\ \text{lb}$

$F_3 = 600\ \text{lb}$

$\theta = 30\ \text{deg}$

$a = 1.5\ \text{ft}$

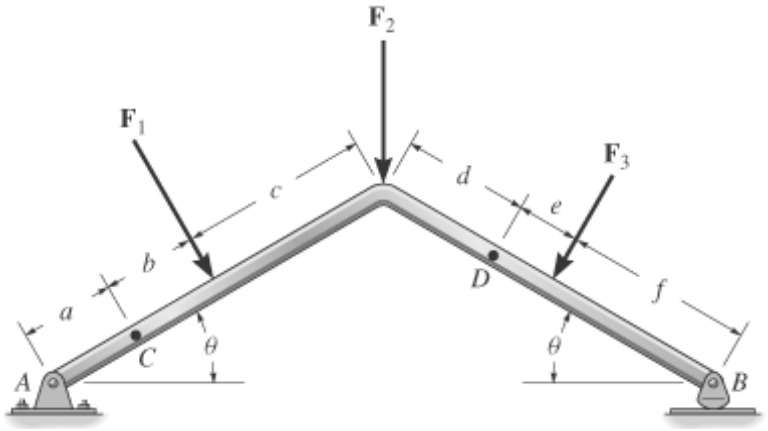
$b = 1.5\ \text{ft}$

$c = 3\ \text{ft}$

$d = 2\ \text{ft}$

$e = 1\ \text{ft}$

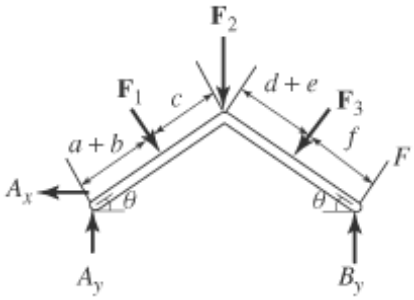
$f = a + b + c - d - e$



Solution:

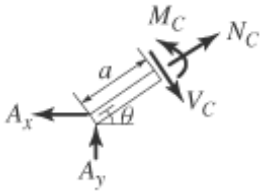
Guesses $B_y = 1\ \text{lb}$ $A_x = 1\ \text{lb}$ $A_y = 1\ \text{lb}$

$N_C = 1\ \text{lb}$ $V_C = 1\ \text{lb}$ $M_C = 1\ \text{lb}\cdot\text{ft}$



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Given



$$-A_x + V_C \sin(\theta) + N_C \cos(\theta) = 0$$

$$A_y - V_C \cos(\theta) + N_C \sin(\theta) = 0$$

$$M_C - A_x\,(a)\sin(\theta) - A_y\,(a)\cos(\theta) = 0$$

$$-A_x + F_1 \sin(\theta) - F_3 \sin(\theta) = 0$$

$$A_y + B_y - F_2 - F_1 \cos(\theta) - F_3 \cos(\theta) = 0$$

$$-F_1(a+b) - F_2(a+b+c)\cos(\theta) - F_3 \cos(\theta)(a+b+c+d+e)\cos(\theta) \ \ldots = 0$$
$$+ F_3 \sin(\theta) f \sin(\theta) + B_y 2(a+b+c)\cos(\theta)$$

$$\begin{pmatrix} A_x \\ A_y \\ B_y \\ N_C \\ V_C \\ M_C \end{pmatrix} = \text{Find}\left(A_x, A_y, B_y, N_C, V_C, M_C\right)$$

$$\begin{pmatrix} A_x \\ A_y \\ B_y \end{pmatrix} = \begin{pmatrix} 100 \\ 985 \\ 927 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} N_C \\ V_C \end{pmatrix} = \begin{pmatrix} -406 \\ 903 \end{pmatrix} \text{lb}$$

$$M_C = 1.355\ \text{kip}\cdot\text{ft}$$

Problem 7-18

Determine the normal force, shear force, and moment acting at a section passing through point *D*.

Units Used: $\text{kip} = 10^3\ \text{lb}$

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Given:

$F_1 = 800$ lb

$F_2 = 700$ lb

$F_3 = 600$ lb

$\theta = 30$ deg

$a = 1.5$ ft

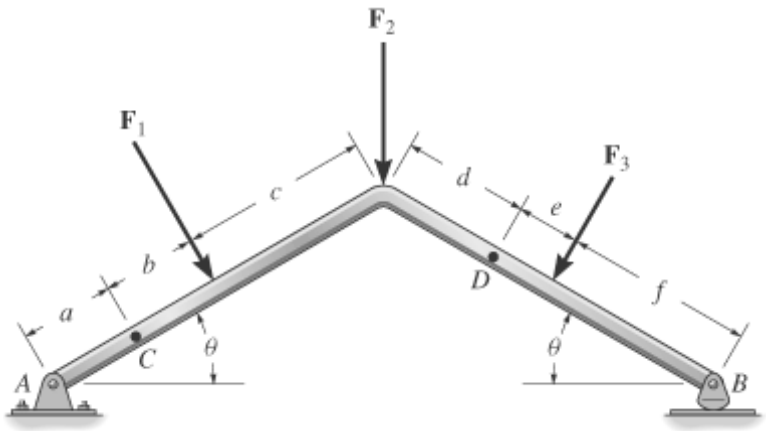
$b = 1.5$ ft

$c = 3$ ft

$d = 2$ ft

$e = 1$ ft

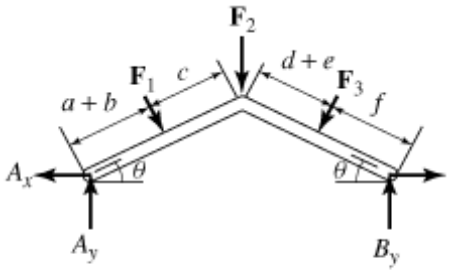
$f = a + b + c - d - e$



Solution:

Guesses $B_y = 1$ lb $A_x = 1$ lb $A_y = 1$ lb

$N_D = 1$ lb $V_D = 1$ lb $M_D = 1$ lb·ft



Given

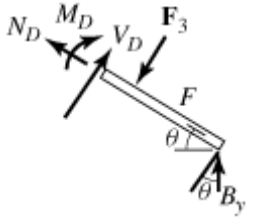
$V_D \sin(\theta) - N_D \cos(\theta) - F_3 \sin(\theta) = 0$

$B_y + V_D \cos(\theta) + N_D \sin(\theta) - F_3 \cos(\theta) = 0$

$-M_D - F_3 e + B_y(e + f)\cos(\theta) = 0$

$A_y + B_y - F_2 - F_1 \cos(\theta) - F_3 \cos(\theta) = 0$

$-A_x + F_1 \sin(\theta) - F_3 \sin(\theta) = 0$



$-F_1(a + b) - F_2(a + b + c)\cos(\theta) - F_3 \cos(\theta)(a + b + c + d + e)\cos(\theta) \ldots = 0$
$+ F_3 \sin(\theta) f \sin(\theta) + B_y 2(a + b + c)\cos(\theta)$

$$\begin{pmatrix} A_x \\ A_y \\ B_y \\ N_D \\ V_D \\ M_D \end{pmatrix} = \text{Find}(A_x, A_y, B_y, N_D, V_D, M_D)$$

$$\begin{pmatrix} A_x \\ A_y \\ B_y \end{pmatrix} = \begin{pmatrix} 100 \\ 985 \\ 927 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} N_D \\ V_D \end{pmatrix} = \begin{pmatrix} -464 \\ -203 \end{pmatrix} \text{lb}$$

$M_D = 2.61$ kip·ft

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Problem 7-19

Determine the normal force, shear force, and moment at a section passing through point *C*.

Units Used:

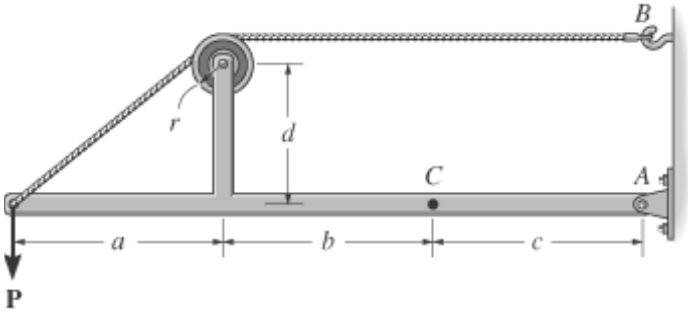
$\text{kN} = 10^3\ \text{N}$

Given:

$P = 8\ \text{kN}$ $c = 0.75\ \text{m}$

$a = 0.75\text{m}$ $d = 0.5\ \text{m}$

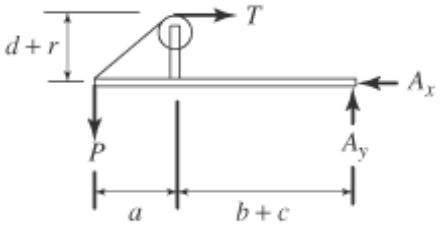
$b = 0.75\text{m}$ $r = 0.1\ \text{m}$



Solution:

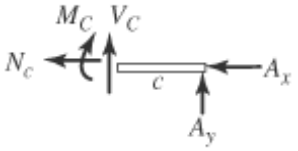
$\curvearrowleft + \Sigma M_A = 0;$ $-T\,(d + r) + P(a + b + c) = 0$

$T = P\left(\dfrac{a + b + c}{d + r}\right)$ $T = 30\,\text{kN}$



$\xrightarrow{+} \Sigma F_x = 0;$ $A_x = T$ $A_x = 30\,\text{kN}$

$+\uparrow \Sigma F_y = 0;$ $A_y = P$ $A_y = 8\,\text{kN}$



$\xrightarrow{+} \Sigma F_x = 0;$ $-N_C - T = 0$

$N_C = -T$ $N_C = -30\,\text{kN}$

$+\uparrow \Sigma F_y = 0;$ $V_C + P = 0$

$V_C = -P$ $V_C = -8\,\text{kN}$

$\curvearrowleft + \Sigma M_C = 0;$ $-M_C + P\,c = 0$

$M_C = P\,c$ $M_C = 6\,\text{kN}\cdot\text{m}$

Problem 7-20

The cable will fail when subjected to a tension T_{max}. Determine the largest vertical load P the frame will support and calculate the internal normal force, shear force, and moment at a section passing through point *C* for this loading.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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Given:

$T_{max} = 2$ kN

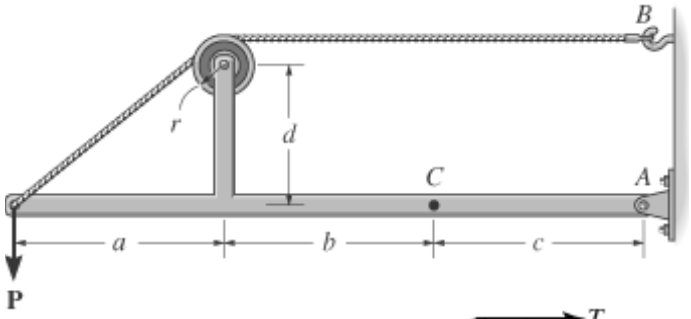
$a = 0.75$ m

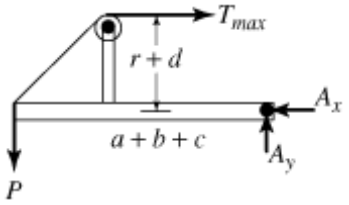
$b = 0.75$ m

$c = 0.75$ m

$d = 0.5$ m

$r = 0.1$ m

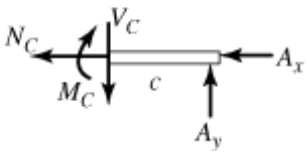




Solution:

$\circlearrowleft \Sigma M_A = 0;$ $\quad -T_{max}(r + d) + P(a + b + c) = 0$

$$P = T_{max}\left(\frac{d + r}{a + b + c}\right) \qquad P = 0.533\,\text{kN}$$



$\xrightarrow{+} \Sigma F_x = 0;$ $\quad T_{max} - A_x = 0$ $\quad A_x = T_{max}$ $\quad A_x = 2\,\text{kN}$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - P = 0$ $\quad A_y = P$ $\quad A_y = 0.533\,\text{kN}$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad -N_C - A_x = 0$ $\quad N_C = -A_x$ $\quad N_C = -2\,\text{kN}$

$+\uparrow \Sigma F_y = 0;$ $\quad -V_C + A_y = 0$ $\quad V_C = A_y$ $\quad V_C = 0.533\,\text{kN}$

$\circlearrowleft \Sigma M_C = 0;$ $\quad -M_C + A_y\, c = 0$ $\quad M_C = A_y\, c$ $\quad M_C = 0.400\,\text{kN}\cdot\text{m}$

Problem 7-21

Determine the internal shear force and moment acting at point *C* of the beam.

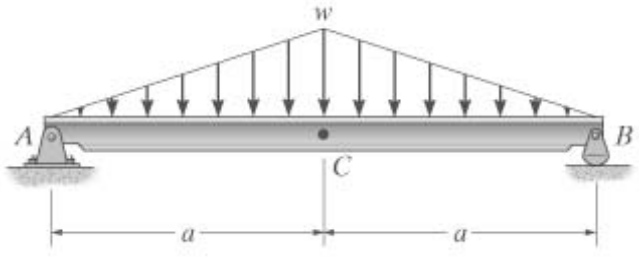
Units Used:

kip $= 10^3$ lb

Given:

$w = 2\ \dfrac{\text{kip}}{\text{ft}}$

$a = 9$ ft



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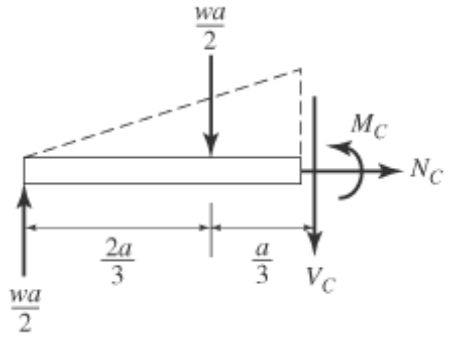
Solution:

$\Sigma F_x = 0;\quad N_C = 0 \qquad N_C = 0$

$\Sigma F_y = 0;\quad \frac{wa}{2} - \frac{wa}{2} - V_C = 0 \qquad V_C = 0$

$\Sigma M_C = 0;\quad M_C - \left(\frac{wa}{2}\right)a + \left(\frac{wa}{2}\right)\left(\frac{a}{3}\right) = 0$

$M_C = \frac{wa^2}{3} \qquad M_C = 54.00\ \text{kip}\cdot\text{ft}$



Problem 7-22

Determine the internal shear force and moment acting at point *D* of the beam.

Units Used:

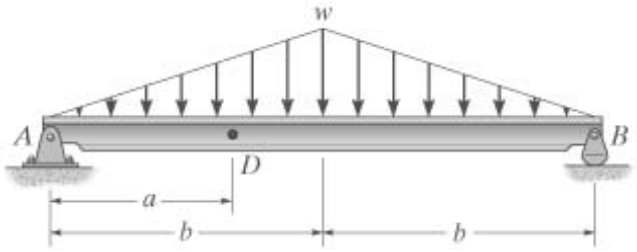
$\text{kip} = 10^3\ \text{lb}$

Given:

$w = 2\ \frac{\text{kip}}{\text{ft}}$

$a = 6\ \text{ft}$

$b = 9\ \text{ft}$



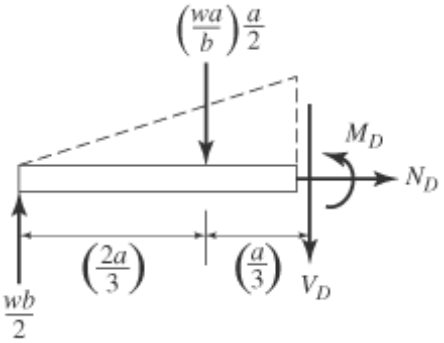
Solution:

$\Sigma F_x = 0;\quad N_D = 0$

$\Sigma F_y = 0;\quad \frac{wb}{2} - w\left(\frac{a}{b}\right)\left(\frac{a}{2}\right) - V_D = 0$

$V_D = \frac{wb}{2} - w\left(\frac{a}{b}\right)\left(\frac{a}{2}\right)$

$V_D = 5.00\ \text{kip}$



$\Sigma M_D = 0;\quad M_D - \left(\frac{wb}{2}\right)a + \left(\frac{wa}{b}\right)\left(\frac{a}{2}\right)\left(\frac{a}{3}\right) = 0$

$M_D = \left(\frac{wb}{2}\right)a - \left(\frac{wa^3}{6b}\right)$

$M_D = 46.00\ \text{kip}\cdot\text{ft}$

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Problem 7-23

The shaft is supported by a journal bearing at *A* and a thrust bearing at *B*. Determine the internal normal force, shear force, and moment at (a) point *C*, which is just to the right of the bearing at *A*, and (b) point *D*, which is just to the left of the $\mathbf{F}_2$ force.

Units Used:

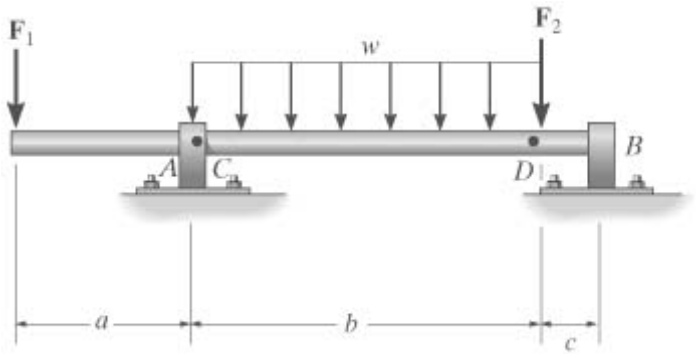
$\text{kip} = 10^3 \text{ lb}$

Given:

$F_1 = 2500 \text{ lb} \quad a = 6 \text{ ft}$

$F_2 = 3000 \text{ lb} \quad b = 12 \text{ ft}$

$w = 75 \dfrac{\text{lb}}{\text{ft}} \quad c = 2 \text{ft}$



Solution:

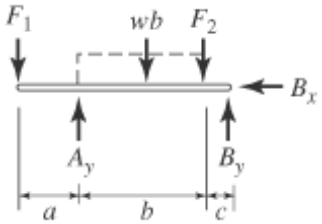
$\Sigma M_B = 0; \quad -A_y(b + c) + F_1(a + b + c) + wb\left(\frac{b}{2} + c\right) + F_2 c = 0$

$A_y = \frac{1}{2}\,\frac{2F_1(a + b + c) + w\left(b^2 + 2cb\right) + 2F_2 c}{b + c} \qquad A_y = 4514 \text{ lb}$

$\Sigma F_x = 0; \quad B_x = 0 \text{ lb}$

$\Sigma F_y = 0; \quad A_y - F_1 - wb - F_2 + B_y = 0$

$B_y = -A_y + F_1 + wb + F_2 \qquad B_y = 1886 \text{ lb}$



Segment *AC* :

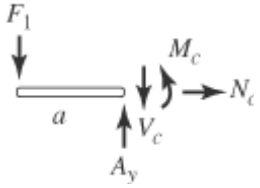
$\Sigma M_C = 0; \quad F_1 a + M_C = 0$

$M_C = -F_1 a \qquad M_C = -15 \text{ kip}\cdot\text{ft}$

$\Sigma F_x = 0; \quad N_C = 0 \qquad N_C = 0$

$\Sigma F_y = 0; \quad -F_1 + A_y - V_C = 0$

$V_C = A_y - F_1 \qquad V_C = 2.01 \text{ kip}$



Segment *BD*:

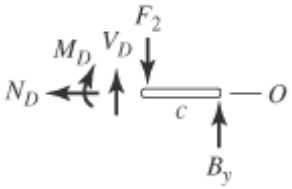
$\Sigma M_D = 0; \quad -M_D + B_y c = 0$

$M_D = B_y c \qquad M_D = 3.77 \text{ kip}\cdot\text{ft}$

$\Sigma F_x = 0; \quad N_D = 0 \qquad N_D = 0$

$\Sigma F_y = 0; \quad V_D - F_2 + B_y = 0$

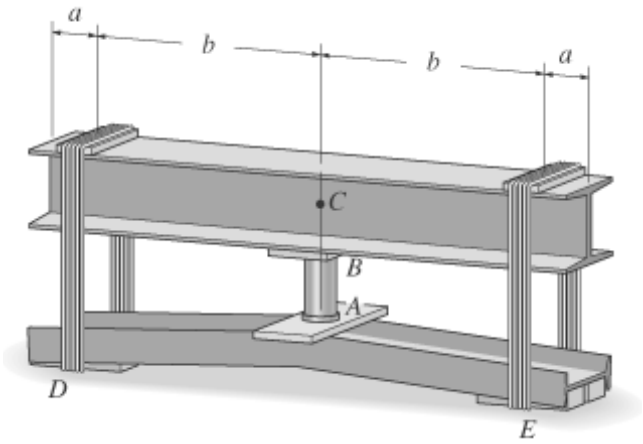
$V_D = F_2 - B_y \qquad V_D = 1.11 \text{ kip}$



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Problem 7-24

The jack *AB* is used to straighten the bent beam *DE* using the arrangement shown. If the axial compressive force in the jack is *P*, determine the internal moment developed at point *C* of the top beam. Neglect the weight of the beams.



Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

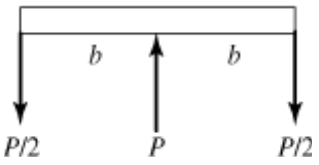
$$P = 5000 \text{ lb}$$

$$a = 2 \text{ ft}$$

$$b = 10 \text{ ft}$$

Solution:

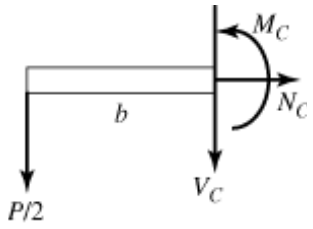
Segment:



$$\circlearrowleft + \Sigma M_C = 0; \quad M_C + \left(\frac{P}{2}\right) b = 0$$

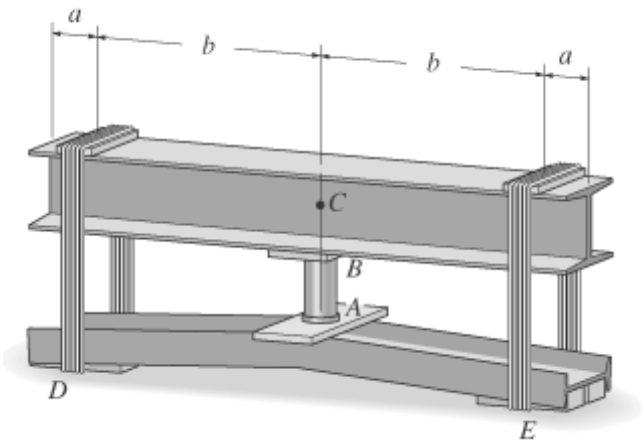
$$M_C = -\frac{P}{2} b$$

$$M_C = -25.00 \text{ kip} \cdot \text{ft}$$



Problem 7-25

The jack *AB* is used to straighten the bent beam *DE* using the arrangement shown. If the axial compressive force in the jack is *P*, determine the internal moment developed at point *C* of the top beam. Assume that each beam has a uniform weight density γ.



Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

$$P = 5000 \text{ lb}$$

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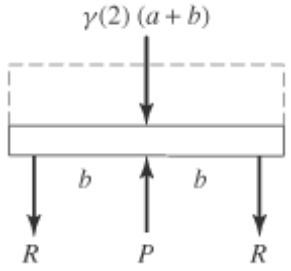
$$\gamma = 150\ \frac{\text{lb}}{\text{ft}}$$

$$a = 2\ \text{ft}$$

$$b = 10\ \text{ft}$$

Solution:

Beam:

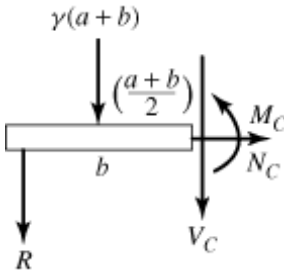


$$+\uparrow \Sigma F_y = 0; \qquad P - 2\gamma(a+b) - 2R = 0$$

$$R = \frac{P}{2} - \gamma(a+b)$$

$$R = 700\,\text{lb}$$

Segment:



$$\circlearrowleft + \Sigma M_C = 0; \qquad M_C + R\,b + \gamma\,(a+b)\left(\frac{a+b}{2}\right) = 0$$

$$M_C = -R\,b - \gamma\,\frac{(a+b)^2}{2}$$

$$M_C = -17.8\,\text{kip}\cdot\text{ft}$$

Problem 7-26

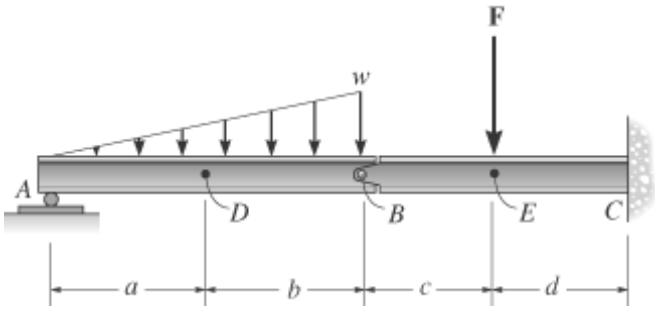
Determine the normal force, shear force, and moment in the beam at sections passing through points *D* and *E*. Point *E* is just to the right of the *F* load.

Units Used:

$$\text{kip} = 10^3\ \text{lb}$$

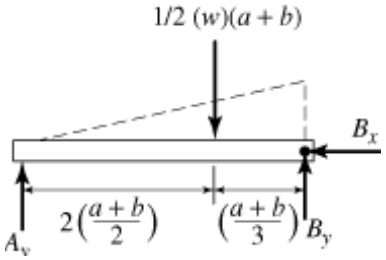
Given:

$$w = 1.5\ \frac{\text{kip}}{\text{ft}} \qquad a = 6\ \text{ft} \qquad b = 6\ \text{ft} \qquad c = 4\ \text{ft} \qquad F = 3\ \text{kip} \qquad d = 4\ \text{ft}$$



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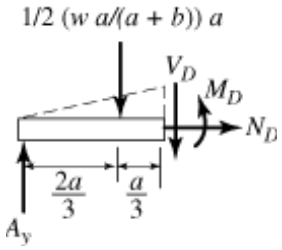
Solution:



$\curvearrowleft \Sigma M_B = 0;$ $\quad \frac{1}{2}w(a+b)\left(\frac{a+b}{3}\right) - A_y(a+b) = 0$

$$A_y = \frac{\frac{1}{2}w(a+b)\left(\frac{a+b}{3}\right)}{a+b}$$

$$A_y = 3\,\text{kip}$$

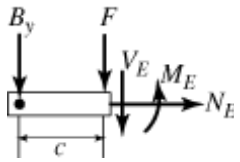


$\xrightarrow{+} \Sigma F_x = 0;$ $\quad B_x = 0$

$+\uparrow \Sigma F_y = 0;$ $\quad B_y + A_y - \frac{1}{2}w(a+b) = 0$

$$B_y = -A_y + \frac{1}{2}w(a+b)$$

$$B_y = 6\,\text{kip}$$



$\xrightarrow{+} \Sigma F_x = 0;$ $\quad N_D = 0$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - \frac{1}{2}\left(\frac{a\,w}{a+b}\right)a - V_D = 0$

$$V_D = A_y - \frac{1}{2}\left(\frac{a\,w}{a+b}\right)a \qquad V_D = 0.75\,\text{kip}$$

$\curvearrowleft \Sigma M_D = 0;$ $\quad M_D + \frac{1}{2}\left(\frac{a\,w}{a+b}\right)a\left(\frac{a}{3}\right) - A_y\,a = 0$

$$M_D = \frac{-1}{2}\left(\frac{a\,w}{a+b}\right)a\left(\frac{a}{3}\right) + A_y\,a \qquad M_D = 13.5\,\text{kip}\cdot\text{ft}$$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad N_E = 0$

$+\uparrow \Sigma F_y = 0;$ $\quad -V_E - F - B_y = 0$

$$V_E = -F - B_y \qquad V_E = -9\,\text{kip}$$

$\Sigma M_E = 0;$ $\quad M_E + B_y\,c = 0$

$$M_E = -B_y\,c \qquad M_E = -24.0\,\text{kip}\cdot\text{ft}$$

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Problem 7-27

Determine the normal force, shear force, and moment at a section passing through point *D* of the two-member frame.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

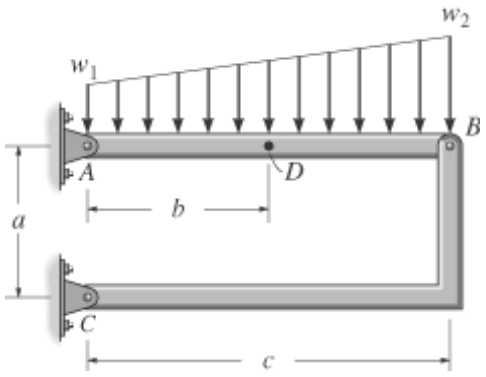
$w_1 = 200\ \frac{\text{N}}{\text{m}}$

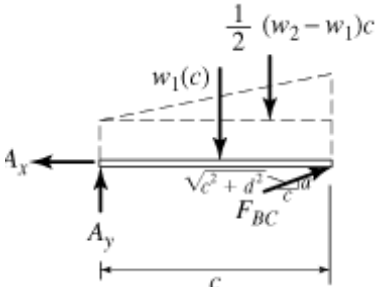
$w_2 = 400\ \frac{\text{N}}{\text{m}}$

$a = 2.5\ \text{m}$

$b = 3\ \text{m}$

$c = 6\ \text{m}$

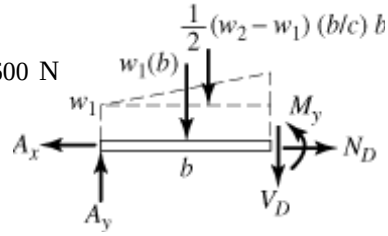




Solution:

$$\circlearrowleft + \Sigma M_A = 0; \qquad -w_1 c\left(\frac{c}{2}\right) - \frac{1}{2}(w_2 - w_1)c\left(\frac{2c}{3}\right) + \frac{a}{\sqrt{a^2 + c^2}}(F_{BC}\ c) = 0$$

$$F_{BC} = \left[w_1 \frac{c^2}{2} + (w_2 - w_1)\frac{c^2}{3}\right]\frac{\sqrt{a^2 + c^2}}{a\,c} \qquad F_{BC} = 2600\ \text{N}$$



$$\xrightarrow{+} \Sigma F_x = 0; \qquad A_x = \left(\frac{c}{\sqrt{a^2 + c^2}}\right)F_{BC} \qquad A_x = 2400\ \text{N}$$

$$+\uparrow \Sigma F_y = 0; \qquad A_y - w_1 c - \frac{1}{2}(w_2 - w_1)c + \left(\frac{a}{\sqrt{a^2 + c^2}}\right)F_{BC} = 0$$

$$A_y = w_1 c + \frac{1}{2}(w_2 - w_1)c - \left(\frac{a}{\sqrt{a^2 + c^2}}\right)F_{BC} \qquad A_y = 800\ \text{N}$$

$$\xrightarrow{+} \Sigma F_x = 0; \qquad -A_x + N_D = 0 \qquad N_D = A_x \qquad N_D = 2.40\ \text{kN}$$

$$+\uparrow \Sigma F_y = 0; \qquad A_y - w_1 b - \frac{1}{2}(w_2 - w_1)\left(\frac{b}{c}\right)b - V_D = 0$$

$$V_D = A_y - w_1 b - \frac{1}{2}(w_2 - w_1)\left(\frac{b^2}{c}\right) \qquad V_D = 50\ \text{N}$$

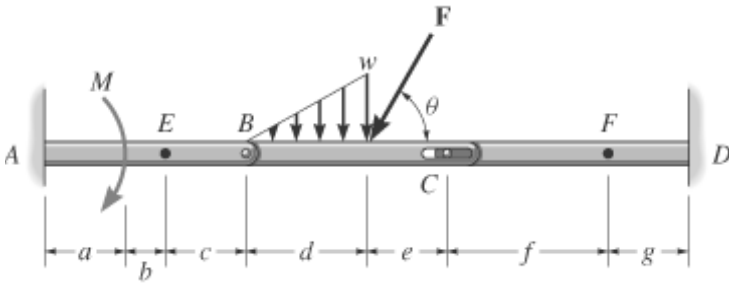
$$\circlearrowleft + \Sigma M_D = 0; \qquad -A_y b + w_1\ b\left(\frac{b}{2}\right) + \frac{1}{2}(w_2 - w_1)\left(\frac{b}{c}\right)\ b\left(\frac{b}{3}\right) + M_D = 0$$

$$M_D = A_y(b) - w_1\left(\frac{b^2}{2}\right) - \frac{1}{2}(w_2 - w_1)\left(\frac{b^3}{3c}\right) \qquad M_D = 1.35\ \text{kN}\cdot\text{m}$$

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Problem 7-28

Determine the normal force, shear force, and moment at sections passing through points E and F. Member BC is pinned at B and there is a smooth slot in it at C. The pin at C is fixed to member CD.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$M = 350 \text{ lb}\cdot\text{ft}$

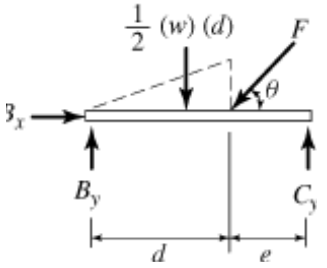
$w = 80 \dfrac{\text{lb}}{\text{ft}}$ $\quad c = 2 \text{ ft}$

$F = 500 \text{ lb}$ $\quad d = 3 \text{ ft}$

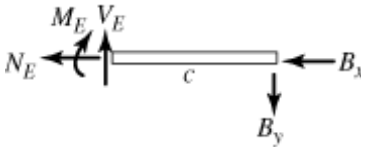
$\theta = 60 \text{ deg}$ $\quad e = 2 \text{ ft}$

$a = 2 \text{ ft}$ $\quad f = 4 \text{ ft}$

$b = 1 \text{ ft}$ $\quad g = 2 \text{ ft}$



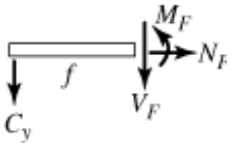
Solution:



$\circlearrowleft + \Sigma M_B = 0;$

$$\frac{-1}{2} w d \left(\frac{2d}{3}\right) - F\sin(\theta) d + C_y(d+e) = 0$$

$$C_y = \frac{\left(\dfrac{w d^2}{3}\right) + F\sin(\theta) d}{d+e} \qquad C_y = 307.8 \text{ lb}$$



$\xrightarrow{+} \Sigma F_x = 0;$ $\quad B_x - F\cos(\theta) = 0$ $\quad B_x = F\cos(\theta)$ $\quad B_x = 250 \text{ lb}$

$+\uparrow \Sigma F_y = 0;$ $\quad B_y - \dfrac{1}{2} w d - F\sin(\theta) + C_y = 0$

$$B_y = \frac{1}{2} w d + F\sin(\theta) - C_y \qquad B_y = 245.2 \text{ lb}$$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad -N_E - B_x = 0$ $\quad N_E = -B_x$ $\quad N_E = -250 \text{ lb}$

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$+\uparrow \Sigma F_y = 0;$	$V_E - B_y = 0$	$V_E = B_y$	$V_E = 245\,\text{lb}$
$\circlearrowleft \Sigma M_E = 0$	$-M_E - B_y c = 0$	$M_E = -B_y c$	$M_E = -490\,\text{lb}\cdot\text{ft}$
$\xrightarrow{+} \Sigma F_x = 0;$	$N_F = 0$	$N_F = 0\text{lb}$	$N_F = 0.00\,\text{lb}$
$+\uparrow \Sigma F_y = 0;$	$-C_y - V_F = 0$	$V_F = -C_y$	$V_F = -308\,\text{lb}$
$\circlearrowleft \Sigma M_F = 0;$	$C_y(f) + M_F = 0$	$M_F = -fC_y$	$M_F = -1.23\,\text{kip}\cdot\text{ft}$

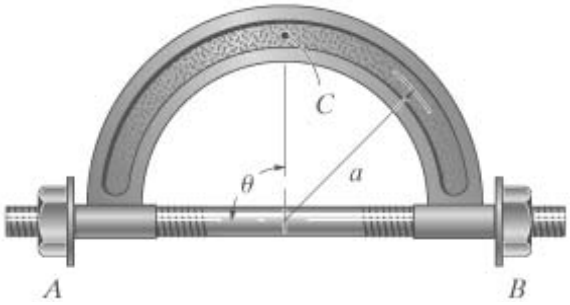
Problem 7-29

The bolt shank is subjected to a tension F. Determine the internal normal force, shear force, and moment at point C.

Given:

$F = 80$ lb

$a = 6$ in



Solution:

$\Sigma F_x = 0; \quad N_C + F = 0$

$N_C = -F$

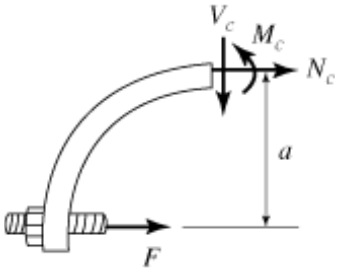
$N_C = -80.00\,\text{lb}$

$\Sigma F_y = 0; \quad V_C = 0$

$\Sigma M_C = 0; \quad M_C + F\,a = 0$

$M_C = -F\,a$

$M_C = -480.00\,\text{lb}\cdot\text{in}$



Problem 7-30

Determine the normal force, shear force, and moment acting at sections passing through points B and C on the curved rod.

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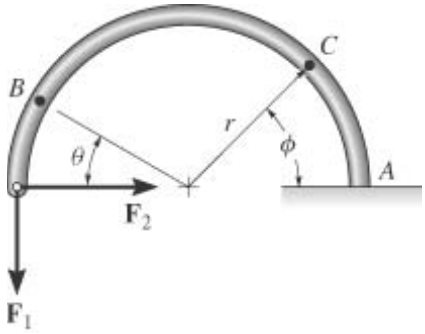
Units Used:

$kip = 10^3\ lb$

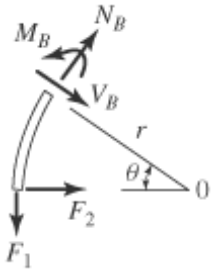
Given:

$F_1 = 300\ lb \quad \theta = 30\ deg \quad r = 2\ ft$

$F_2 = 400\ lb \quad \phi = 45\ deg$



Solution:



$\nearrow \Sigma F_x = 0; \quad F_2\sin(\theta) - F_1\cos(\theta) + N_B = 0$

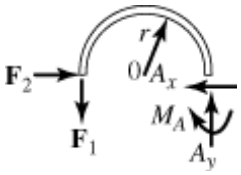
$N_B = -F_2\sin(\theta) + F_1\cos(\theta)$

$N_B = 59.8\ lb$

$\searrow \Sigma F_y = 0; \quad V_B + F_2\cos(\theta) + F_1\sin(\theta) = 0$

$V_B = -F_2\cos(\theta) - F_1\sin(\theta)$

$V_B = -496\ lb$



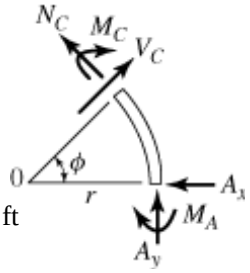
$\circlearrowleft \Sigma M_B = 0; \quad M_B + F_2 r\sin(\theta) + F_1(r - r\cos(\theta)) = 0$

$M_B = -F_2 r\sin(\theta) - F_1 r(1 - \cos(\theta))$

$M_B = -480\ lb\cdot ft$

$\rightarrow \Sigma F_x = 0; \quad F_2 - A_x = 0 \quad A_x = F_2 \quad A_x = 400\ lb$

$+\uparrow \Sigma F_y = 0; \quad A_y - F_1 = 0 \quad A_y = F_1 \quad A_y = 300\ lb$



$\circlearrowleft \Sigma M_A = 0; \quad -M_A + F_1 2r = 0 \quad M_A = 2F_1 r \quad M_A = 1200\ lb\cdot ft$

$\searrow \Sigma F_x = 0; \quad N_C + A_x\sin(\phi) + A_y\cos(\phi) = 0$

$N_C = -A_x\sin(\phi) - A_y\cos(\phi) \quad N_C = -495\ lb$

$\nearrow \Sigma F_y = 0; \quad V_C - A_x\cos(\phi) + A_y\sin(\phi) = 0$

$V_C = A_x\cos(\phi) - A_y\sin(\phi) \quad V_C = 70.7\ lb$

$\circlearrowleft \Sigma M_C = 0; \quad -M_C - M_A - A_x r\sin(\phi) + A_y(r - r\cos(\phi)) = 0$

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$$M_C = -M_A - A_x r\sin(\phi) + A_y r(1 - \cos(\phi)) \qquad M_C = -1.59\,\text{kip}\cdot\text{ft}$$

Problem 7-31

The cantilevered rack is used to support each end of a smooth pipe that has total weight W. Determine the normal force, shear force, and moment that act in the arm at its fixed support A along a vertical section.

Units Used:

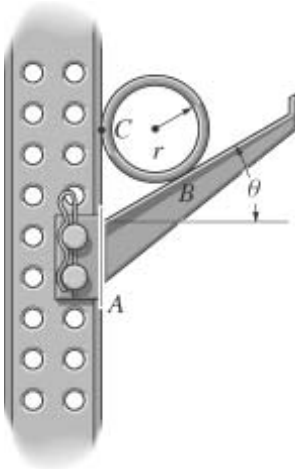
$\text{kip} = 10^3\ \text{lb}$

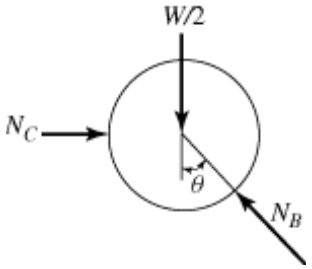
Given:

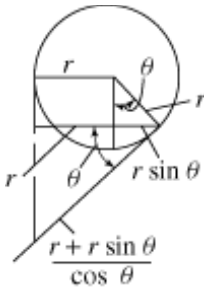
$W = 300\ \text{lb}$

$r = 6\ \text{in}$

$\theta = 30\ \text{deg}$





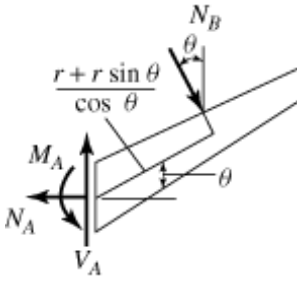


Solution:

Pipe:

$$+\uparrow \Sigma F_y = 0; \qquad N_B\cos(\theta) - \frac{W}{2} = 0$$

$$N_B = \frac{1}{2}\left(\frac{W}{\cos(\theta)}\right) \qquad N_B = 173.205\,\text{lb}$$



Rack:

$$\rightarrow \Sigma F_x = 0; \qquad -N_A + N_B\sin(\theta) = 0$$

$$N_A = N_B\sin(\theta) \qquad N_A = 86.6\,\text{lb}$$

$$+\uparrow \Sigma F_y = 0; \qquad V_A - N_B\cos(\theta) = 0$$

$$V_A = N_B\cos(\theta) \qquad V_A = 150\,\text{lb}$$

$$+\circlearrowleft \Sigma M_A = 0; \qquad M_A - N_B\left(\frac{r + r\sin(\theta)}{\cos(\theta)}\right) = 0$$

$$M_A = N_B\left(\frac{r + r\sin(\theta)}{\cos(\theta)}\right) \qquad M_A = 1.800\,\text{kip}\cdot\text{in}$$

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Problem 7-32

Determine the normal force, shear force, and moment at a section passing through point *D* of the two-member frame.

Units Used:

$$\text{kN} = 10^3\ \text{N}$$

Given:

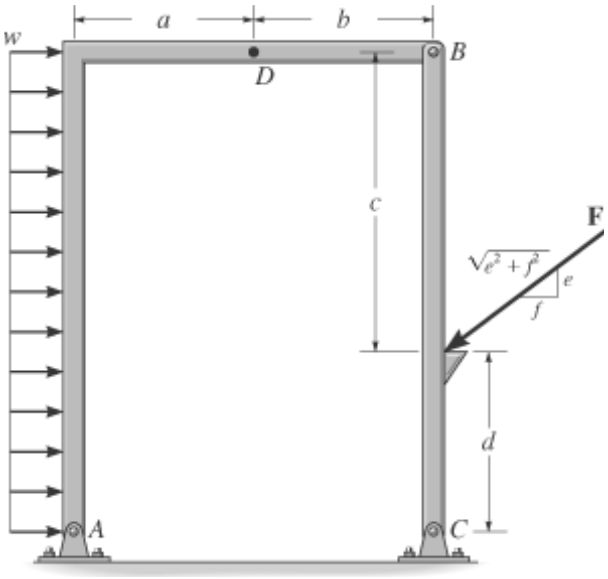
$$w = 0.75\ \frac{\text{kN}}{\text{m}}$$

$$F = 4\ \text{kN}$$

$$a = 1.5\ \text{m} \qquad d = 1.5\ \text{m}$$

$$b = 1.5\ \text{m} \qquad e = 3$$

$$c = 2.5\ \text{m} \qquad f = 4$$

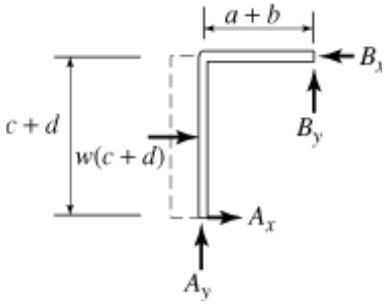


Solution:

$$\circlearrowright + \Sigma M_C = 0;$$

$$-B_x(c+d) + \left(\frac{f}{\sqrt{e^2+f^2}}\right) F\,d = 0$$

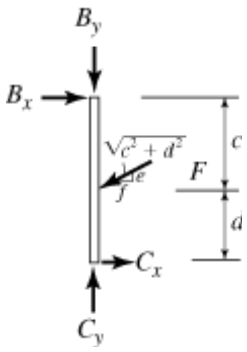
$$B_x = \frac{f\,d\,F}{\sqrt{e^2+f^2}\,(c+d)} \qquad B_x = 1.2\,\text{kN}$$



$$\circlearrowright + \Sigma M_A = 0;$$

$$-w(c+d)\left(\frac{c+d}{2}\right) + B_y(a+b) + B_x(c+d) = 0$$

$$B_y = \frac{w\left[\frac{(c+d)^2}{2}\right] - B_x(c+d)}{a+b} \qquad B_y = 0.40\,\text{kN}$$

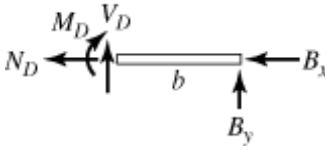


$$\xrightarrow{+} \Sigma F_x = 0; \qquad -N_D - B_x = 0 \qquad N_D = -B_x \qquad N_D = -1.2\,\text{kN}$$

$$+\uparrow \Sigma F_y = 0; \qquad V_D + B_y = 0 \qquad V_D = -B_y \qquad V_D = -0.4\,\text{kN}$$

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$\circlearrowleft + \Sigma M_D = 0; \quad -M_D + B_y\, b = 0 \quad M_D = B_y\, b \quad M_D = 0.6\,\text{kN}\cdot\text{m}$



Problem 7-33

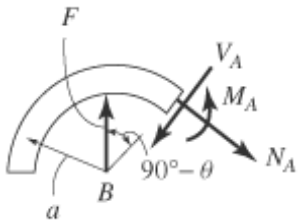
Determine the internal normal force, shear force, and moment acting at point *A* of the smooth hook.

Given:

$\theta = 45$ deg

$a = 2$ in

$F = 20$ lb





Solution:

$\Sigma F_x = 0; \quad N_A - F\cos(\theta) = 0$

$N_A = F\cos(\theta) \quad N_A = 14.1\,\text{lb}$

$\Sigma F_y = 0; \quad V_A - F\sin(\theta) = 0$

$V_A = F\sin(\theta) \quad V_A = 14.1\,\text{lb}$

$\Sigma M_B = 0; \quad M_A - N_A\, a = 0$

$M_A = N_A\, a \quad M_A = 28.3\,\text{lb}\cdot\text{in}$

Problem 7-34

Determine the internal normal force, shear force, and moment acting at points *B* and *C* on the curved rod.

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Units Used:

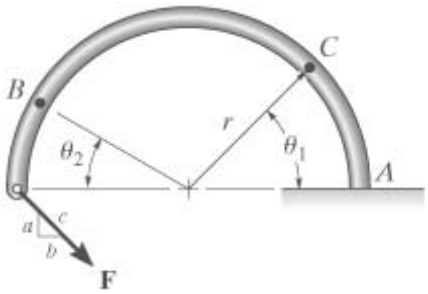
$kip = 10^3\ lb$

Given:

$F = 500\ lb$ $\theta_2 = 30\ deg$

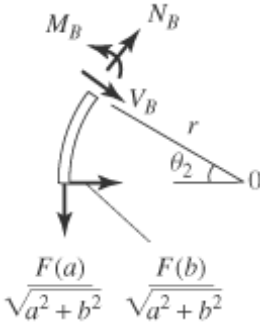
$r = 2\ ft$ $a = 3$

$\theta_1 = 45\ deg$ $b = 4$



Solution:

$$\Sigma F_N = 0; \quad \left(\frac{F\,b}{\sqrt{a^2+b^2}}\right)\sin(\theta_2) - \left(\frac{F\,a}{\sqrt{a^2+b^2}}\right)\cos(\theta_2) + N_B = 0$$



$$N_B = F\left[\frac{(a)\cos(\theta_2) - b\,\sin(\theta_2)}{\sqrt{a^2+b^2}}\right] \qquad N_B = 59.8\ lb$$

$$\Sigma F_V = 0; \quad V_B + \left(\frac{F\,b}{\sqrt{a^2+b^2}}\right)\cos(\theta_2) + \left(\frac{F\,a}{\sqrt{a^2+b^2}}\right)\sin(\theta_2) = 0$$

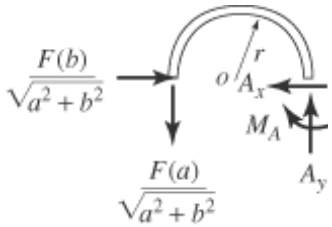
$$V_B = -F\left[\frac{b\cos(\theta_2) + (a)\sin(\theta_2)}{\sqrt{a^2+b^2}}\right] \qquad V_B = -496\ lb$$

$$\Sigma M_B = 0; \quad M_B + \left(\frac{F\,b}{\sqrt{a^2+b^2}}\right)r\sin(\theta_2) + F\left(\frac{a}{\sqrt{a^2+b^2}}\right)\left(r - r\cos(\theta_2)\right) = 0$$

$$M_B = F\,r\left[\frac{-b\sin(\theta_2) - a + (a)\cos(\theta_2)}{\sqrt{a^2+b^2}}\right] \qquad M_B = -480\ lb\cdot ft$$

Also,

$$\Sigma F_x = 0; \quad -A_x + \frac{F\,b}{\sqrt{a^2+b^2}} = 0$$

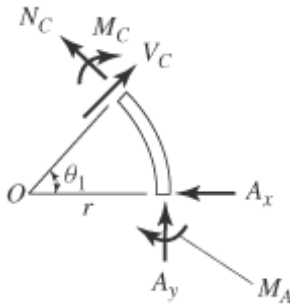


$$A_x = F\left(\frac{b}{\sqrt{a^2+b^2}}\right) \qquad A_x = 400.00\ lb$$

$$\Sigma F_y = 0; \quad A_y - \frac{F\,a}{\sqrt{a^2+b^2}} = 0$$

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$$A_y = F\left(\frac{a}{\sqrt{a^2 + b^2}}\right) \qquad A_y = 300.00\,\text{lb}$$



$$\Sigma M_A = 0; \quad -M_A + \left(\frac{F\,a}{\sqrt{a^2 + b^2}}\right)2r = 0$$

$$M_A = \frac{2F\,r\,a}{\sqrt{a^2 + b^2}} \qquad M_A = 1200\,\text{lb}\cdot\text{ft}$$

$$\Sigma F_x = 0; \quad N_C + A_x\sin(\theta_1) + A_y\cos(\theta_1) = 0$$

$$N_C = -A_x\sin(\theta_1) - A_y\cos(\theta_1) \qquad N_C = -495\,\text{lb}$$

$$\Sigma F_y = 0; \quad V_C - A_x\cos(\theta_1) + A_y\sin(\theta_1) = 0$$

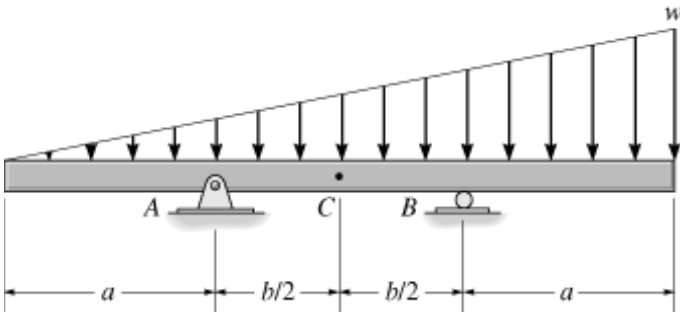
$$V_C = A_x\cos(\theta_1) - A_y\sin(\theta_1) \qquad V_C = 70.7\,\text{lb}$$

$$\Sigma M_C = 0; \quad -M_C - M_A + A_y(r - r\cos(\theta_1)) - A_x\,r\sin(\theta_1) = 0$$

$$M_C = -M_A + A_y(r - r\cos(\theta_1)) - A_x\,r\sin(\theta_1) \qquad M_C = -1.59\,\text{kip}\cdot\text{ft}$$

Problem 7-35

Determine the ratio *a/b* for which the shear force will be zero at the midpoint *C* of the beam.



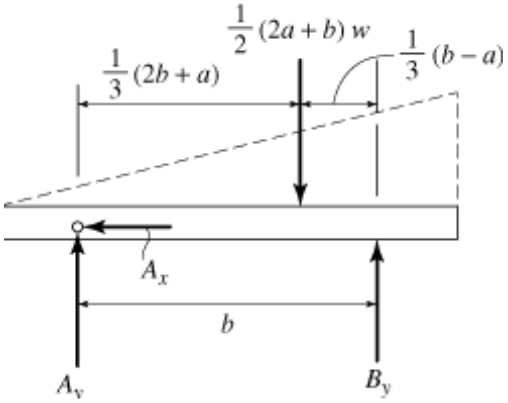
Solution:

Find A_y: $\circlearrowleft + \Sigma M_B = 0;$

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$$\frac{1}{2}(2a+b)\ w\left[\frac{1}{3}(b-a)\right] - A_y b = 0$$

$$A_y = \frac{w}{6b}(2a+b)(b-a)$$



This problem requires $V_C = 0$.

Summing forces vertically for the section, we have

$$+\uparrow \Sigma F_y = 0;$$

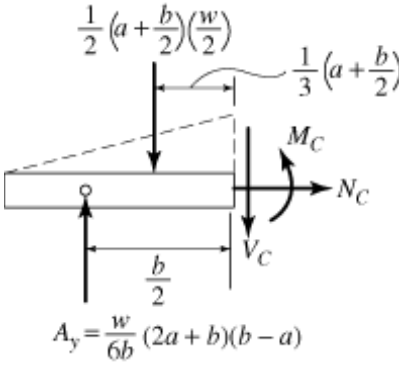
$$\frac{w}{6b}(2a+b)(b-a) - \frac{1}{2}\left(a+\frac{b}{2}\right)\frac{w}{2} = 0$$

$$\frac{w}{6b}(2a+b)(b-a) = \frac{w}{8}(2a+b)$$

$$4(b-a) = 3b$$

$$b = 4a$$

$$\frac{a}{b} = \frac{1}{4}$$

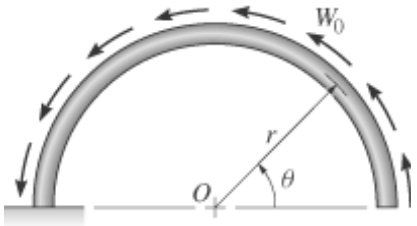


Problem 7-36

The semicircular arch is subjected to a uniform distributed load along its axis of w_0 per unit length. Determine the internal normal force, shear force, and moment in the arch at angle θ.

Given:

$\theta = 45$ deg

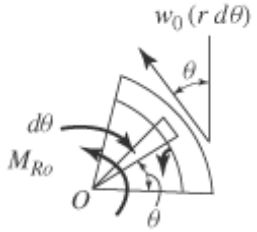


Solution:

Resultants of distributed load:

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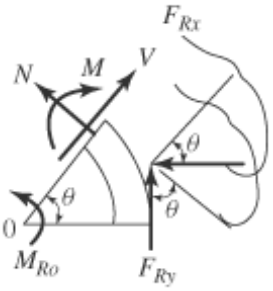
$$F_{Rx} = \int_0^{\theta} w_0 r \, d\theta \sin(\theta) = r w_0 (1 - \cos(\theta))$$



$$F_{Rx} = r w_0 (1 - \cos(\theta))$$

$$F_{Ry} = \int_0^{\theta} w_0 r \, d\theta \cos(\theta) = r w_0 \sin(\theta)$$

$$F_{Rx} = r w_0 (\sin(\theta))$$



$$M_{Ro} = \int_0^{\theta} w_0 r \, d\theta r = r^2 w_0 \theta$$

$\Sigma F_x = 0;$ $\quad -V + F_{Rx}\cos(\theta) - F_{Ry}\sin(\theta) = 0$

$$V = \left[r w_0 (1 - \cos(\theta))\right]\cos(\theta) - \left[r w_0 (\sin(\theta))\right]\sin(\theta)$$

$$V = w_0 r (\cos(\theta) - 1)$$

$$a = \cos(\theta) - 1$$

$a = -0.293$ $\qquad V = a r w_0$

$\Sigma F_y = 0;$ $\quad N + F_{Ry}\cos(\theta) + F_{Rx}\sin(\theta) = 0$

$$N = -\left[r w_0 (1 - \cos(\theta))\right]\sin(\theta) - \left[r w_0 (\sin(\theta))\right]\cos(\theta)$$

$$N = -w_0 r \sin(\theta)$$

$$b = -\sin(\theta)$$

$b = -0.707$ $\qquad N = w_0\ r\ b$

$\Sigma M_o = 0;$ $\quad -M + r^2 w_0(\theta) + b r w_0 r = 0$

$$M = w_0 r^2 (\theta + b)$$

$$c = \theta + b$$

$c = 0.0783$ $\qquad M = c r^2 w_0$

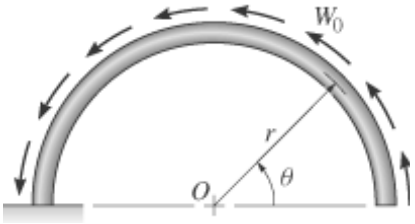
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Problem 7-37

The semicircular arch is subjected to a uniform distributed load along its axis of w_0 per unit length. Determine the internal normal force, shear force, and moment in the arch at angle θ.

Given:

$\theta = 120 \text{ deg}$

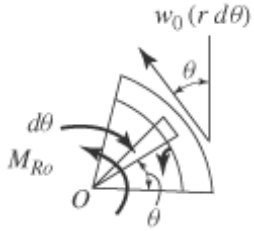


Solution:

Resultants of distributed load:

$$F_{Rx} = \int_0^\theta w_0 r \, d\theta \sin(\theta) = r w_0 (1 - \cos(\theta))$$

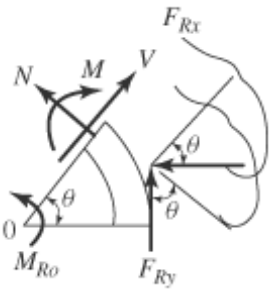
$$F_{Rx} = r w_0 (1 - \cos(\theta))$$



$$F_{Ry} = \int_0^\theta w_0 r \, d\theta \cos(\theta) = r w_0 \sin(\theta)$$

$$F_{Rx} = r w_0 (\sin(\theta))$$

$$M_{Ro} = \int_0^\theta w_0 r \, d\theta r = r^2 w_0 \theta$$



$\Sigma F_x = 0;$ $\quad -V + F_{Rx}\cos(\theta) - F_{Ry}\sin(\theta) = 0$

$$V = \left[r w_0 (1 - \cos(\theta))\right]\cos(\theta) - \left[r w_0 (\sin(\theta))\right]\sin(\theta)$$

$$V = w_0 r (\cos(\theta) - 1)$$

$$a = \cos(\theta) - 1$$

$a = -1.500$ $\qquad V = a r w_0$

$\Sigma F_y = 0;$ $\quad N + F_{Ry}\cos(\theta) + F_{Rx}\sin(\theta) = 0$

$$N = -\left[r w_0 (1 - \cos(\theta))\right]\sin(\theta) - \left[r w_0 (\sin(\theta))\right]\cos(\theta)$$

$$N = -w_0 r \sin(\theta)$$

$$b = -\sin(\theta)$$

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$b = -0.866$ $N = w_0 \cdot r \cdot b$

$\circlearrowleft + \Sigma M_o = 0; \quad -M + r^2 w_0(\theta) + b\, r\, w_0\, r = 0$

$M = w_0\, r^2(\theta + b)$

$c = \theta + b$

$c = 1.2284$ $M = c\, r^2 w_0$

Problem 7-38

Determine the *x*, *y*, *z* components of internal loading at a section passing through point *C* in the pipe assembly. Neglect the weight of the pipe.

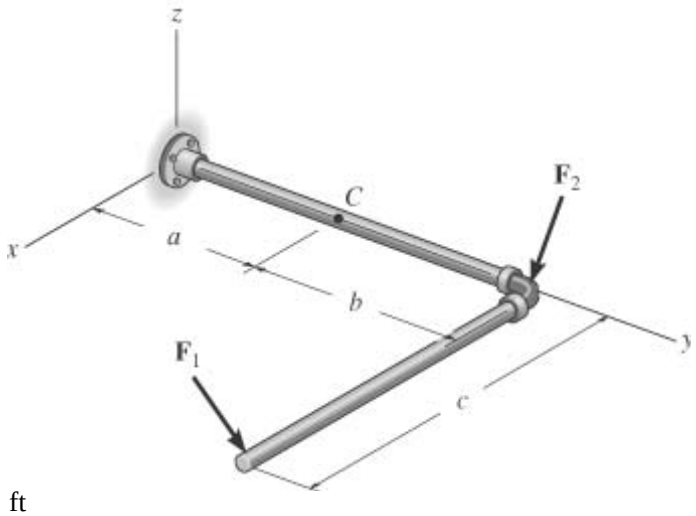
Units Used:

$\text{kip} = 10^3\ \text{lb}$

Given:

$$\mathbf{F_1} = \begin{pmatrix} 0 \\ 350 \\ -400 \end{pmatrix} \text{lb}$$

$$\mathbf{F_2} = \begin{pmatrix} 150 \\ 0 \\ -300 \end{pmatrix} \text{lb}$$

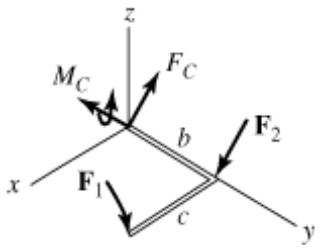


$a = 1.5\ \text{ft}$ $b = 2\ \text{ft}$ $c = 3\ \text{ft}$

Solution:

$$\mathbf{r_1} = \begin{pmatrix} c \\ b \\ 0 \end{pmatrix} \quad \mathbf{r_2} = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix}$$

$\mathbf{F_C} = -\mathbf{F_1} - \mathbf{F_2}$ $\mathbf{F_C} = \begin{pmatrix} -150.00 \\ -350.00 \\ 700.00 \end{pmatrix} \text{lb}$



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$$\mathbf{M_C} = -\mathbf{r_1} \times \mathbf{F_1} - \mathbf{r_2} \times \mathbf{F_2} \qquad \mathbf{M_C} = \begin{pmatrix} 1400.00 \\ -1200.00 \\ -750.00 \end{pmatrix} \text{lb}\cdot\text{ft}$$

Problem 7-39

Determine the *x*, *y*, *z* components of internal loading at a section passing through point *C* in the pipe assembly. Neglect the weight of the pipe.

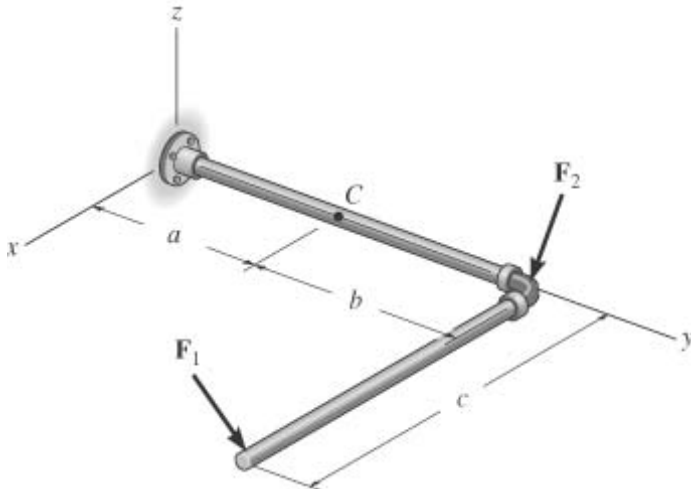
Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

$$\mathbf{F_1} = \begin{pmatrix} -80 \\ 200 \\ -300 \end{pmatrix} \text{lb}$$

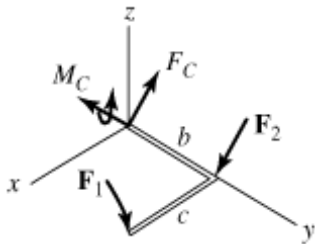
$$\mathbf{F_2} = \begin{pmatrix} 250 \\ -150 \\ -200 \end{pmatrix} \text{lb}$$



$a = 1.5$ ft $\quad b = 2$ ft $\quad c = 3$ ft

Solution:

$$\mathbf{r_1} = \begin{pmatrix} c \\ b \\ 0 \end{pmatrix} \qquad \mathbf{r_2} = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix}$$



$$\mathbf{F_C} = -\mathbf{F_1} - \mathbf{F_2} \qquad \mathbf{F_C} = \begin{pmatrix} -170.00 \\ -50.00 \\ 500.00 \end{pmatrix} \text{lb}$$

$$\mathbf{M_C} = -\mathbf{r_1} \times \mathbf{F_1} - \mathbf{r_2} \times \mathbf{F_2} \qquad \mathbf{M_C} = \begin{pmatrix} 1000.00 \\ -900.00 \\ -260.00 \end{pmatrix} \text{lb}\cdot\text{ft}$$

Problem 7-40

Determine the *x*, *y*, *z* components of internal loading in the rod at point *D*.

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Units Used:

$kN = 10^3 N$

Given:

$M = 3\ kN \cdot m$

$\mathbf{F} = \begin{pmatrix} 7 \\ -12 \\ -5 \end{pmatrix} kN$

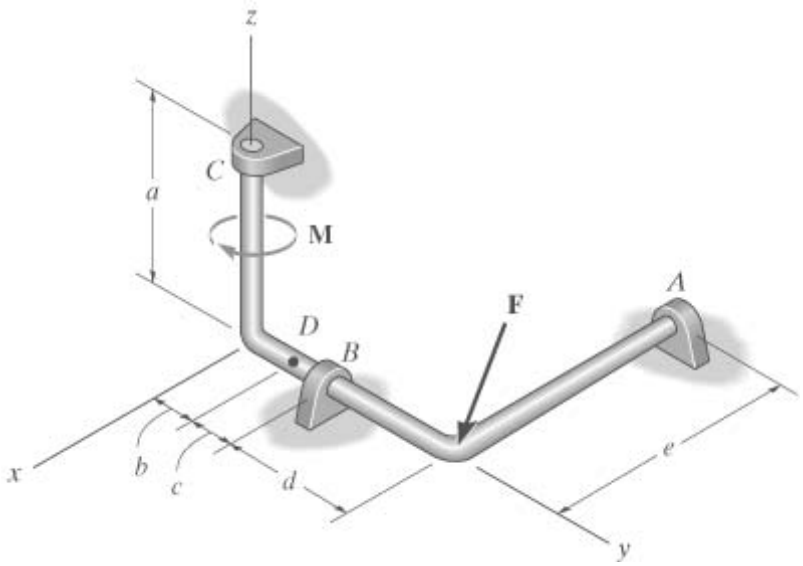
$a = 0.75\ m$

$b = 0.2\ m$

$c = 0.2\ m$

$d = 0.6\ m$

$e = 1\ m$

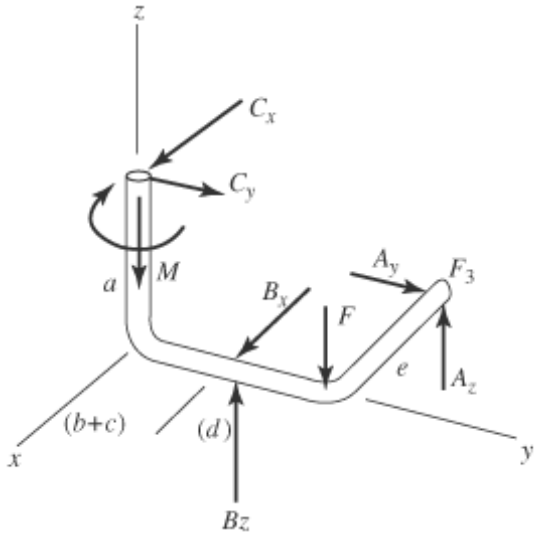


Solution:

Guesses

$C_x = 1\ N \quad C_y = 1\ N \quad B_x = 1\ N$

$B_z = 1\ N \quad A_y = 1\ N \quad A_z = 1\ N$



Given

$$\begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} B_x \\ 0 \\ B_z \end{pmatrix} + \begin{pmatrix} C_x \\ C_y \\ 0 \end{pmatrix} + \mathbf{F} = \mathbf{0}$$

$$\begin{pmatrix} -e \\ b+c+d \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} 0 \\ b+c \\ 0 \end{pmatrix} \times \begin{pmatrix} B_x \\ 0 \\ B_z \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ a \end{pmatrix} \times \begin{pmatrix} C_x \\ C_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ b+c+d \\ 0 \end{pmatrix} \times \mathbf{F} + \begin{pmatrix} 0 \\ 0 \\ -M \end{pmatrix} = \mathbf{0}$$

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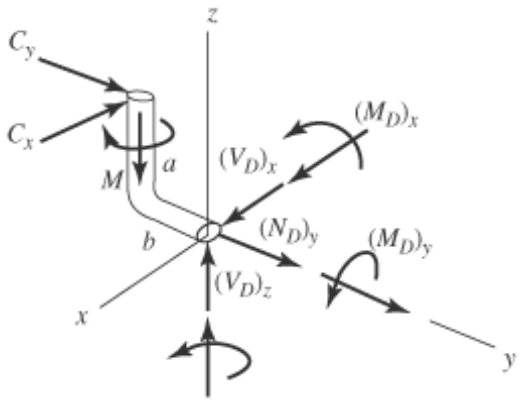
$$\begin{pmatrix} A_y \\ A_z \\ B_x \\ B_z \\ C_x \\ C_y \end{pmatrix} = \text{Find}\left(A_y, A_z, B_x, B_z, C_x, C_y\right) \qquad \begin{pmatrix} A_y \\ A_z \\ B_x \\ B_z \\ C_x \\ C_y \end{pmatrix} = \begin{pmatrix} -53.60 \\ 87.00 \\ 109.00 \\ -82.00 \\ -116.00 \\ 65.60 \end{pmatrix} \text{kN}$$

Guesses

$V_{Dx} = 1\ \text{N}$ $N_{Dy} = 1\ \text{N}$

$V_{Dz} = 1\text{N}$ $M_{Dx} = 1\ \text{N}\cdot\text{m}$

$M_{Dy} = 1\ \text{N}\cdot\text{m}$ $M_{Dz} = 1\ \text{N}\cdot\text{m}$



Given

$$\begin{pmatrix} C_x \\ C_y \\ 0 \end{pmatrix} + \begin{pmatrix} V_{Dx} \\ N_{Dy} \\ V_{Dz} \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} 0 \\ -b \\ a \end{pmatrix} \times \begin{pmatrix} C_x \\ C_y \\ 0 \end{pmatrix} + \begin{pmatrix} M_{Dx} \\ M_{Dy} \\ M_{Dz} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -M \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} V_{Dx} \\ N_{Dy} \\ V_{Dz} \\ M_{Dx} \\ M_{Dy} \\ M_{Dz} \end{pmatrix} = \text{Find}\left(V_{Dx}, N_{Dy}, V_{Dz}, M_{Dx}, M_{Dy}, M_{Dz}\right)$$

$$\begin{pmatrix} V_{Dx} \\ N_{Dy} \\ V_{Dz} \end{pmatrix} = \begin{pmatrix} 116.00 \\ -65.60 \\ 0.00 \end{pmatrix} \text{kN}$$

$$\begin{pmatrix} M_{Dx} \\ M_{Dy} \\ M_{Dz} \end{pmatrix} = \begin{pmatrix} 49.20 \\ 87.00 \\ 26.20 \end{pmatrix} \text{kN}\cdot\text{m}$$

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Problem 7-41

Determine the x, y, z components of internal loading in the rod at point E.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$M = 3\ \text{kN}\cdot\text{m}$

$\mathbf{F} = \begin{pmatrix} 7 \\ -12 \\ -5 \end{pmatrix}\ \text{kN}$

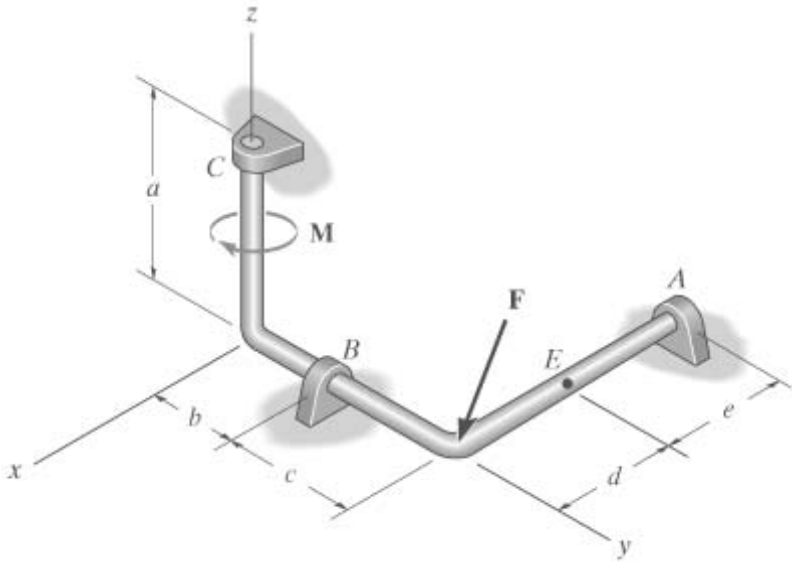
$a = 0.75\ \text{m}$

$b = 0.4\ \text{m}$

$c = 0.6\ \text{m}$

$d = 0.5\ \text{m}$

$e = 0.5\ \text{m}$

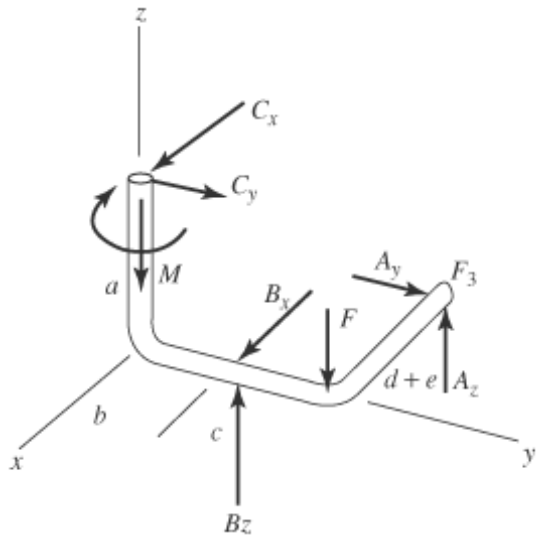


Solution:

Guesses

$C_x = 1\ \text{N}$ $C_y = 1\ \text{N}$ $B_x = 1\ \text{N}$

$B_z = 1\ \text{N}$ $A_y = 1\ \text{N}$ $A_z = 1\ \text{N}$



Given

$$\begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} B_x \\ 0 \\ B_z \end{pmatrix} + \begin{pmatrix} C_x \\ C_y \\ 0 \end{pmatrix} + \mathbf{F} = \mathbf{0}$$

$$\begin{pmatrix} -d-e \\ b+c \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} B_x \\ 0 \\ B_z \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ a \end{pmatrix} \times \begin{pmatrix} C_x \\ C_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ b+c \\ 0 \end{pmatrix} \times \mathbf{F} + \begin{pmatrix} 0 \\ 0 \\ -M \end{pmatrix} = \mathbf{0}$$

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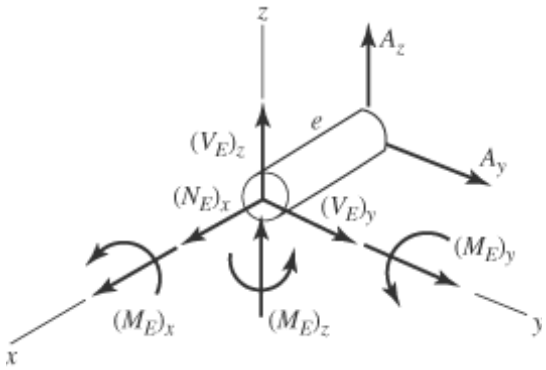
$$\begin{pmatrix} A_y \\ A_z \\ B_x \\ B_z \\ C_x \\ C_y \end{pmatrix} = \text{Find}\left(A_y, A_z, B_x, B_z, C_x, C_y\right) \qquad \begin{pmatrix} A_y \\ A_z \\ B_x \\ B_z \\ C_x \\ C_y \end{pmatrix} = \begin{pmatrix} -53.60 \\ 87.00 \\ 109.00 \\ -82.00 \\ -116.00 \\ 65.60 \end{pmatrix} \text{kN}$$

Guesses

$N_{Ex} = 1\ \text{N}$ $V_{Ey} = 1\ \text{N}$

$V_{Ez} = 1\ \text{N}$ $M_{Ex} = 1\ \text{N·m}$

$M_{Ey} = 1\ \text{N·m}$ $M_{Ez} = 1\ \text{N·m}$



Given

$$\begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} N_{Ex} \\ V_{Ey} \\ V_{Ez} \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} -e \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} M_{Ex} \\ M_{Ey} \\ M_{Ez} \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} N_{Ex} \\ V_{Ey} \\ V_{Ez} \\ M_{Ex} \\ M_{Ey} \\ M_{Ez} \end{pmatrix} = \text{Find}\left(N_{Ex}, V_{Ey}, V_{Ez}, M_{Ex}, M_{Ey}, M_{Ez}\right)$$

$$\begin{pmatrix} N_{Ex} \\ V_{Ey} \\ V_{Ez} \end{pmatrix} = \begin{pmatrix} 0.00 \\ 53.60 \\ -87.00 \end{pmatrix} \text{kN}$$

$$\begin{pmatrix} M_{Ex} \\ M_{Ey} \\ M_{Ez} \end{pmatrix} = \begin{pmatrix} 0.00 \\ -43.50 \\ -26.80 \end{pmatrix} \text{kN·m}$$

Problem 7-42

Draw the shear and moment diagrams for the shaft in terms of the parameters shown; There is

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a thrust bearing at *A* and a journal bearing at *B*.

Units Used:

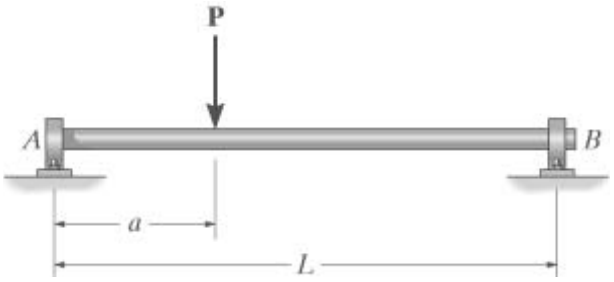
$kN = 10^3 N$

Given:

$P = 9\ kN$

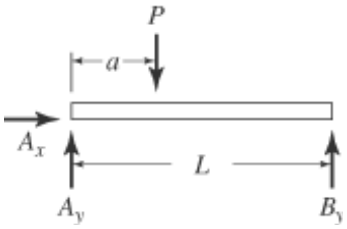
$a = 2\ m$

$L = 6\ m$



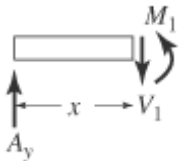
Solution:

$$P(L-a) - A_y L = 0 \qquad A_y = P\,\frac{L-a}{L}$$



$x_1 = 0, 0.01a .. a$

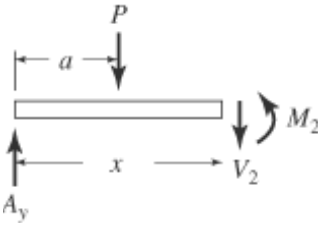
$$A_y - V_1(x) = 0 \qquad V_1(x) = \frac{A_y}{kN}$$



$$M_1(x) - A_y x = 0 \qquad M_1(x) = \frac{A_y x}{kN \cdot m}$$

$x_2 = a, 1.01a .. L$

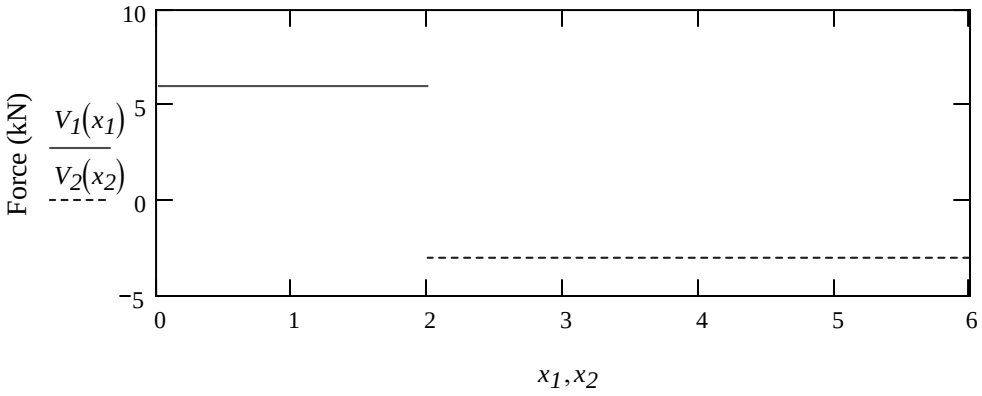
$$A_y - P - V_2(x) = 0 \qquad V_2(x) = \frac{A_y - P}{kN}$$

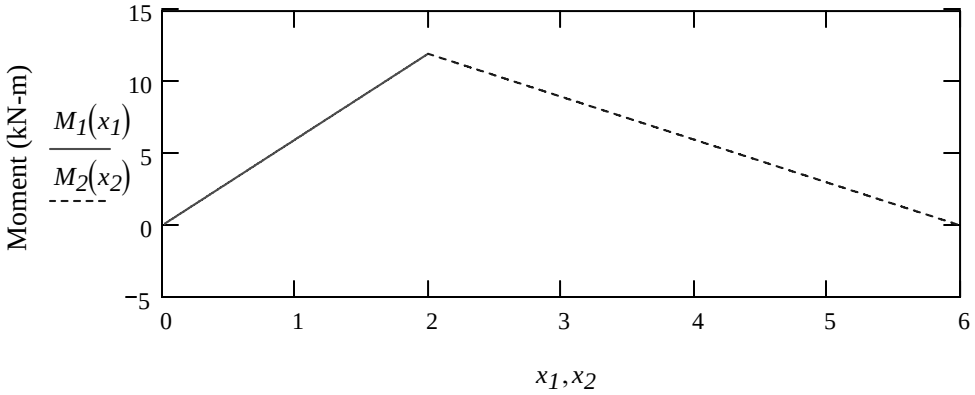


$$M_2(x) - A_y x + P(x-a) = 0$$

$$M_2(x) = \frac{A_y x - P(x-a)}{kN \cdot m}$$

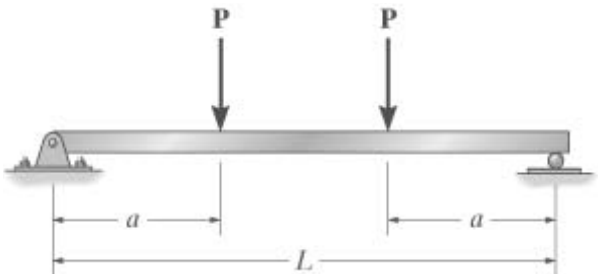
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Problem 7-43

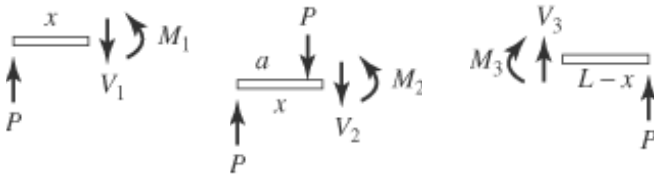
Draw the shear and moment diagrams for the beam in terms of the parameters shown.



Given:

$P = 800$ lb $a = 5$ ft $L = 12$ ft

Solution:

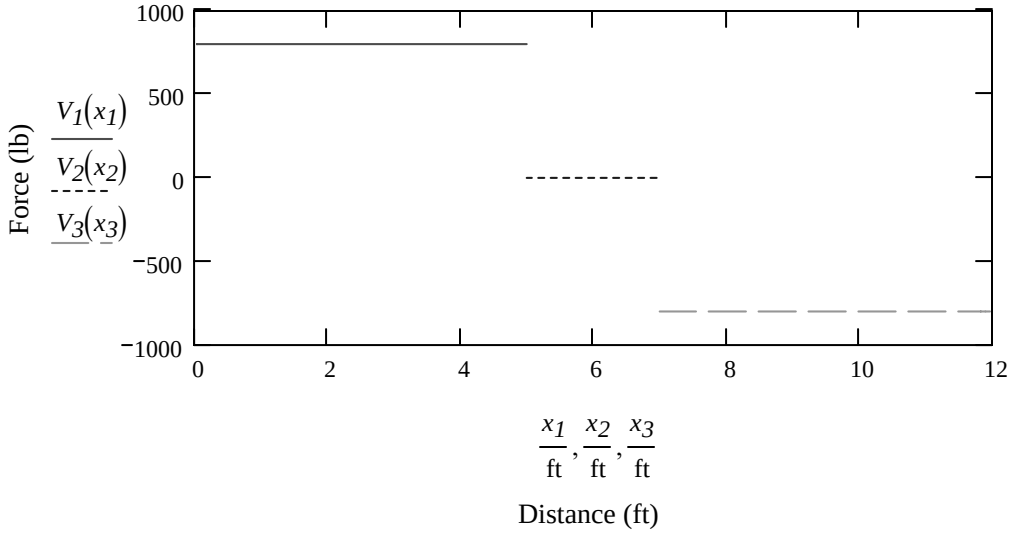


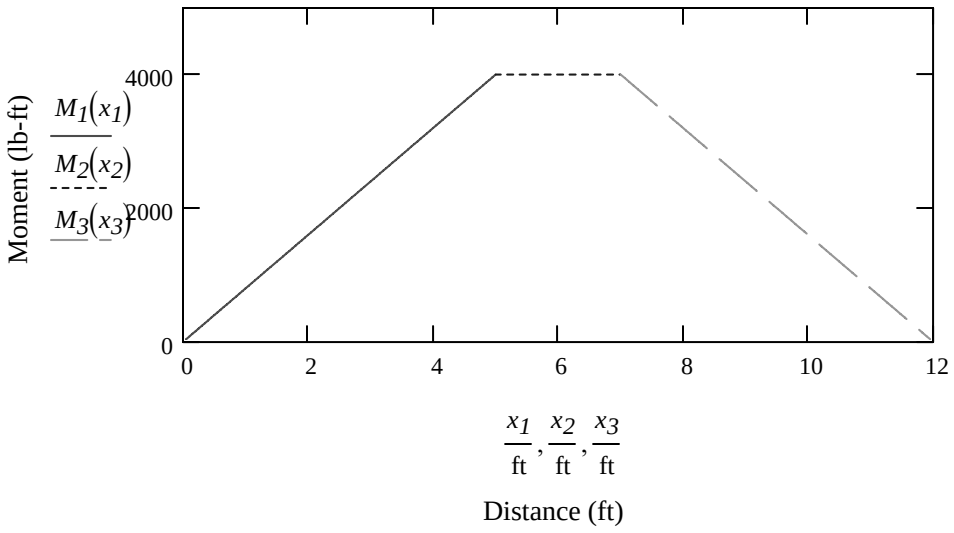
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$x_1 = 0, 0.01a .. a$ $x_2 = a, 1.01a .. L - a$ $x_3 = L - a, 1.01(L - a) .. L$

$V_1(x) = \frac{P}{\text{lb}}$ $V_2(x) = 0$ $V_3(x) = \frac{-P}{\text{lb}}$

$M_1(x) = \frac{P\,x}{\text{lb}\cdot\text{ft}}$ $M_2(x) = \frac{P\,a}{\text{lb}\cdot\text{ft}}$ $M_3(x) = \frac{P(L - x)}{\text{lb}\cdot\text{ft}}$





Problem 7-44

Draw the shear and moment diagrams for the beam (a) in terms of the parameters

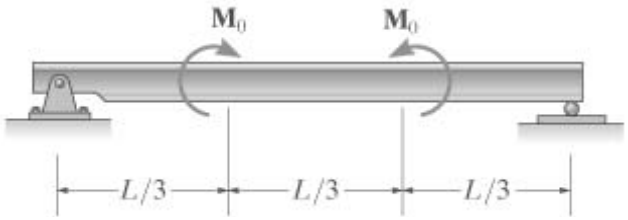
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shown; (b) set M_0 and L as given.

Given:

$M_0 = 500\ \text{N}\cdot\text{m}$

$L = 8\ \text{m}$

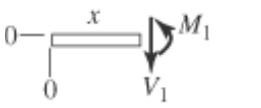


Solution:

For $0 \le x \le \frac{L}{3}$

$+\uparrow \Sigma F_y = 0; \quad V_1 = 0$

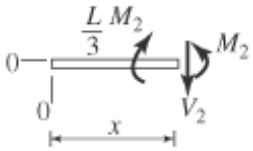
$\circlearrowleft \Sigma M_x = 0; \quad M_1 = 0$



For $\frac{L}{3} \le x \le \frac{2L}{3}$

$+\uparrow \Sigma F_y = 0; \quad V_2 = 0$

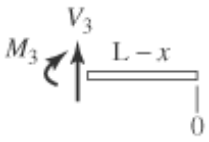
$\circlearrowleft \Sigma M_x = 0; \quad M_2 = M_0$

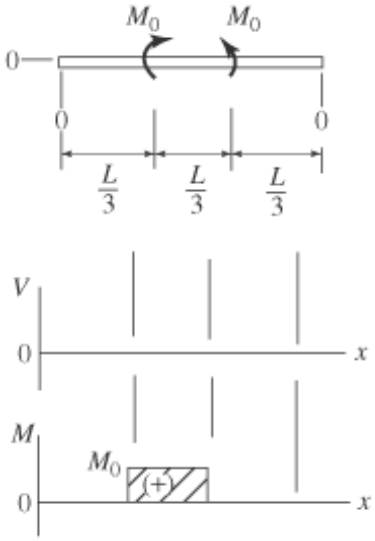


For $\frac{2L}{3} \le x \le L$

$+\uparrow \Sigma F_y = 0; \quad V_3 = 0$

$\circlearrowleft \Sigma M_x = 0; \quad M_3 = 0$



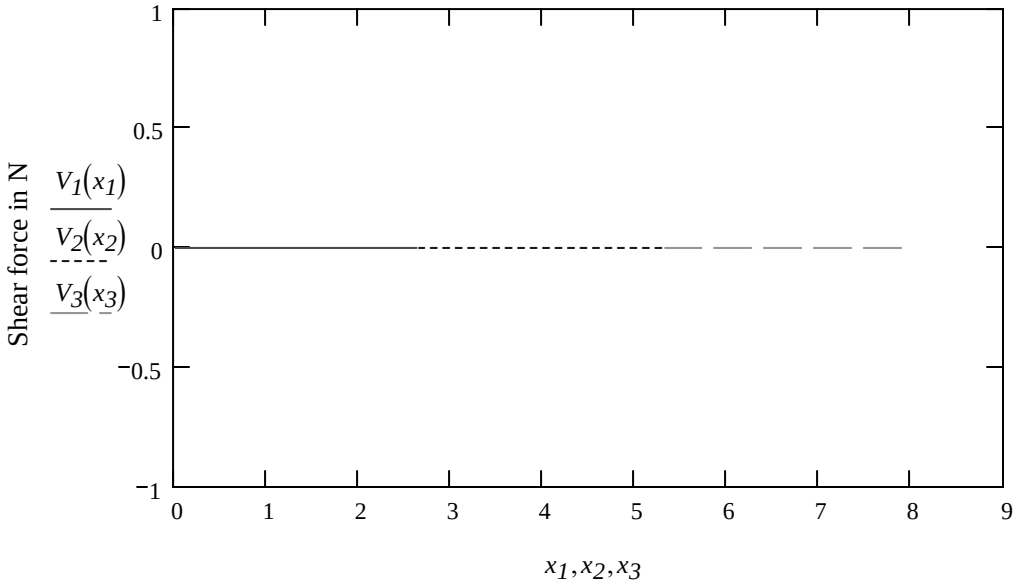


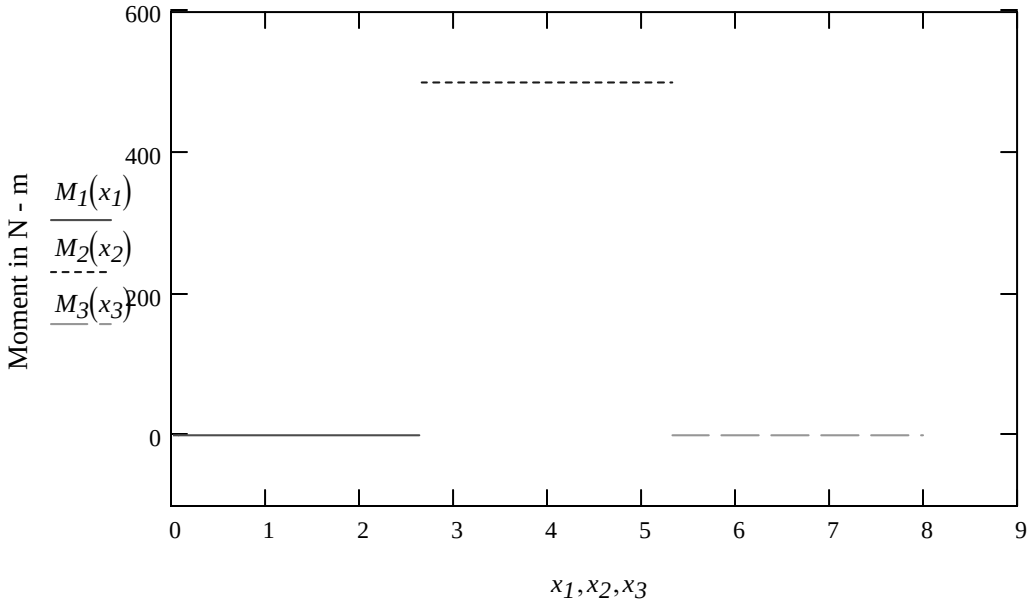
(b) $x_1 = 0, 0.01L .. \frac{L}{3}$ $\quad x_2 = \frac{L}{3}, \frac{L}{3}1.01 .. \frac{2L}{3}$ $\quad x_3 = \frac{2L}{3}, \frac{2L}{3}1.01 .. L$

$V_1(x_1) = 0$ $\quad V_2(x_2) = 0$ $\quad V_3(x_3) = 0$

$M_1(x_1) = 0$ $\quad M_2(x_2) = M_0$ $\quad M_3(x_3) = 0$

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Problem 7-45

The beam will fail when the maximum shear force is V_{max} or the maximum bending moment is M_{max}. Determine the magnitude M_0 of the largest couple moments it will support.

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Units Used:

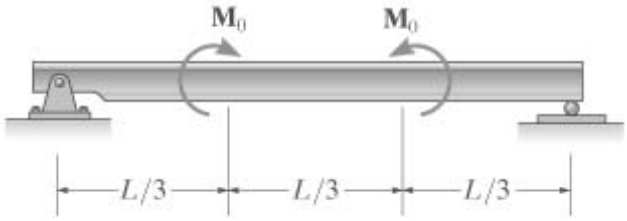
$\text{kN} = 10^3\ \text{N}$

Given:

$L = 9\ \text{m}$

$V_{max} = 5\ \text{kN}$

$M_{max} = 2\ \text{kN}\cdot\text{m}$



Solution:

The shear force is zero everywhere in the beam.

The moment is zero in the first third and the last third of the bam.

In the middle section of the beam the moment is $M = M_0$

Thus the beam will fail when $M_0 = M_{max}$ $M_0 = 2.00\,\text{kN}\cdot\text{m}$

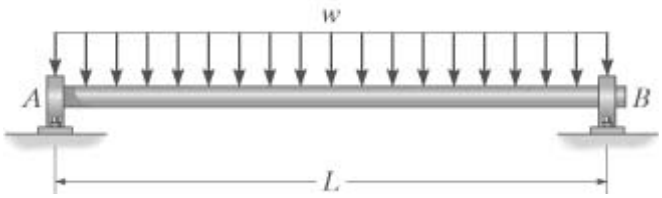
Problem 7-46

The shaft is supported by a thrust bearing at *A* and a journal bearing at *B*. Draw the shear and moment diagrams for the shaft in terms of the parameters shown.

Given:

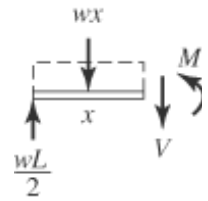
$w = 500\ \frac{\text{lb}}{\text{ft}}$

$L = 10\ \text{ft}$



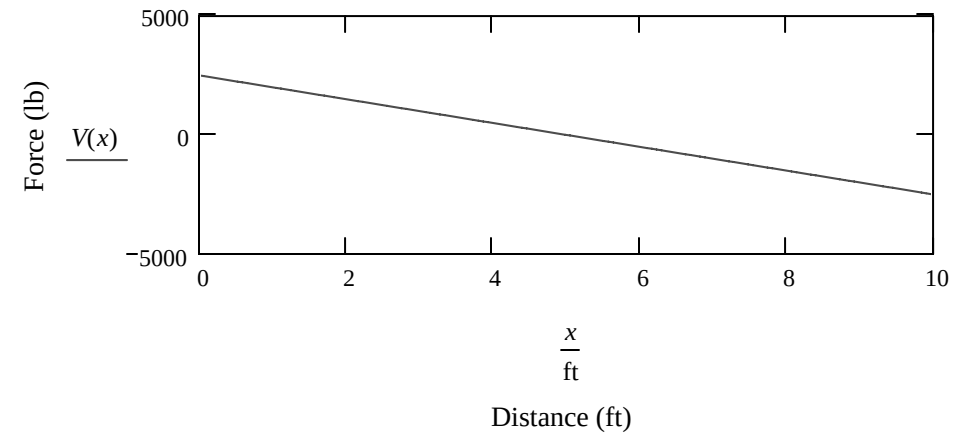
Solution: For $0 \le x < L$

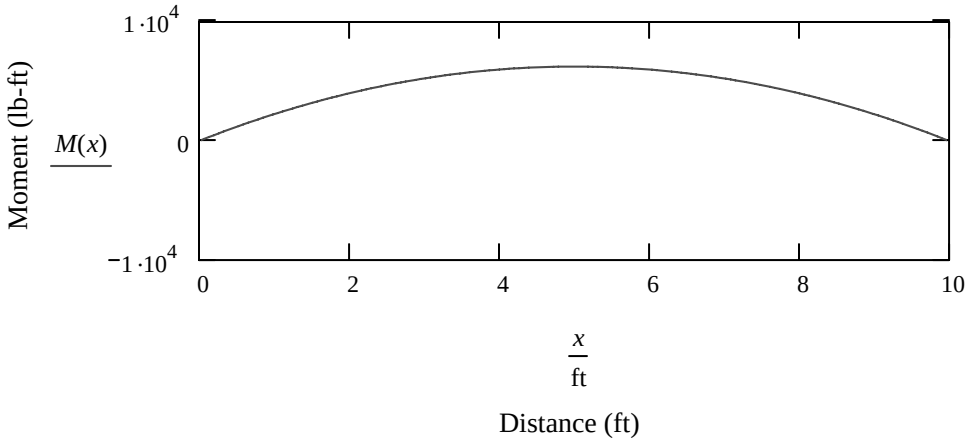
$\Sigma F_y = 0; \quad \frac{wL}{2} - wx - V = 0 \qquad V(x) = \frac{w}{2}(L - 2x)\frac{1}{\text{lb}}$



$\Sigma M = 0; \quad \frac{-wL}{2}x + wx\left(\frac{x}{2}\right) + M = 0 \qquad M(x) = \frac{w}{2}\left(Lx - x^2\right)\frac{1}{\text{lb}\cdot\text{ft}}$

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Problem 7-47

The shaft is supported by a thrust bearing at *A* and a journal bearing at *B*. The shaft will fail when the maximum moment is M_{max}. Determine the largest uniformly distributed load *w* the shaft will support.

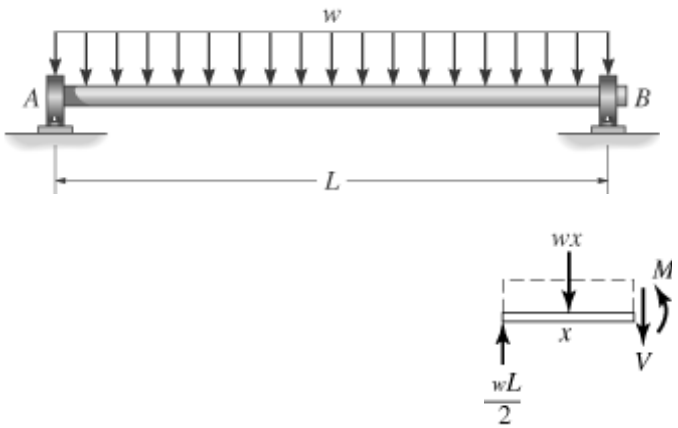
Units Used:

$kip = 10^3$ lb

Given:

$L = 10$ ft

$M_{max} = 5$ kip·ft

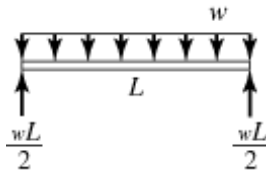


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Solution:

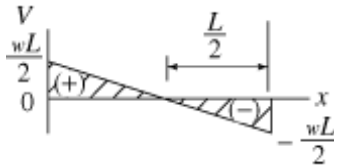
$$\frac{wL}{2} - wx - V = 0 \qquad V = -wx + \frac{wL}{2}$$

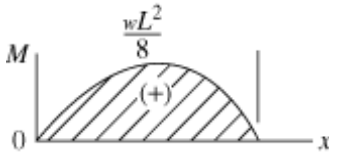
$$-\frac{wL}{2}x + wx\left(\frac{x}{2}\right) + M = 0 \qquad M = \left(\frac{wL}{2}\right)x - \frac{wx^2}{2}$$



From the Moment Diagram,

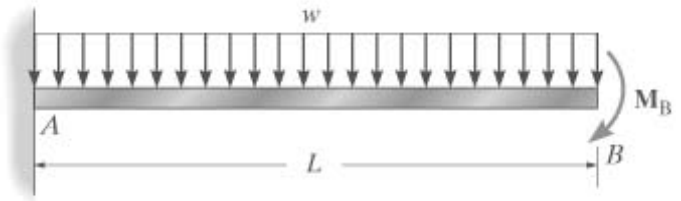
$$M_{max} = \frac{wL^2}{8} \qquad w = \frac{8M_{max}}{L^2} \qquad w = 400.00\,\frac{\text{lb}}{\text{ft}}$$





Problem 7-48

Draw the shear and moment diagrams for the beam.



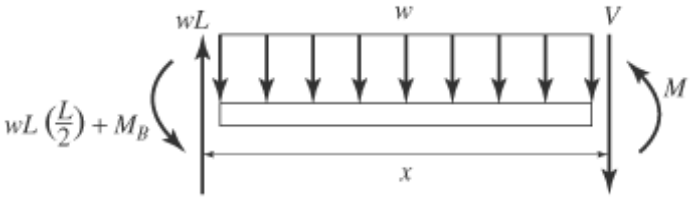
Units Used: $\text{kN} = 10^3\ \text{N}$

Given:

$$w = 2\,\frac{\text{kN}}{\text{m}}$$

$$L = 5\ \text{m}$$

$$M_B = 5\ \text{kN}\cdot\text{m}$$



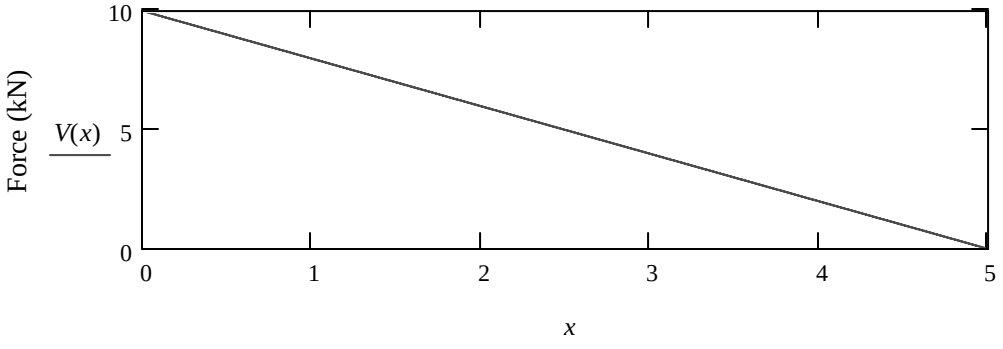
Solution:

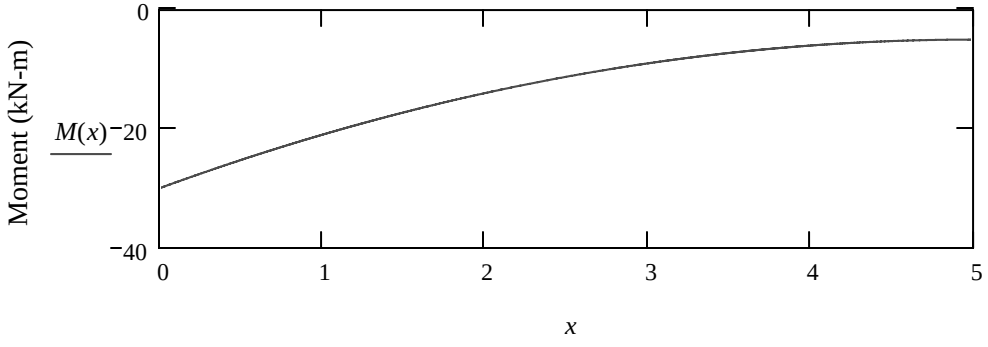
$$\Sigma F_y = 0; \quad -V(x) + \mathbf{w}L - \mathbf{w}x = 0 \qquad V(x) = (wL - wx)\frac{1}{\text{kN}}$$

$$\Sigma M = 0; \quad M(x) + wL\left(\frac{L}{2}\right) + M_B - wLx + wx\left(\frac{x}{2}\right) = 0$$

$$M(x) = \left[wLx - wx\left(\frac{x}{2}\right) - wL\left(\frac{L}{2}\right) - M_B\right]\frac{1}{\text{kN}\cdot\text{m}}$$

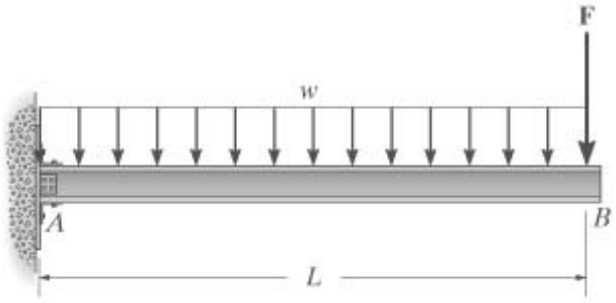
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Problem 7-49

Draw the shear and moment diagrams for the beam.



Units Used: $\text{kN} = 10^3\ \text{N}$

Given:

$w = 3\ \frac{\text{kN}}{\text{m}} \qquad F = 10\ \text{kN}$

$L = 6\ \text{m}$

Solution:

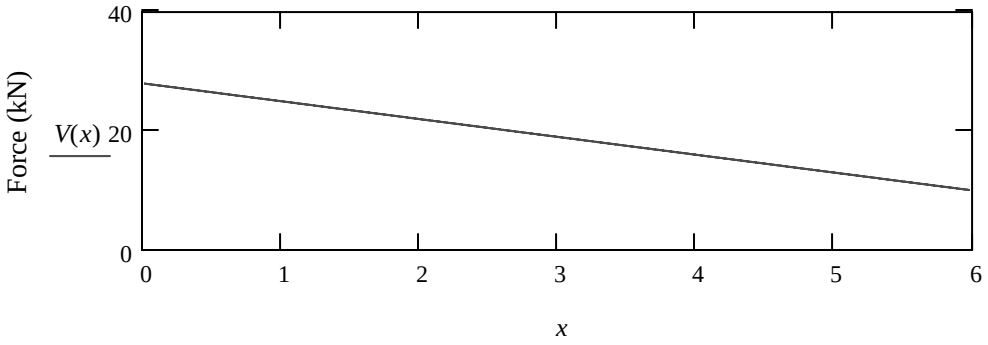
$V(x) - w(L - x) - F = 0$

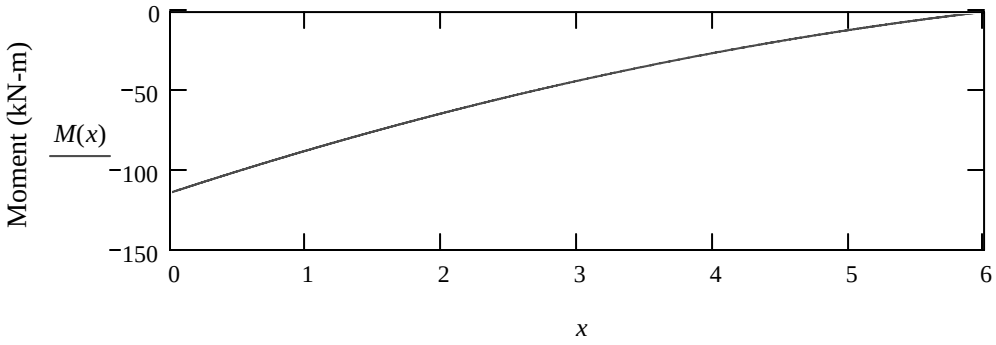
$V(x) = [w(L - x) + F]\frac{1}{\text{kN}}$

$-M(x) - w(L - x)\left(\frac{L - x}{2}\right) - F(L - x) = 0$

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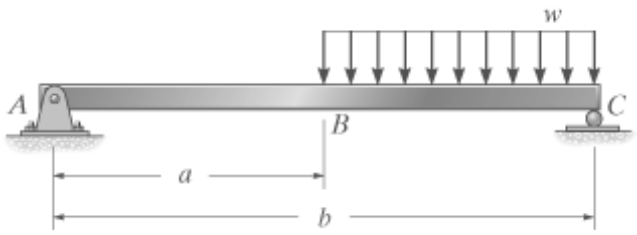
$$M(x) = \left[-w\frac{(L-x)^2}{2} - F(L-x)\right]\frac{1}{\text{kN}\cdot\text{m}}$$





Problem 7-50

Draw the shear and moment diagrams for the beam.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$a = 2\ \text{m} \quad b = 4\ \text{m} \quad w = 1.5\ \frac{\text{kN}}{\text{m}}$

Solution:

$$w(b-a)\left(\frac{b-a}{2}\right) - A_y b = 0 \qquad A_y = \frac{w(b-a)^2}{2b} \qquad A_y = 0.75\,\text{kN}$$

$x_1 = 0, 0.01a .. a$

$$A_y - V_1(x) = 0 \qquad V_1(x) = A_y\frac{1}{\text{kN}}$$

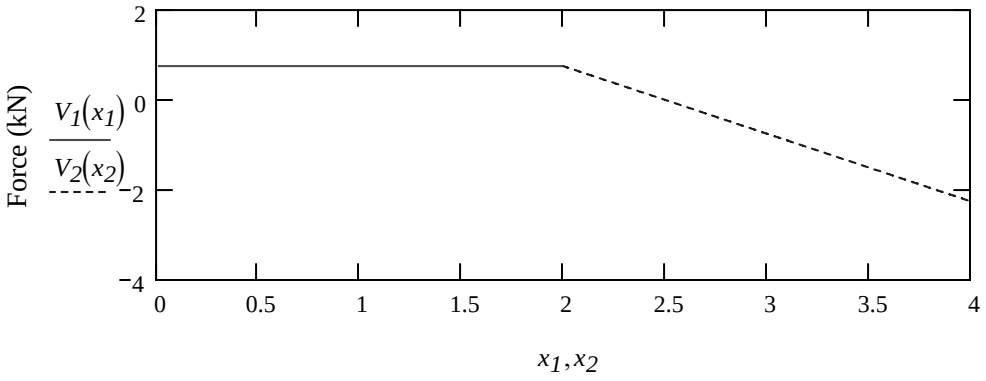
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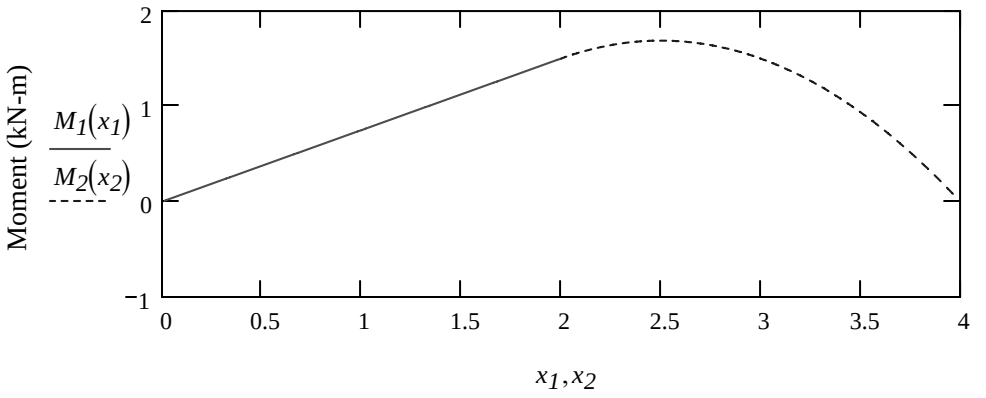
$$-A_y x + M_1(x) = 0 \qquad M_1(x) = A_y x \frac{1}{\text{kN}\cdot\text{m}}$$

$x_2 = b - a, 1.01(b-a) .. b$

$$A_y - w(x-a) - V_2(x) = 0 \qquad V_2(x) = \left[A_y - w(x-a)\right] \frac{1}{\text{kN}}$$

$$-A_y x + w(x-a)\left(\frac{x-a}{2}\right) + M_2(x) = 0 \qquad M_2(x) = \left[A_y x - w\frac{(x-a)^2}{2}\right] \frac{1}{\text{kN}\cdot\text{m}}$$

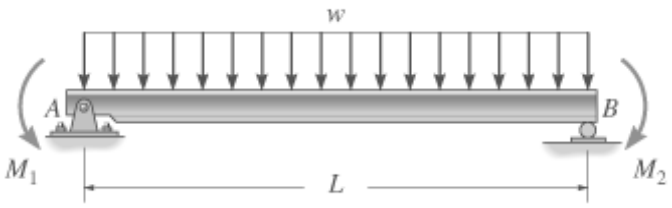




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Problem 7-51

Draw the shear and moment diagrams for the beam.



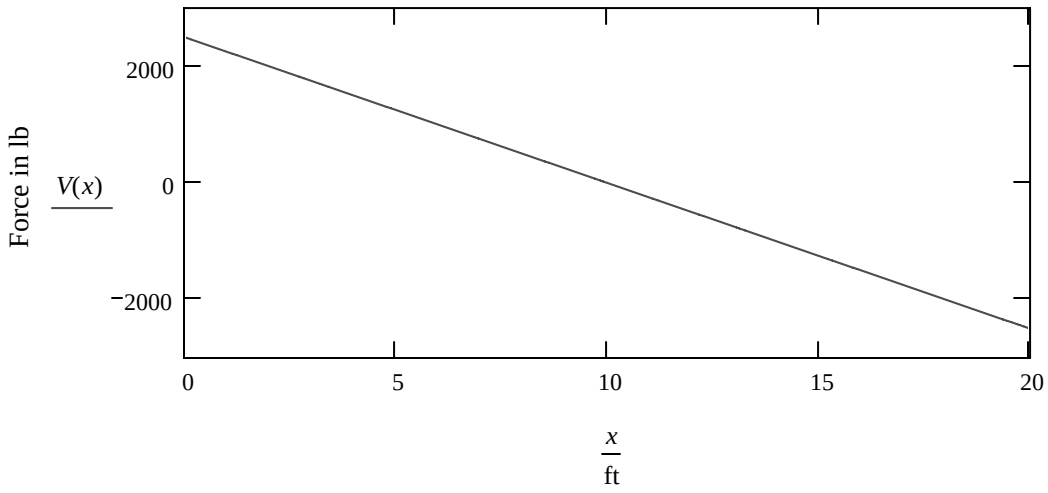
Given:

$L = 20\ \text{ft} \qquad w = 250\ \frac{\text{lb}}{\text{ft}} \qquad M_1 = 150\ \text{lb}\cdot\text{ft} \qquad M_2 = 150\ \text{lb}\cdot\text{ft}$

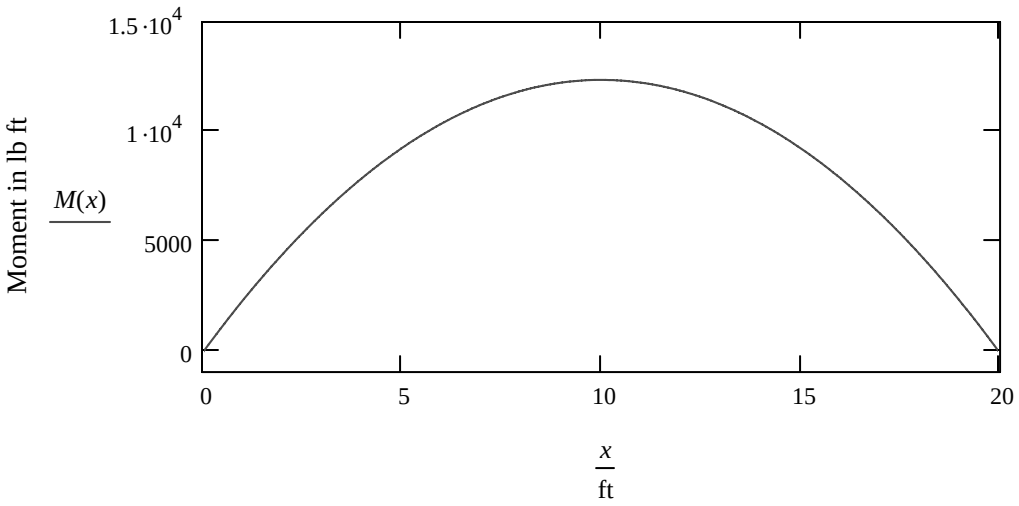
Solution:

$M_1 + wL\left(\frac{L}{2}\right) - M_2 - A_y\text{l} = 0 \qquad A_y = \frac{M_1 - M_2 + w\left(\frac{L^2}{2}\right)}{L} \qquad A_y = 2500\,\text{lb}$

$V(x) = \left(A_y - wx\right)\frac{1}{\text{lb}} \qquad M(x) = \left[A_y x - w\left(\frac{x^2}{2}\right) - M_1\right]\frac{1}{\text{lb}\cdot\text{ft}}$

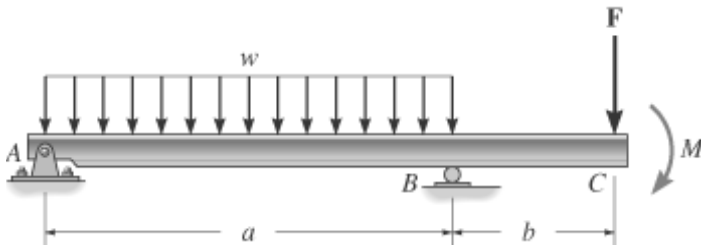


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Problem 7-52

Draw the shear and moment diagrams for the beam.



Units Used:

$$\text{kN} = 10^3\ \text{N}$$

Given:

$$w = 40\ \frac{\text{kN}}{\text{m}} \qquad F = 20\ \text{kN} \qquad M = 150\ \text{kN·m} \qquad a = 8\ \text{m} \qquad b = 3\ \text{m}$$

Solution:

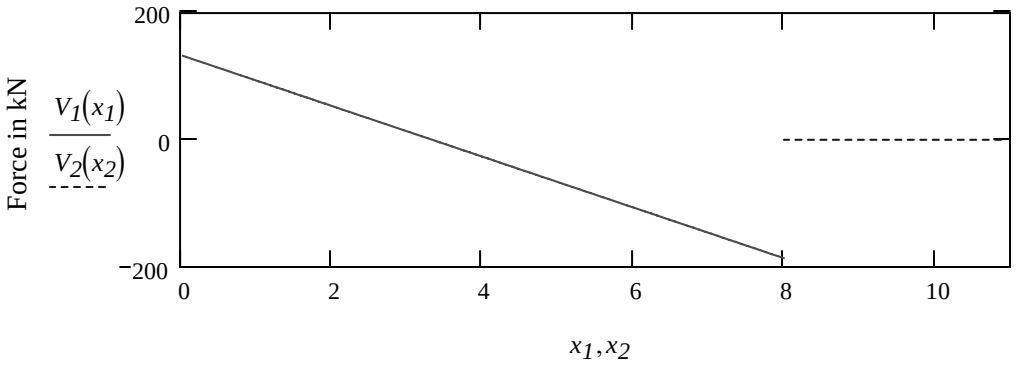
$$-A_y a + w a\left(\frac{a}{2}\right) - F b - M = 0 \qquad A_y = \left[w\left(\frac{a^2}{2}\right) - F b - M\right]\left(\frac{1}{a}\right) \qquad A_y = 133.75\,\text{kN}$$

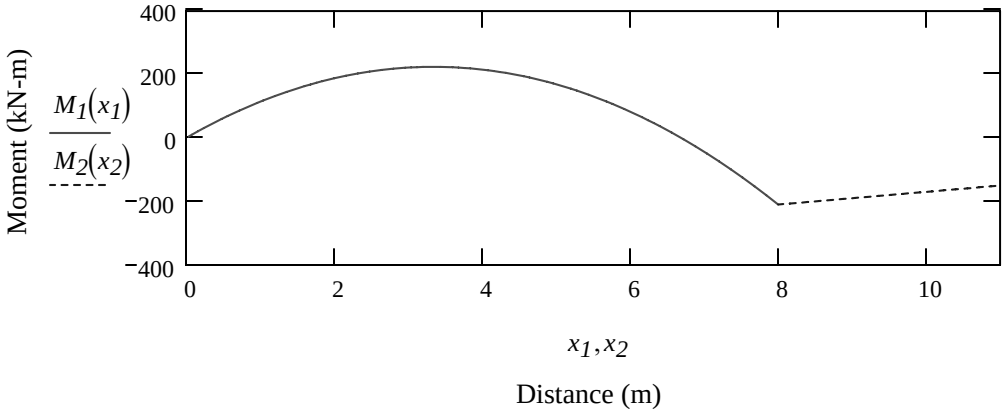
$$x_1 = 0, 0.01a .. a \qquad x_2 = a, 1.01a .. a + b$$

$$V_1(x) = \left(A_y - w x\right)\frac{1}{\text{kN}} \qquad V_2(x) = \text{F}\ \frac{1}{\text{kN}}$$

$$M_1(x) = \left[A_y x - w\left(\frac{x^2}{2}\right)\right]\frac{1}{\text{kN·m}} \qquad M_2(x) = [-F(a + b - x) - M]\ \frac{1}{\text{kN·m}}$$

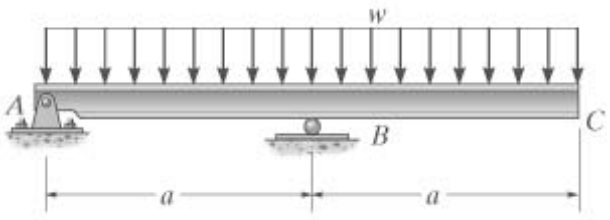
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Problem 7-53

Draw the shear and moment diagrams for the beam.



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Solution:

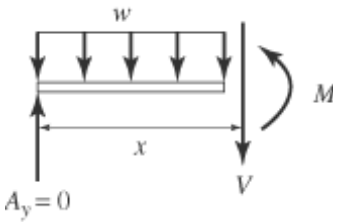
$0 \le x < a$

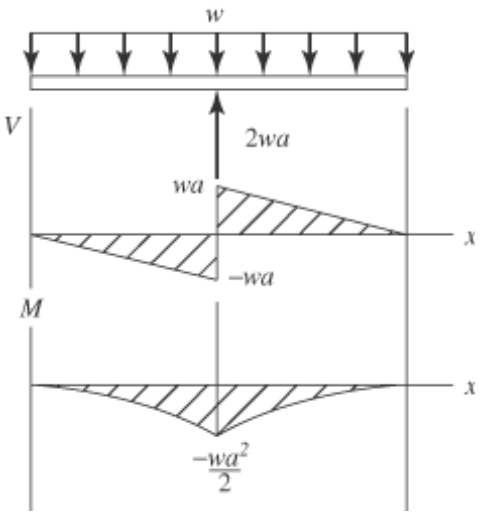
$\Sigma F_y = 0; \quad -V - wx = 0$

$$V = -wx$$

$\Sigma M = 0; \quad M + wx\left(\frac{x}{2}\right) = 0$

$$M = -w\frac{x^2}{2}$$





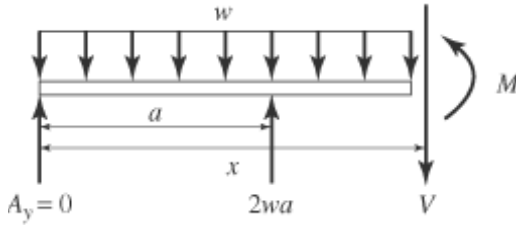
$a < x \le 2a$

$\Sigma F_y = 0; \quad -V + 2wa - wx = 0$

$$V = w(2a - x)$$

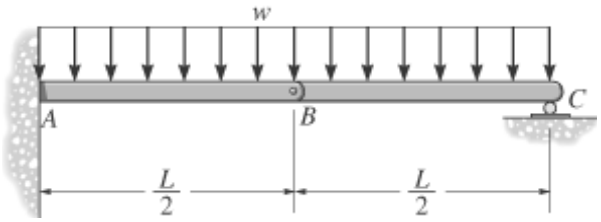
$\Sigma M = 0; \quad M + wx\left(\frac{x}{2}\right) - 2wa(x - a) = 0$

$$M = 2wa(x - a) - \frac{wx^2}{2}$$



Problem 7-54

Draw the shear and bending-moment diagrams for beam *ABC*. Note that there is a pin at *B*.



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Solution:

Support Reactions: From *FBD* (b),

$$\frac{wL}{2}\left(\frac{L}{4}\right) - B_y\left(\frac{L}{2}\right) = 0$$

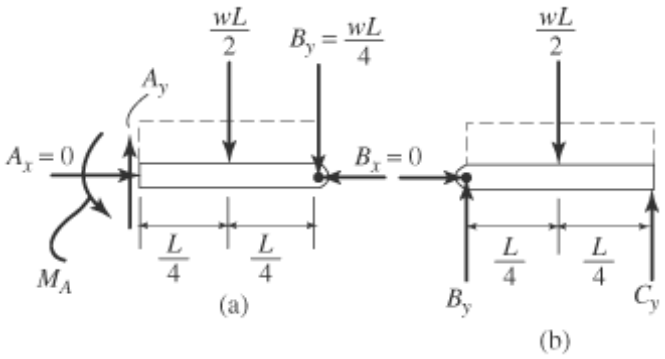
$$B_y = \frac{wL}{4}$$

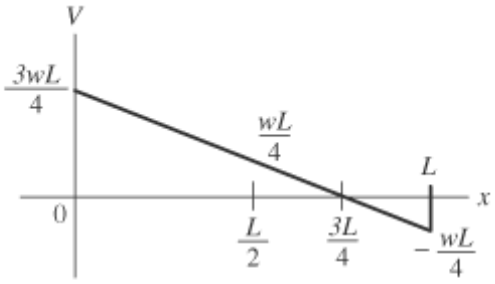
From *FBD* (a), $A_y - \frac{wL}{2} - \frac{B}{y} = 0$

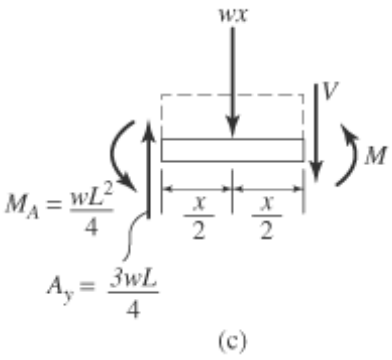
$$A_y = \frac{3wL}{4}$$

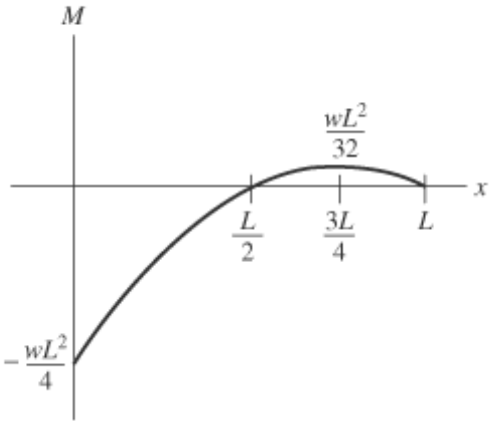
$$-B_y\,\frac{L}{2} - w\left(\frac{L}{2}\right)\left(\frac{L}{4}\right) + M_A = 0$$

$$M_A = w\left(\frac{L^2}{4}\right)$$









Shear and Moment Functions:
From *FBD* (c) For

$0 \le x \le L$ $\qquad A_y - wx - V = 0$

$$V = \frac{w}{4}(3L - 4x)$$

$$M_A - A_y x + wx\left(\frac{x}{2}\right) + M = 0 \qquad M = \frac{w}{4}\left(3Lx - 2x^2 - L^2\right)$$

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Problem 7-55

The beam has depth a and is subjected to a uniform distributed loading w which acts at an angle θ from the vertical as shown. Determine the internal normal force, shear force, and moment in the beam as a function of x. *Hint*:The moment loading is to be determined from a point along the centerline of the beam (x axis).

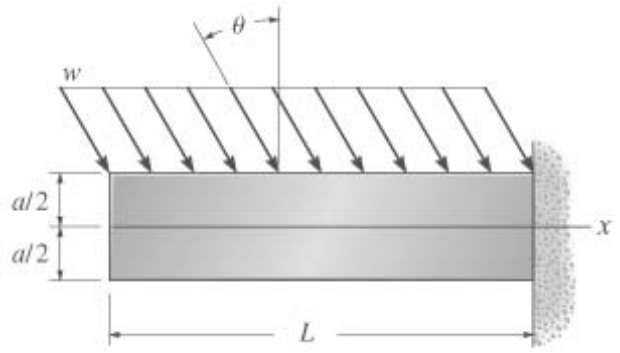
Given:

$a = 2$ ft

$L = 10$ ft

$\theta = 30$ deg

$w = 50 \dfrac{\text{lb}}{\text{ft}}$

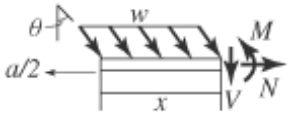


Solution:

$0 \le x \le L$

$\Sigma F_x = 0; \quad N + w\sin(\theta)x = 0$

$N(x) = -w\sin(\theta)x$



$\Sigma F_y = 0; \quad -V - w\cos(\theta)x = 0$

$V = -w\cos(\theta)x$

$\Sigma M = 0; \quad w\cos(\theta)x\left(\dfrac{x}{2}\right) - w\sin(\theta)x\left(\dfrac{a}{2}\right) + M = 0$

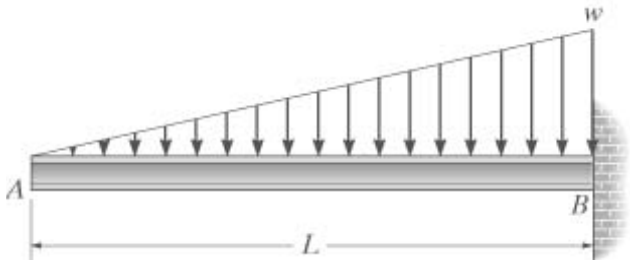
$M(x) = -w\cos(\theta)\left(\dfrac{x^2}{2}\right) + w\sin(\theta)\left(\dfrac{xa}{2}\right)$

Problem 7-56

Draw the shear and moment diagrams for the beam.

Given:

$w = 250 \dfrac{\text{lb}}{\text{ft}} \qquad L = 12$ ft

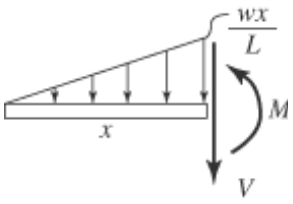


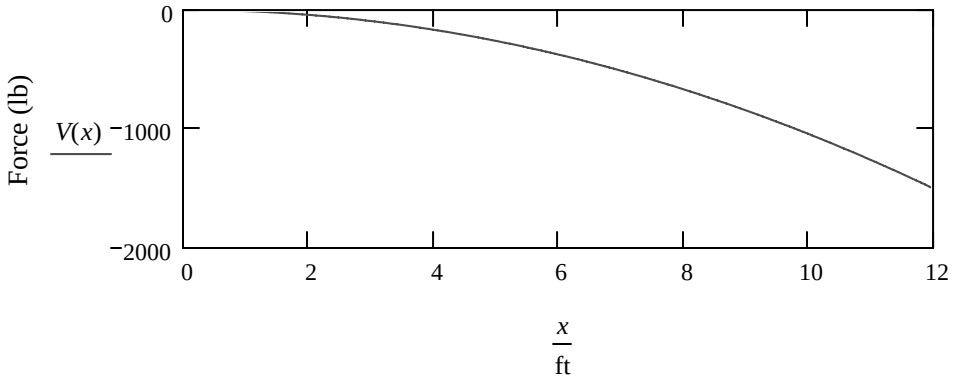
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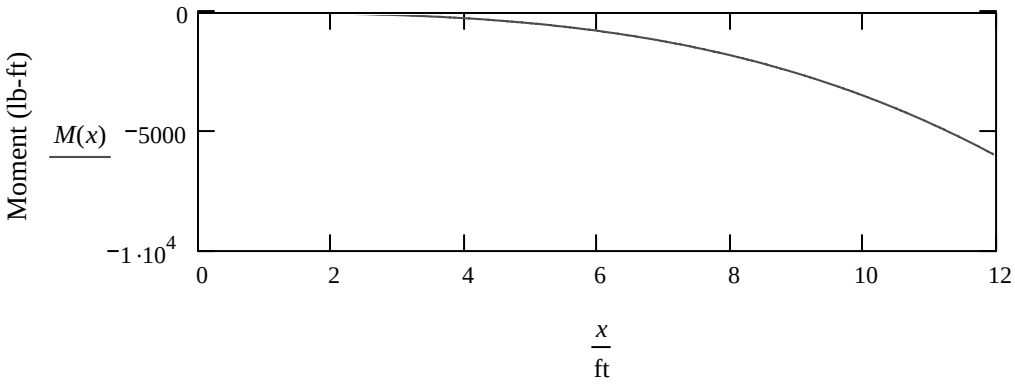
Solution:

$$\Sigma F_y = 0; \qquad -V - \frac{1}{2}x\,\frac{wx}{L} = 0 \qquad V(x) = -\frac{wx^2}{2L}\,\frac{1}{\text{lb}}$$

$$\Sigma M = 0; \qquad M + \frac{x}{2}\left(\frac{wx}{\text{L}}\right)\left(\frac{1}{3}x\right) = 0 \qquad M(x) = \frac{-wx^3}{6L}\,\frac{1}{\text{lb}\cdot\text{ft}}$$







Problem 7-57

The beam will fail when the maximum shear force is V_{max} or the maximum moment is M_{max}. Determine the largest intensity w of the distributed loading it will support.

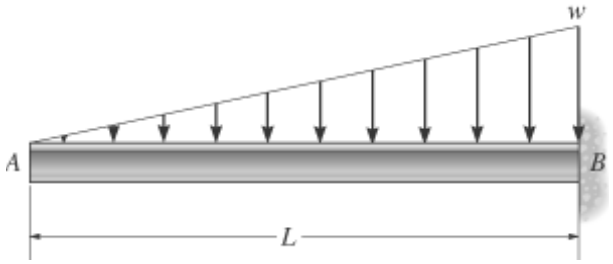
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Given:

$L = 18\ \text{ft}$

$V_{max} = 800\ \text{lb}$

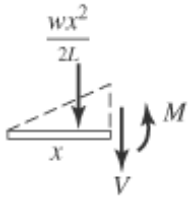
$M_{max} = 1200\ \text{lb}\cdot\text{ft}$



Solution:

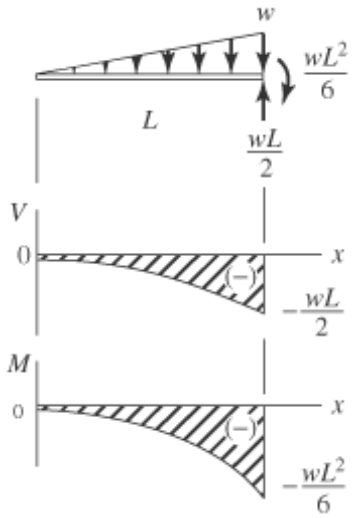
For $0 \le x \le L$

$$V = \frac{-wx^2}{2L} \qquad M = \frac{-wx^3}{6L}$$



$$\left|V_{max}\right| = \frac{wL}{2}$$

$$w_1 = 2\left(\frac{V_{max}}{L}\right) \qquad w_1 = 88.9\,\frac{\text{lb}}{\text{ft}}$$



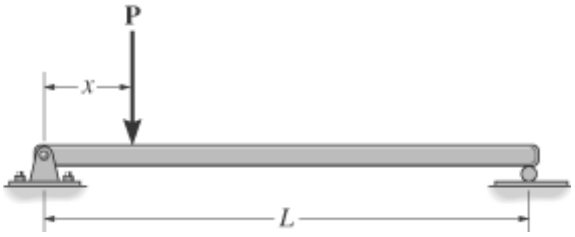
$$\left|M_{max}\right| = \frac{w_2\,L^2}{6}$$

$$w_2 = 6\left(\frac{M_{max}}{L^2}\right) \qquad w_2 = 22.2\,\frac{\text{lb}}{\text{ft}}$$

Now choose the critical case $w = \min\left(w_1, w_2\right)$ $\quad w = 22.22\,\frac{\text{lb}}{\text{ft}}$

Problem 7-58

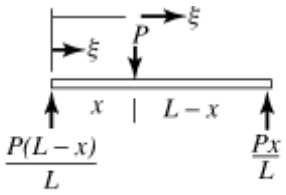
The beam will fail when the maximum internal moment is M_{max}. Determine the position x of the concentrated force **P** and its smallest magnitude that will cause failure.



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Solution:

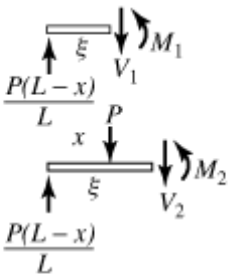
For $\xi < x$, $\quad M_1 = \dfrac{P\xi(L-x)}{L}$



For $\xi > x$, $\quad M_2 = \dfrac{Px(L-\xi)}{L}$

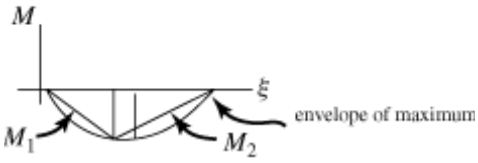
Note that $M_1 = M_2$ when $x = \xi$

$$M_{max} = M_1 = M_2 = \frac{Px(L-x)}{L} = \frac{P}{L}\left(Lx - x^2\right)$$



$$\frac{d}{dx}\left(Lx - x^2\right) = L - 2x \qquad x = \frac{L}{2}$$

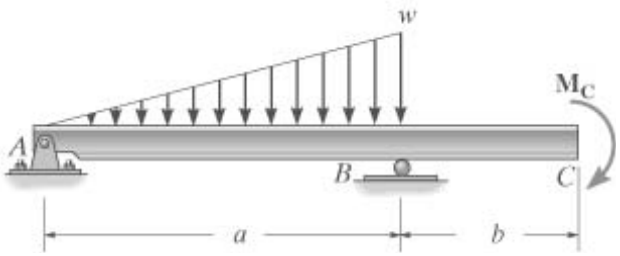
Thus, $\quad M_{max} = \dfrac{P}{L}\left(\dfrac{L}{2}\right)\left(L - \dfrac{L}{2}\right) = \dfrac{P}{2}\left(\dfrac{L}{2}\right)$



$$P = \frac{4M_{max}}{L}$$

Problem 7-59

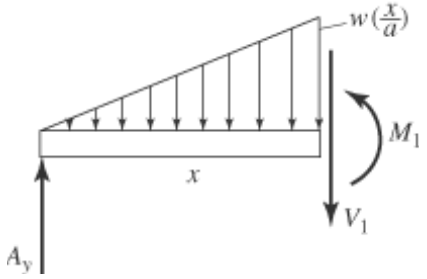
Draw the shear and moment diagrams for the beam.



Given:

$w = 30 \dfrac{\text{lb}}{\text{ft}} \qquad M_C = 180 \text{ lb}\cdot\text{ft}$

$a = 9 \text{ ft} \qquad b = 4.5 \text{ ft}$



Solution:

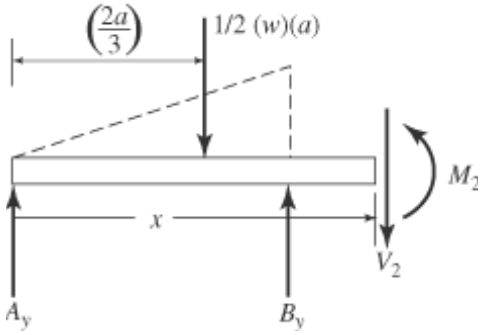
$$-A_y a + \frac{1}{2}\, w a \left(\frac{a}{3}\right) - M_C = 0$$

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$$A_y = \frac{wa}{6} - \frac{M_C}{a}$$

$$A_y + B_y - \frac{1}{2}wa = 0$$

$$B_y = \frac{wa}{2} - A_y$$



$x_1 = 0, 0.01a .. a$

$$A_y - \frac{1}{2}\, w\left(\frac{x}{a}\right)x - V_1(x) = 0 \qquad V_1(x) = \left(A_y - \frac{wx^2}{2a}\right)\frac{1}{\text{lb}}$$

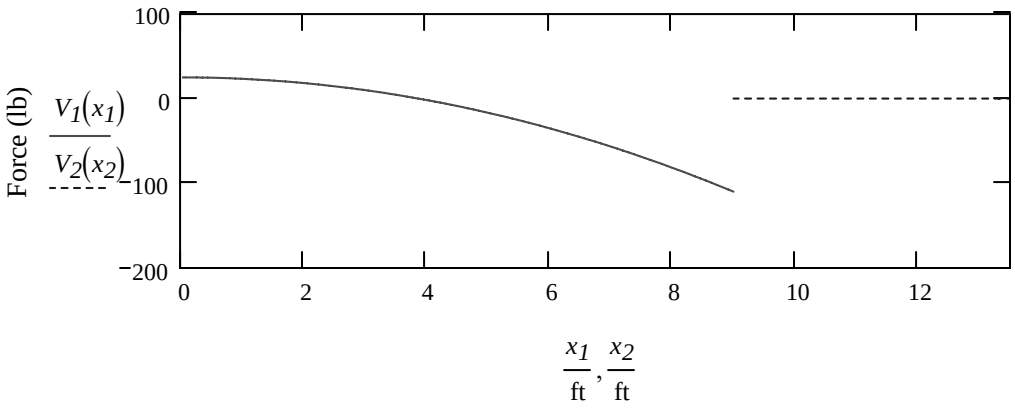
$$-A_y x + \frac{1}{2}w\left(\frac{x}{a}\right)x\left(\frac{x}{3}\right) + M_1(x) = 0 \qquad M_1(x) = \left(A_y x - \frac{wx^3}{6a}\right)\frac{1}{\text{lb}\cdot\text{ft}}$$

$x_2 = a, 1.01a .. a + b$

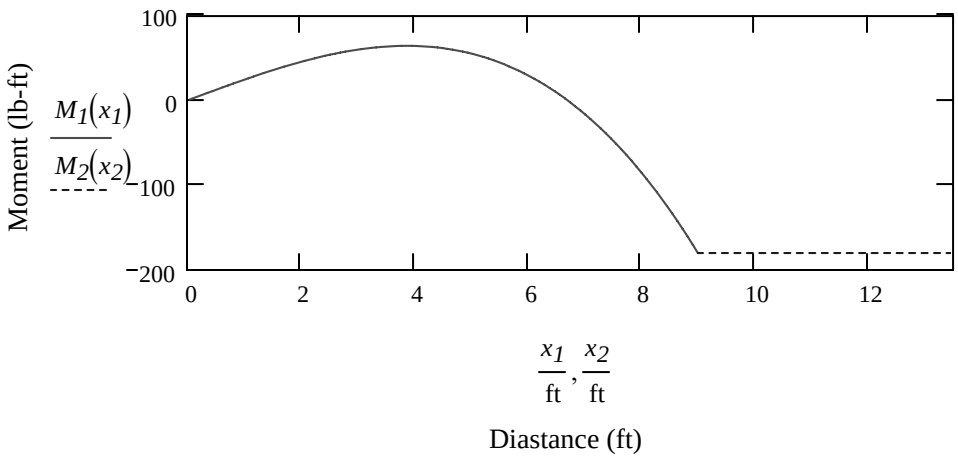
$$A_y - \frac{1}{2}wa + B_y - V_2(x) = 0 \qquad V_2(x) = \left(A_y + B_y - \frac{wa}{2}\right)\frac{1}{\text{lb}}$$

$$-A_y x + \frac{1}{2}wa\left(x - \frac{2a}{3}\right) - B_y(x-a) + M_2(x) = 0$$

$$M_2(x) = \left[A_y x + B_y(x-a) - \frac{wa}{2}\cdot\left(x - \frac{2a}{3}\right)\right]\frac{1}{\text{lb}\cdot\text{ft}}$$



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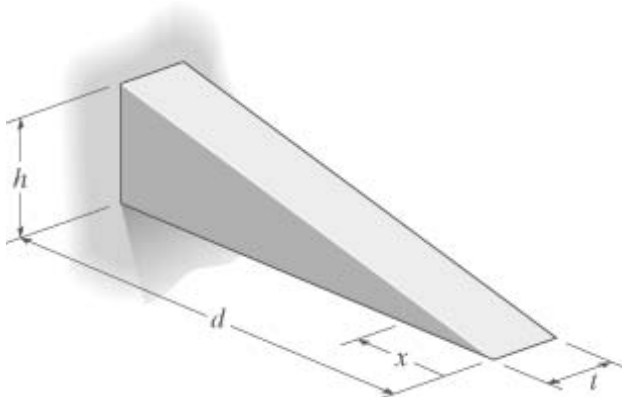
Problem 7-60

The cantilevered beam is made of material having a specific weight γ. Determine the shear and moment in the beam as a function of x.

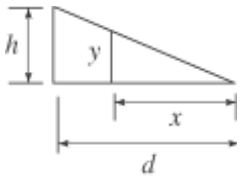
Solution:

By similar triangles

$$\frac{y}{x} = \frac{h}{d} \qquad y = \frac{h}{d}\,x$$

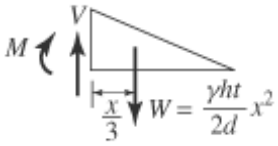


$$W = \gamma V = \gamma\left(\frac{1}{2}yxt\right) = \gamma\left[\frac{1}{2}\left(\frac{h}{d}x\right)xt\right] = \left(\frac{\gamma h t}{2d}\right)x^2$$



$$\Sigma F_y = 0; \qquad V - \left(\frac{\gamma h t}{2d}\right)x^2 = 0$$

$$V = \left(\frac{\gamma h t}{2d}\right)x^2$$



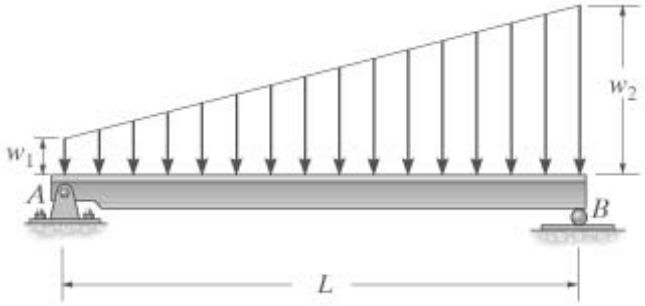
$$\Sigma M = 0; \qquad -M - \left(\frac{\gamma h t}{2d}\right)x^2\left(\frac{x}{3}\right) = 0$$

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$$M = -\frac{\gamma h t}{6d} x^3$$

Problem 7-61

Draw the shear and moment diagrams for the beam.

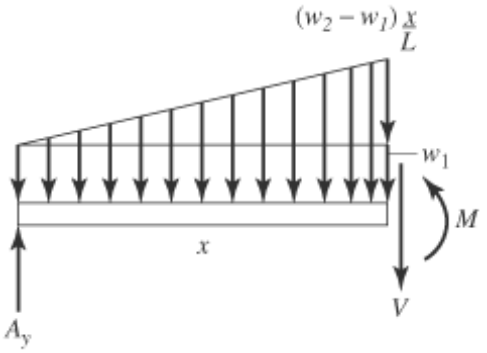


Given: $\text{kip} = 10^3 \text{ lb}$

$$w_1 = 30\frac{\text{lb}}{\text{ft}} \qquad w_2 = 120\ \frac{\text{lb}}{\text{ft}}$$

$$L = 12 \text{ ft}$$

Solution:



$$w_1 L\left(\frac{L}{2}\right) + \frac{1}{2}(w_2 - w_1)L\left(\frac{L}{3}\right) - A_y L = 0$$

$$A_y = w_1\left(\frac{L}{2}\right) + \left(\frac{w_2 - w_1}{2}\right)\left(\frac{L}{3}\right)$$

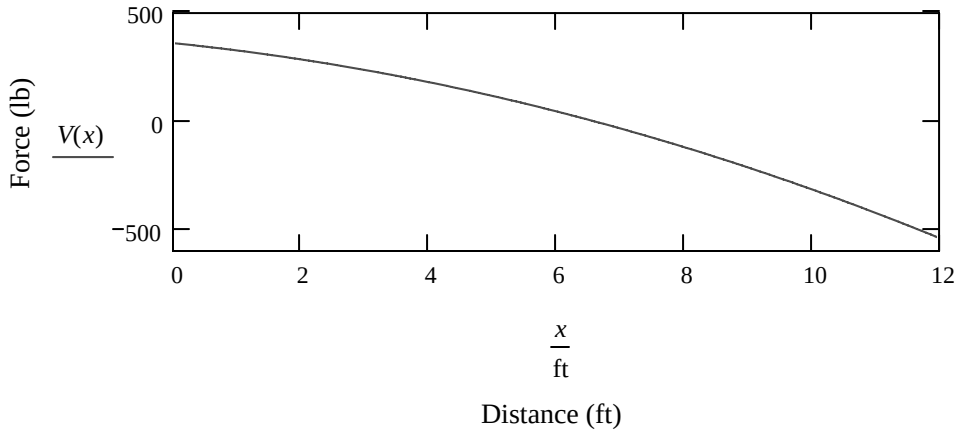
$$A_y - w_1 x - \frac{1}{2}(w_2 - w_1)\left(\frac{x}{L}\right)x - V(x) = 0$$

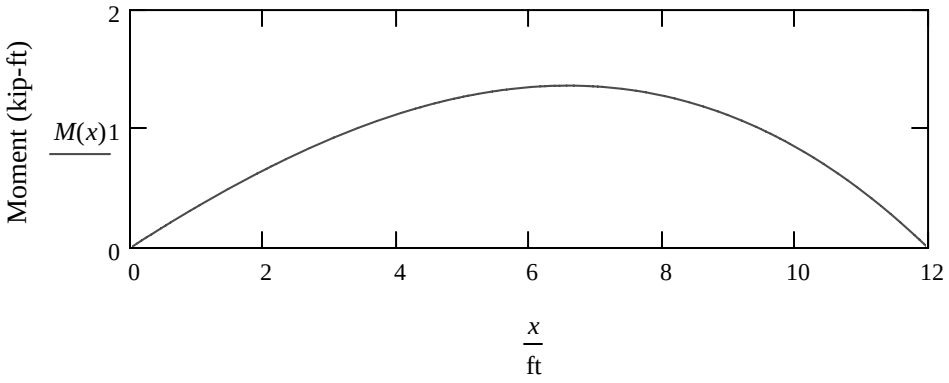
$$V(x) = \left[A_y - w_1 x - \frac{1}{2}(w_2 - w_1)\left(\frac{x^2}{L}\right)\right]\frac{1}{\text{lb}}$$

$$-A_y x + w_1 x\left(\frac{x}{2}\right) + \frac{1}{2}(w_2 - w_1)\left(\frac{x}{L}\right)x\left(\frac{x}{3}\right) + M(x) = 0$$

$$M(x) = \left[A_y x - w_1\left(\frac{x^2}{2}\right) - (w_2 - w_1)\left(\frac{x^3}{6L}\right)\right]\frac{1}{\text{kip}\cdot\text{ft}}$$

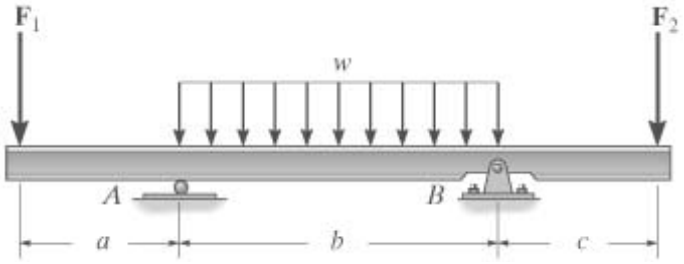
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Problem 7-62

Draw the shear and moment diagrams for the beam.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$F_1 = 20 \text{ kip} \quad F_2 = 20 \text{ kip} \quad w = 4 \frac{\text{kip}}{\text{ft}} \quad a = 15 \text{ ft} \quad b = 30 \text{ ft} \quad c = 15 \text{ ft}$

Solution:

$$F_1(a+b) + wb\left(\frac{b}{2}\right) - F_2 c - A_y b = 0 \qquad A_y = F_1\left(\frac{a+b}{b}\right) + \frac{wb}{2} - F_2\left(\frac{c}{b}\right)$$

$$A_y + B_y - F_1 - F_2 - wb = 0 \qquad B_y = F_1 + F_2 + wb - A_y$$

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$x_1 = 0, 0.01a .. a$

$-F_1 - V_1(x) = 0 \qquad V_1(x) = -F_1 \frac{1}{\text{kip}}$

$F_1 x + M_1(x) = 0 \qquad M_1(x) = -F_1 x \frac{1}{\text{kip}\cdot\text{ft}}$

$x_2 = a, 1.01a .. a + b$

$-F_1 - w(x-a) + A_y - V_2(x) = 0 \qquad V_2(x) = \left[-F_1 - w(x-a) + A_y\right] \frac{1}{\text{kip}}$

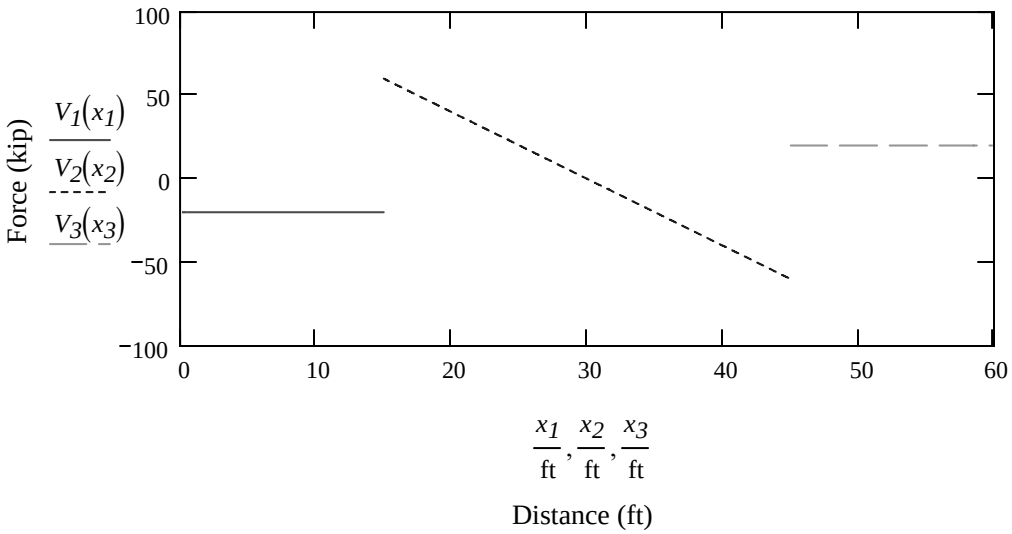
$F_1 x - A_y(x-a) + w(x-a)\left(\frac{x-a}{2}\right) + M_2(x) = 0$

$M_2(x) = \left[-F_1 x + A_y(x-a) - w\frac{(x-a)^2}{2}\right]\frac{1}{\text{kip}\cdot\text{ft}}$

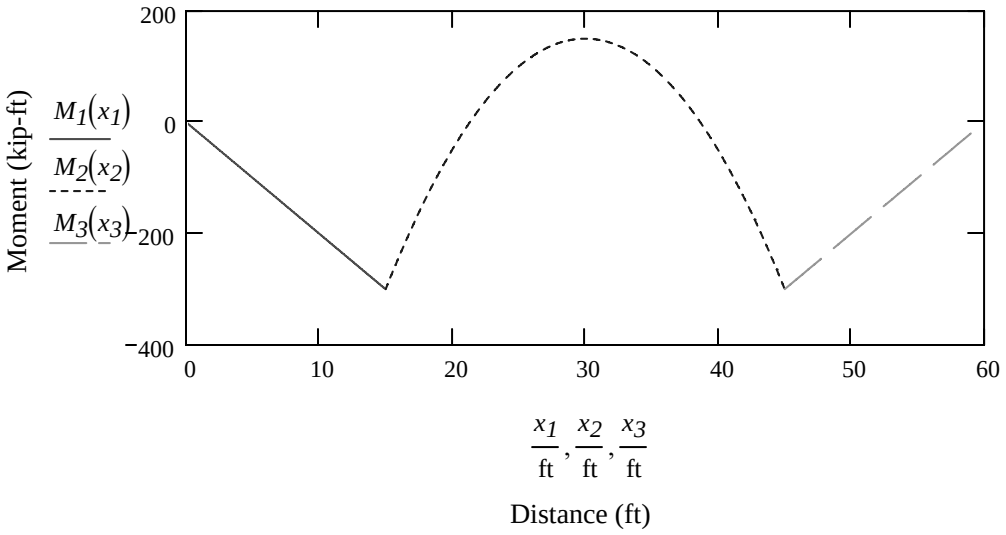
$x_3 = a + b, 1.01(a+b) .. a + b + c$

$V_3(x) - F_2 = 0 \qquad V_3(x) = F_2 \frac{1}{\text{kip}}$

$-M_3(x) - F_2(a + b + c - x) = 0 \qquad M_3(x) = -F_2(a + b + c - x) \frac{1}{\text{kip}\cdot\text{ft}}$

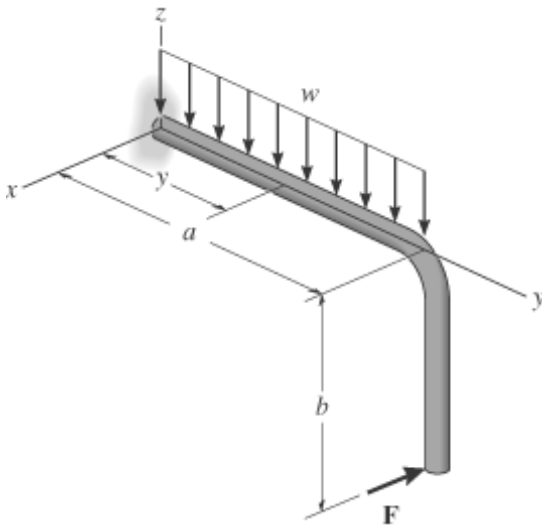


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Problem 7-63

Express the *x*, *y*, *z* components of internal loading in the rod at the specific value for *y*, where $0 < y < a$



Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given: $y = 2.5$ ft

$w = 800 \dfrac{\text{lb}}{\text{ft}}$ $F = 1500$ lb

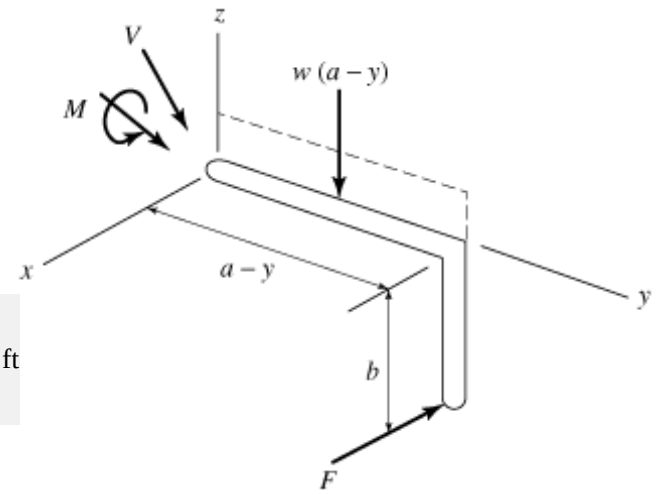
$a = 4$ ft $b = 2$ ft

Solution: In general we have

$$\mathbf{V} = \begin{bmatrix} F \\ 0 \\ w(a - y) \end{bmatrix}$$

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$$\mathbf{M} = \begin{bmatrix} w(a-y)\left(\dfrac{a-y}{2}\right) \\ -F\,b \\ -F(a-y) \end{bmatrix}$$

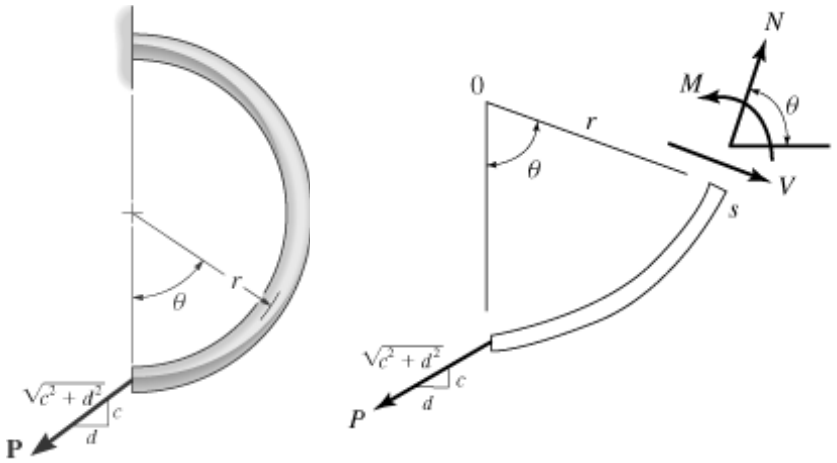


For these values we have

$$\mathbf{V} = \begin{pmatrix} 1500.00 \\ 0.00 \\ 1200.00 \end{pmatrix} \text{lb} \qquad \mathbf{M} = \begin{pmatrix} 900.00 \\ -3000.00 \\ -2250.00 \end{pmatrix} \text{lb}\cdot\text{ft}$$

Problem 7-64

Determine the normal force, shear force, and moment in the curved rod as a function of θ.



Given:

$c = 3$

$d = 4$

Solution:

For $0 \le \theta \le \pi$

$\nearrow \Sigma F_x = 0;$ $\quad N - \left(\dfrac{d}{\sqrt{c^2+d^2}}\right)P\cos(\theta) - \left(\dfrac{c}{\sqrt{c^2+d^2}}\right)P\sin(\theta) = 0$

$$N = \left(\frac{P}{\sqrt{c^2+d^2}}\right)\left(d\cos(\theta) + c\sin(\theta)\right)$$

$+\searrow \Sigma F_y = 0;$ $\quad V - \left(\dfrac{d}{\sqrt{c^2+d^2}}\right)P\sin(\theta) + \left(\dfrac{c}{\sqrt{c^2+d^2}}\right)P\cos(\theta) = 0$

$$V = \left(\frac{P}{\sqrt{c^2+d^2}}\right)\left(d\sin(\theta) - c\cos(\theta)\right)$$

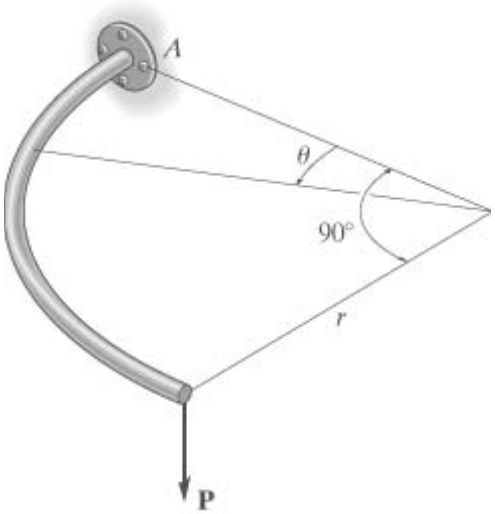
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$\circlearrowleft + \Sigma M = 0;$ $\left(\frac{-d}{\sqrt{c^2+d^2}}\right)P(r - r\cos(\theta)) + \left(\frac{c}{\sqrt{c^2+d^2}}\right)P\,r\sin(\theta) + M = 0$

$$M = \left(\frac{P\,r}{\sqrt{c^2+d^2}}\right)(d - d\cos(\theta) - c\sin(\theta))$$

Problem 7-65

The quarter circular rod lies in the horizontal plane and supports a vertical force **P** at its end. Determine the magnitudes of the components of the internal shear force, moment, and torque acting in the rod as a function of the angle θ.



Solution:

$\Sigma F_z = 0;$ $V = |P|$

$\Sigma M_x = 0;$ $M + P\,r\cos(\theta) = 0$

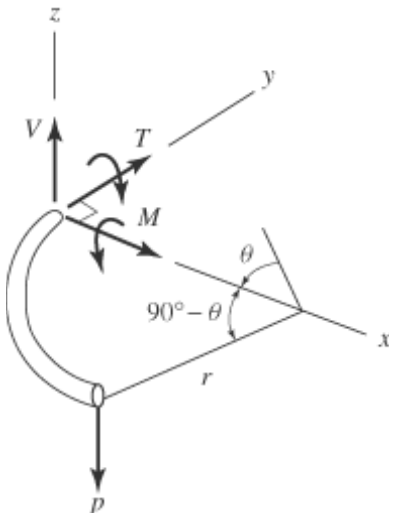
$M = -P\,r\cos(\theta)$

$M = |P\,r\cos(\theta)|$

$\Sigma M_y = 0;$ $T + P\,r(1 - \sin(\theta)) = 0$

$T = -P\,r(1 - \sin(\theta)$

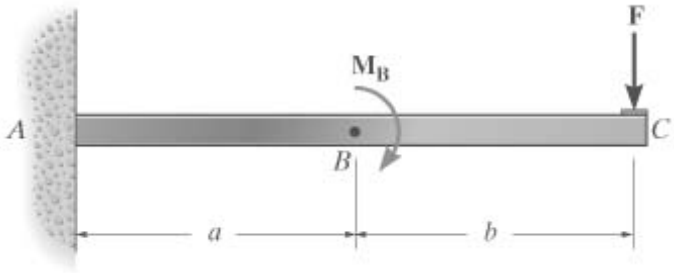
$T = |P\,r(1 - \sin(\theta)\ |$



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Problem 7-66

Draw the shear and moment diagrams for the beam.



Given:

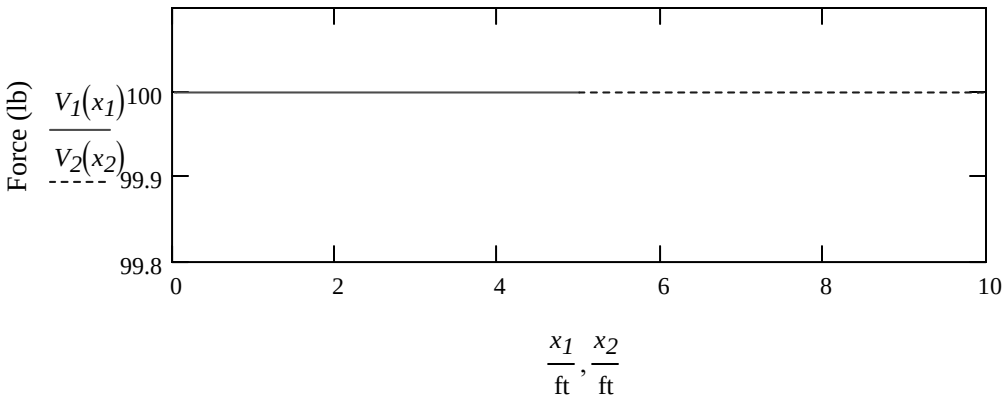
$M_B = 800\ \text{lb}\cdot\text{ft} \quad a = 5\ \text{ft}$

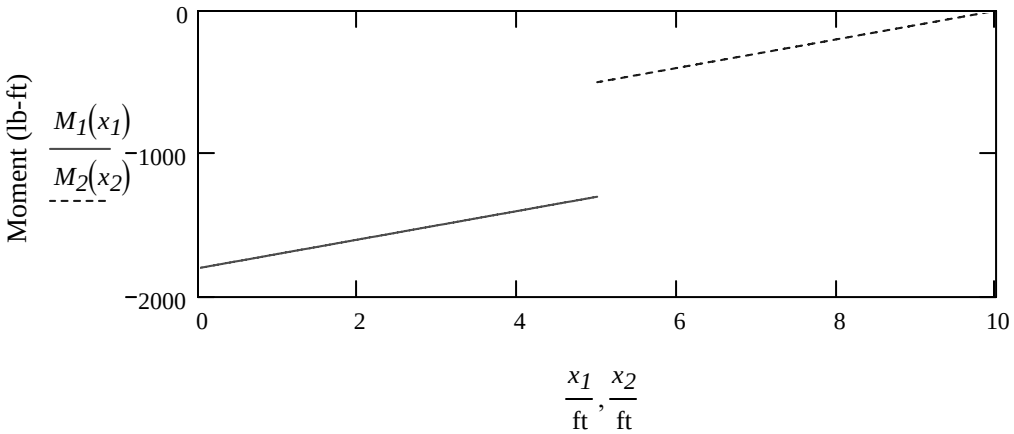
$F = 100\ \text{lb} \quad b = 5\ \text{ft}$

Solution:

$x_1 := 0, 0.01a .. a$ $\quad V_1(x) := F \dfrac{1}{\text{lb}}$ $\quad M_1(x) := \left[-F(a+b-x) - M_B\right] \dfrac{1}{\text{lb}\cdot\text{ft}}$

$x_2 := a, 1.01a .. a+b$ $\quad V_2(x) := F \dfrac{1}{\text{lb}}$ $\quad M_2(x) := -F(a+b-x) \dfrac{1}{\text{lb}\cdot\text{ft}}$

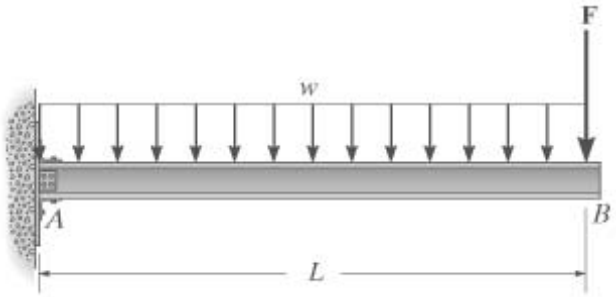




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Problem 7-67

Draw the shear and moment diagrams for the beam.



Units Used: $\text{kN} = 10^3\ \text{N}$

Given:

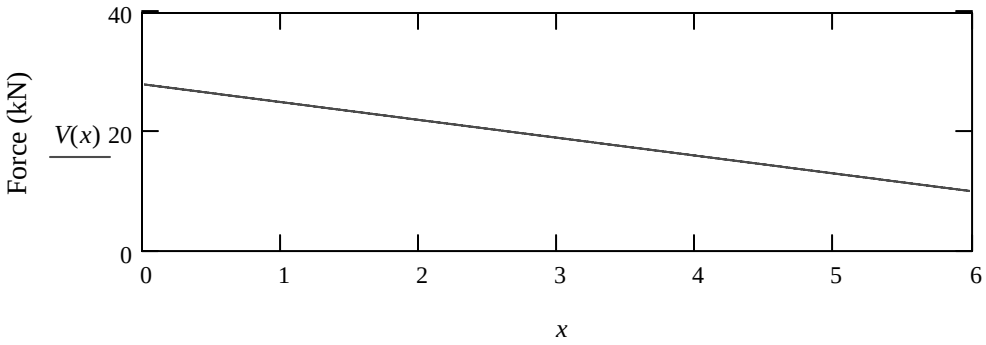
$w = 3\ \frac{\text{kN}}{\text{m}}$ $F = 10\ \text{kN}$

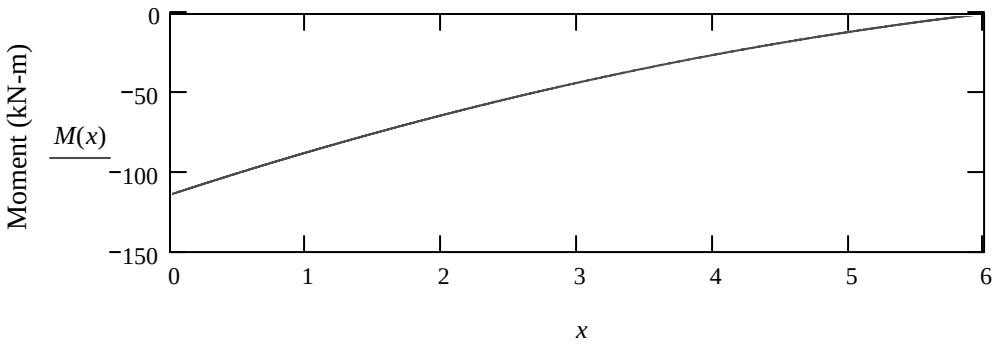
$L = 6\ \text{m}$

Solution:

$$V(x) = [w(L - x) + F]\ \frac{1}{\text{kN}}$$

$$M(x) = \left[-w\frac{(L - x)^2}{2} - F(L - x)\right]\ \frac{1}{\text{kN}\cdot\text{m}}$$

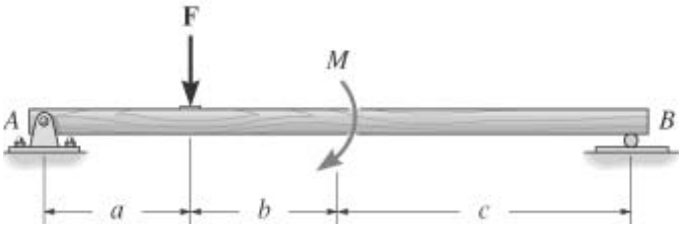




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Problem 7-68

Draw the shear and moment diagrams for the beam.



Units Used: $\text{kN} = 10^3\ \text{N}$

Given: $F = 7\ \text{kN}$ $M = 12\ \text{kN}\cdot\text{m}$ $a = 2\ \text{m}$ $b = 2\ \text{m}$ $c = 4\ \text{m}$

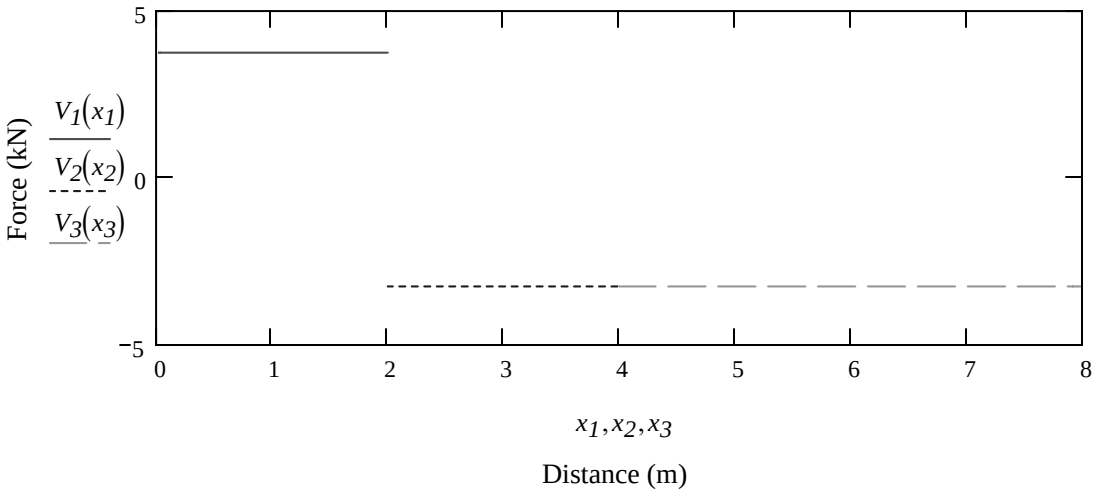
Solution: Guesses $A = 1\ \text{N}$ $B = 1\ \text{N}$

Given $A + B - F = 0$ $-F\,a - M + B(a + b + c) = 0$ $\begin{pmatrix} A \\ B \end{pmatrix} = \text{Find}(A, B)$

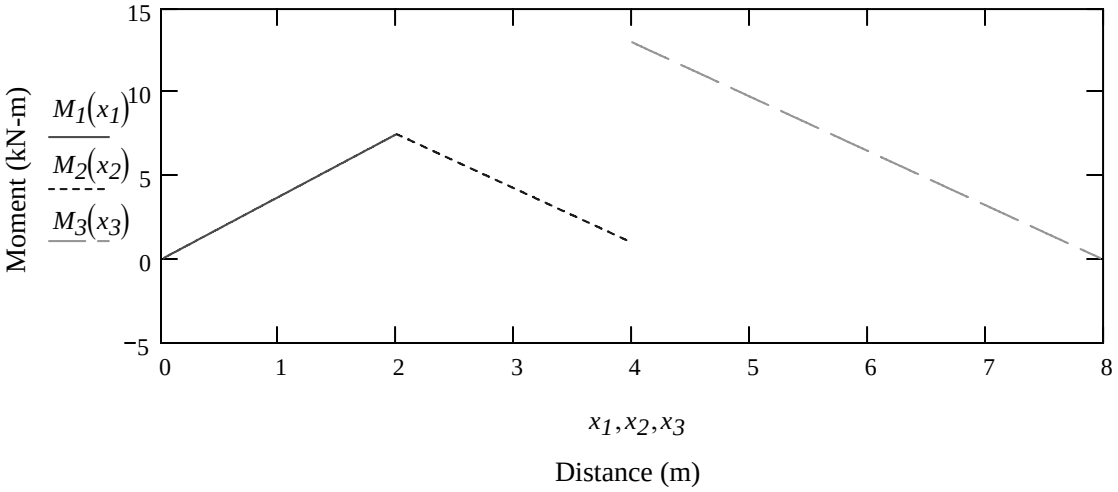
$x_1 = 0, 0.01a .. a$ $x_2 = a, 1.01a .. a + b$ $x_3 = a + b, 1.01(a + b) .. a + b + c$

$V_1(x_1) = A\,\dfrac{1}{\text{kN}}$ $V_2(x_2) = (A - F)\,\dfrac{1}{\text{kN}}$ $V_3(x_3) = -B\,\dfrac{1}{\text{kN}}$

$M_1(x_1) = \dfrac{A\,x_1}{\text{kN}\cdot\text{m}}$ $M_2(x_2) = \dfrac{A\,x_2 - F(x_2 - a)}{\text{kN}\cdot\text{m}}$ $M_3(x_3) = B(a + b + c - x_3)\,\dfrac{1}{\text{kN}\cdot\text{m}}$

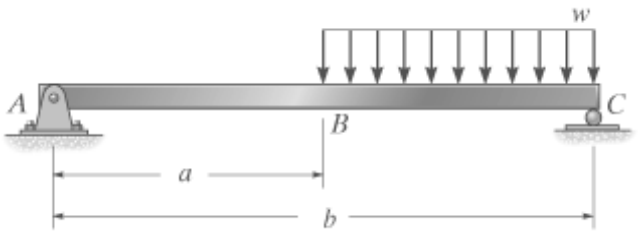


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Problem 7-69

Draw the shear and moment diagrams for the beam.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$a = 2\ \text{m} \quad b = 4\ \text{m} \quad w = 1.5\ \frac{\text{kN}}{\text{m}}$

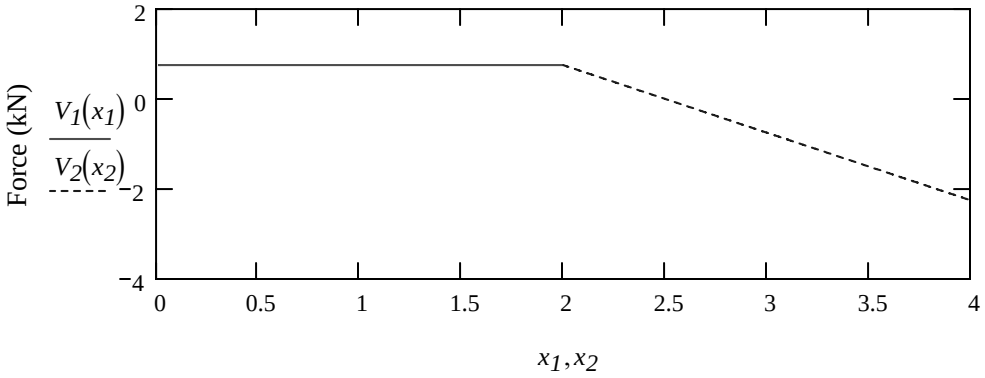
Solution:

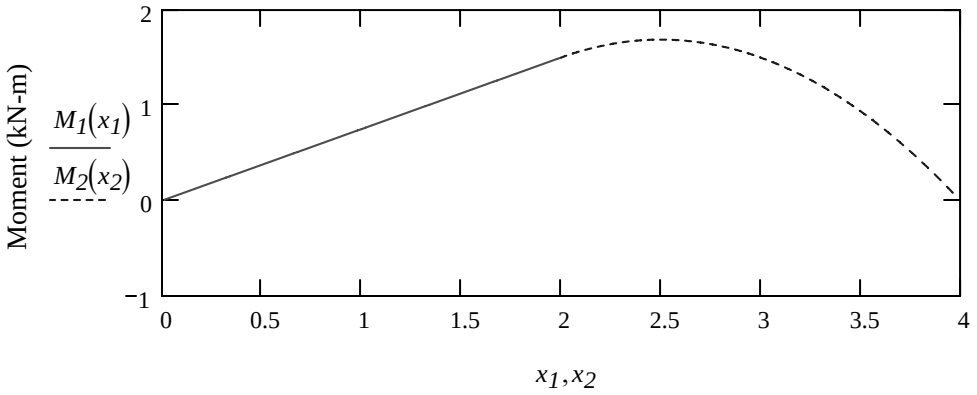
$w(b-a)\left(\frac{b-a}{2}\right) - A_y b = 0 \qquad A_y = \frac{w(b-a)^2}{2b} \qquad A_y = 0.75\,\text{kN}$

$x_1 = 0, 0.01a .. a \qquad V_1(x) = A_y \frac{1}{\text{kN}} \qquad M_1(x) = A_y x \frac{1}{\text{kN}\cdot\text{m}}$

$x_2 = b-a, 1.01(b-a) .. b \qquad V_2(x) = \left[A_y - w(x-a)\right]\frac{1}{\text{kN}} \qquad M_2(x) = \left[A_y x - w\frac{(x-a)^2}{2}\right]\frac{1}{\text{kN}\cdot\text{m}}$

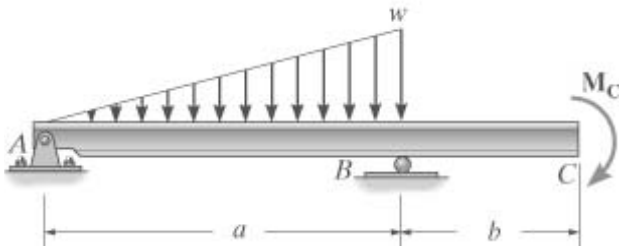
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Problem 7-70

Draw the shear and moment diagrams for the beam.



Given:

$w = 30 \dfrac{\text{lb}}{\text{ft}}$ $M_C = 180$ lb·ft

$a = 9$ ft $b = 4.5$ ft

Solution:

$$-A_y a + \frac{1}{2} w a \left(\frac{a}{3}\right) - M_C = 0$$

$$A_y = \frac{w a}{6} - \frac{M_C}{a}$$

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$$A_y + B_y - \frac{1}{2} w a = 0$$

$$B_y = \frac{w a}{2} - A_y$$

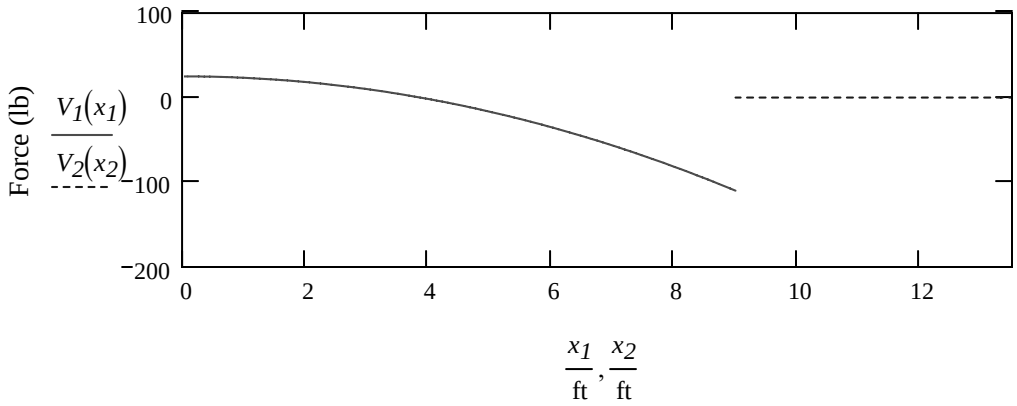
$x_1 = 0, 0.01a .. a$

$$V_1(x) = \left(A_y - \frac{w x^2}{2a}\right)\frac{1}{\text{lb}} \qquad M_1(x) = \left(A_y x - \frac{w x^3}{6a}\right)\frac{1}{\text{lb}\cdot\text{ft}}$$

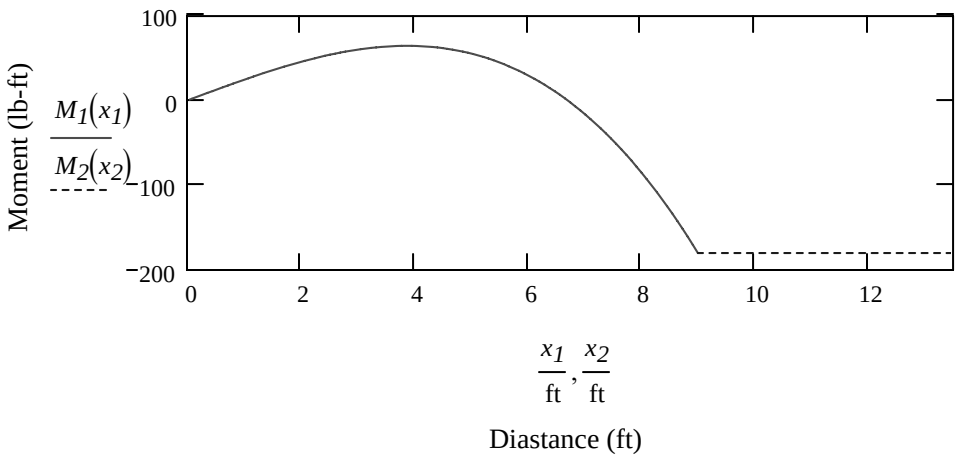
$x_2 = a, 1.01a .. a + b$

$$V_2(x) = \left(A_y + B_y - \frac{w a}{2}\right)\frac{1}{\text{lb}}$$

$$M_2(x) = \left[A_y x + B_y (x - a) - \frac{w a}{2}\left(x - \frac{2a}{3}\right)\right]\frac{1}{\text{lb}\cdot\text{ft}}$$

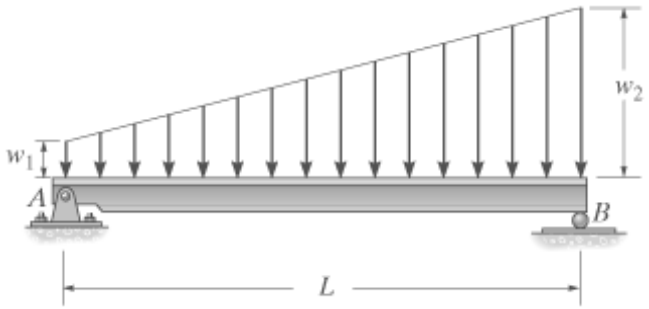


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Problem 7-71

Draw the shear and moment diagrams for the beam.



Given: $\text{kip} = 10^3\ \text{lb}$

$w_1 = 30\ \frac{\text{lb}}{\text{ft}} \qquad w_2 = 120\ \frac{\text{lb}}{\text{ft}}$

$L = 12\ \text{ft}$

Solution:

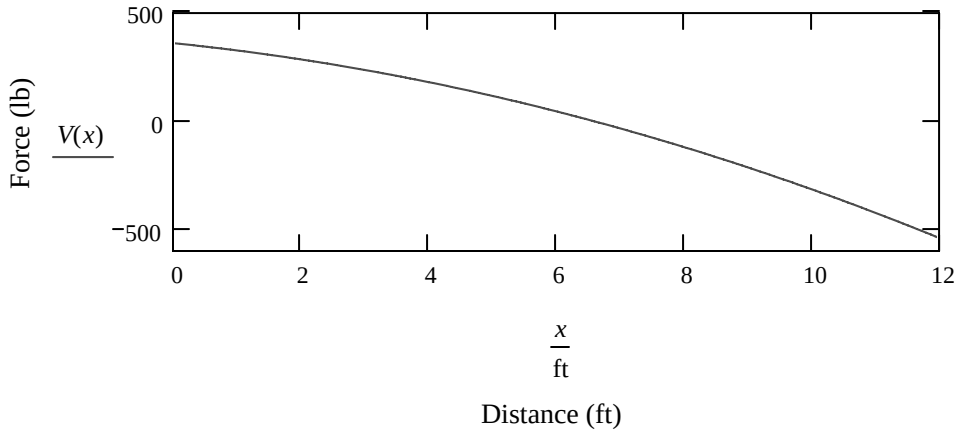
$$w_1\ L\left(\frac{L}{2}\right) + \frac{1}{2}(w_2 - w_1)L\left(\frac{L}{3}\right) - A_y L = 0$$

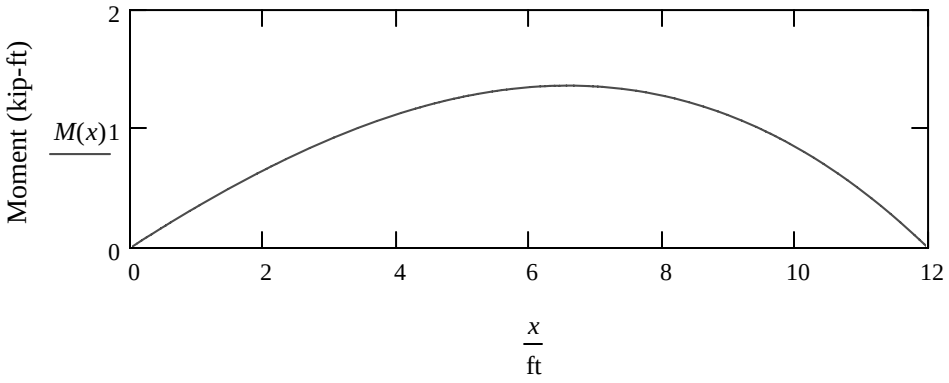
$$A_y = w_1\left(\frac{L}{2}\right) + \left(\frac{w_2 - w_1}{2}\right)\left(\frac{L}{3}\right)$$

$$V(x) = \left[A_y - w_1 x - \frac{1}{2}(w_2 - w_1)\frac{x^2}{L}\right]\frac{1}{\text{lb}}$$

$$M(x) = \left[A_y x - w_1 \frac{x^2}{2} - (w_2 - w_1)\frac{x^3}{6L}\right]\frac{1}{\text{kip}\cdot\text{ft}}$$

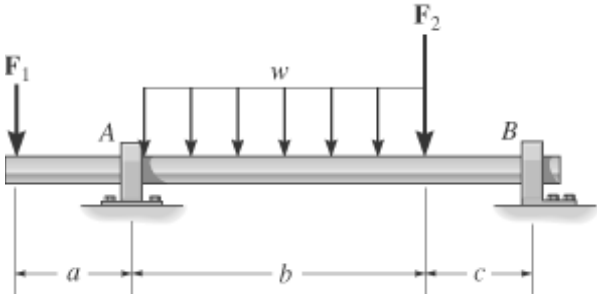
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Problem 7-72

Draw the shear and moment diagrams for the shaft. The support at *A* is a journal bearing and at *B* it is a thrust bearing.



Given:

$F_1 = 400\ \text{lb} \qquad F_2 = 800\ \text{lb}$

$w = 100\ \frac{\text{lb}}{\text{in}} \qquad a = 4\ \text{in} \qquad b = 12\ \text{in} \qquad c = 4\ \text{in}$

Solution: $F_1 a - w b\left(\frac{b}{2}\right) - F_2 b + B(b + c) = 0 \qquad B = \frac{w\left(\frac{b^2}{2}\right) + F_2 b - F_1 a}{b + c} \qquad B = 950.00\,\text{lb}$

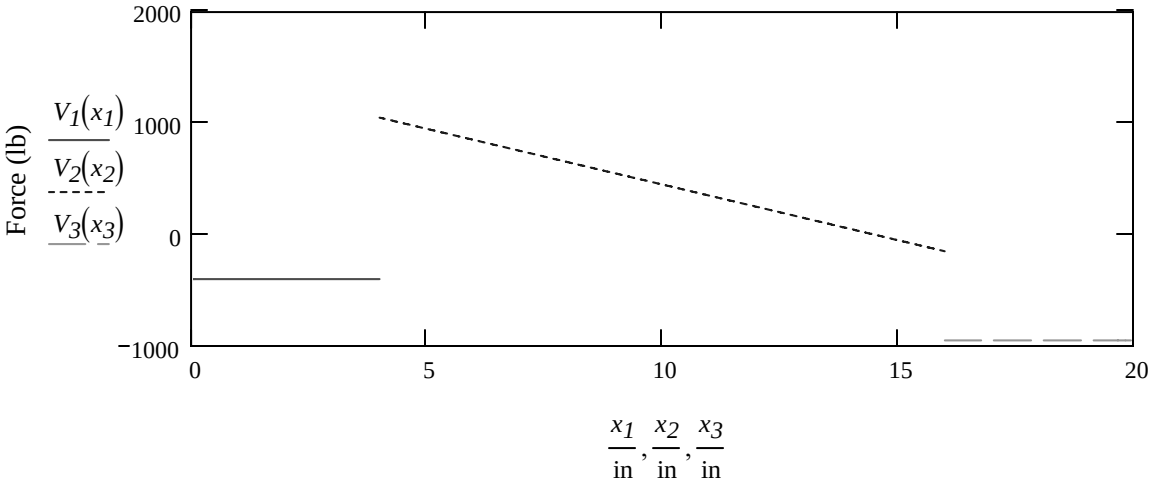
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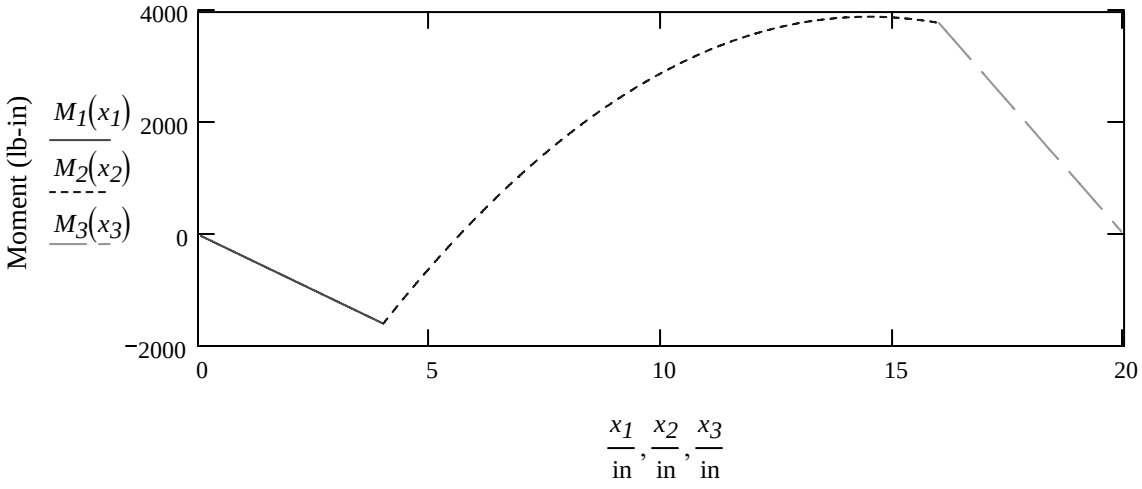
$x_1 = 0, 0.01a .. a \qquad V_1(x) = -F_1 \frac{1}{\text{lb}} \qquad M_1(x) = -F_1 x \frac{1}{\text{lb}\cdot\text{in}}$

$x_2 = a, 1.01a .. a + b \qquad V_2(x_2) = \left[-B + F_2 + w(a + b - x_2)\right]\frac{1}{\text{lb}}$

$$M_2(x_2) = \left[B(a + b + c - x_2) - F_2(a + b - x_2) - w\cdot\frac{(a + b - x_2)^2}{2}\right]\frac{1}{\text{lb}\cdot\text{in}}$$

$x_3 = a + b, 1.01(a + b) .. a + b + c \qquad V_3(x_3) = \frac{-B}{\text{lb}} \qquad M_3(x_3) = B(a + b + c - x_3)\frac{1}{\text{lb}\cdot\text{in}}$

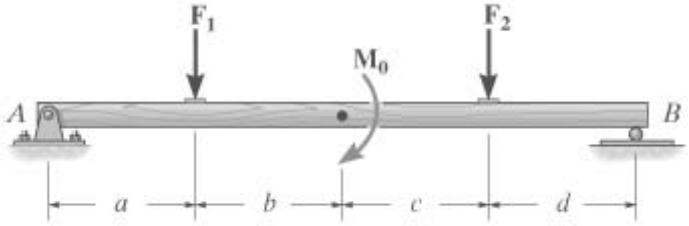




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Problem 7-73

Draw the shear and moment diagrams for the beam.



Units Used:

$kN = 10^3\ N$

Given:

$F_1 = 10\ kN \quad F_2 = 10\ kN \quad M_0 = 12\ kN\cdot m$

$a = 2\ m \quad b = 2\ m \quad c = 2\ m \quad d = 2\ m$

Solution:

$$F_1(b + c + d) - M_0 + F_2 d - A_y(a + b + c + d) = 0$$

$$A_y = \frac{F_1(b + c + d) - M_0 + F_2 d}{a + b + c + d} \qquad A_y = 8.50\,kN$$

$$A_y + B_y - F_1 - F_2 = 0 \qquad B_y = F_1 + F_2 - A_y \qquad B_y = 11.50\,kN$$

$x_1 = 0, 0.01a .. a$

$$V_1(x) = A_y \frac{1}{kN} \qquad M_1(x) = A_y x \frac{1}{kN\cdot m}$$

$x_2 = a, 1.01a .. a + b$

$$V_2(x) = (A_y - F_1)\frac{1}{kN} \qquad M_2(x) = \left[A_y x - F_1(x - a)\right]\frac{1}{kN\cdot m}$$

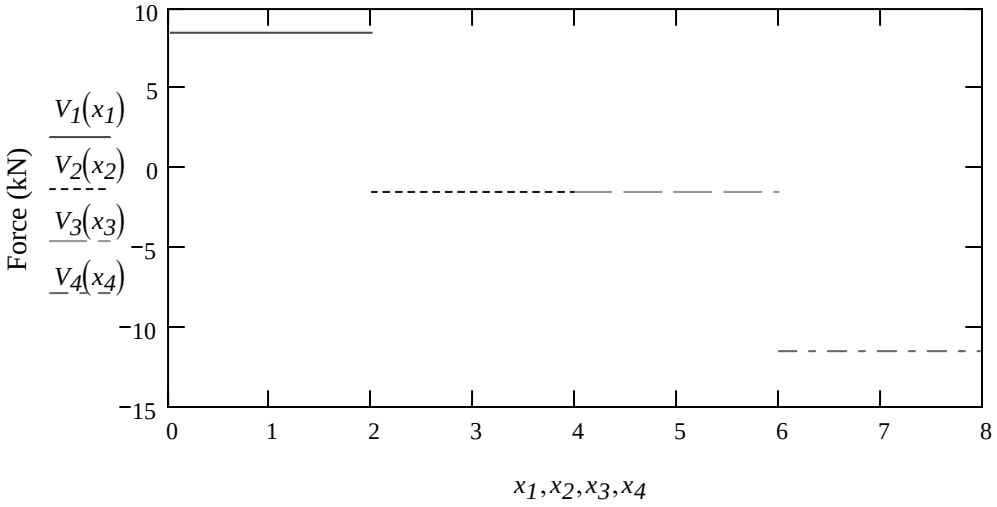
$x_3 = a + b, 1.01(a + b) .. a + b + c$

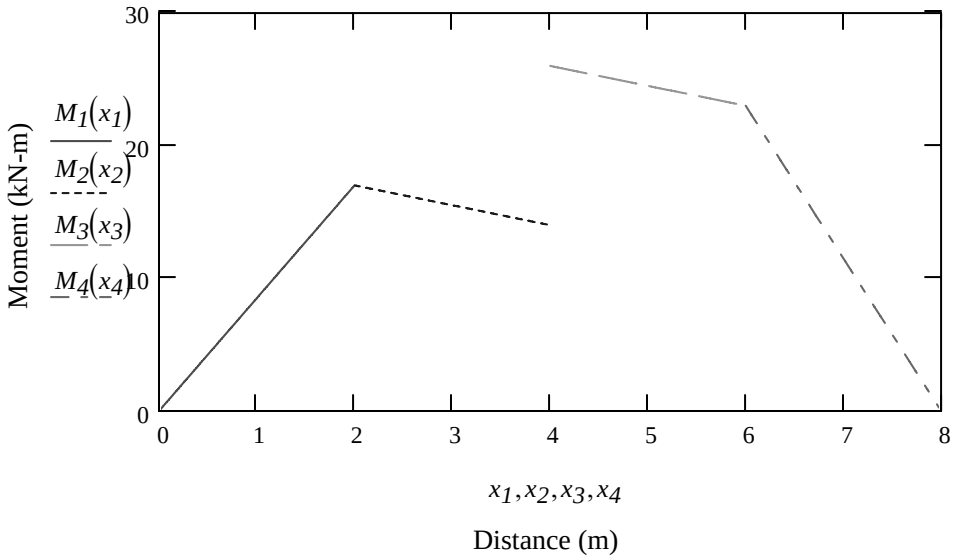
$$V_3(x) = (A_y - F_1)\frac{1}{kN} \qquad M_3(x) = \left[A_y x - F_1(x - a) + M_0\right]\frac{1}{kN\cdot m}$$

$x_4 = a + b + c, 1.01(a + b + c) .. a + b + c + d$

$$V_4(x) = -B_y \frac{1}{kN} \qquad M_4(x) = B_y(a + b + c + d - x)\frac{1}{kN\cdot m}$$

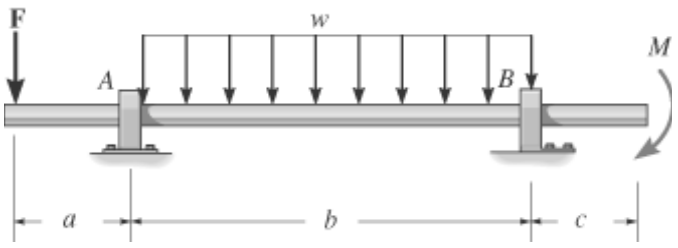
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Problem 7-74

Draw the shear and moment diagrams for the shaft. The support at *A* is a journal bearing and at *B* it is a thrust bearing.



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Given:

$F = 200\ \text{lb}$ $w = 100\ \frac{\text{lb}}{\text{ft}}$ $M = 300\ \text{lb}\cdot\text{ft}$ $a = 1\ \text{ft}$ $b = 4\ \text{ft}$ $c = 1\ \text{ft}$

Solution:

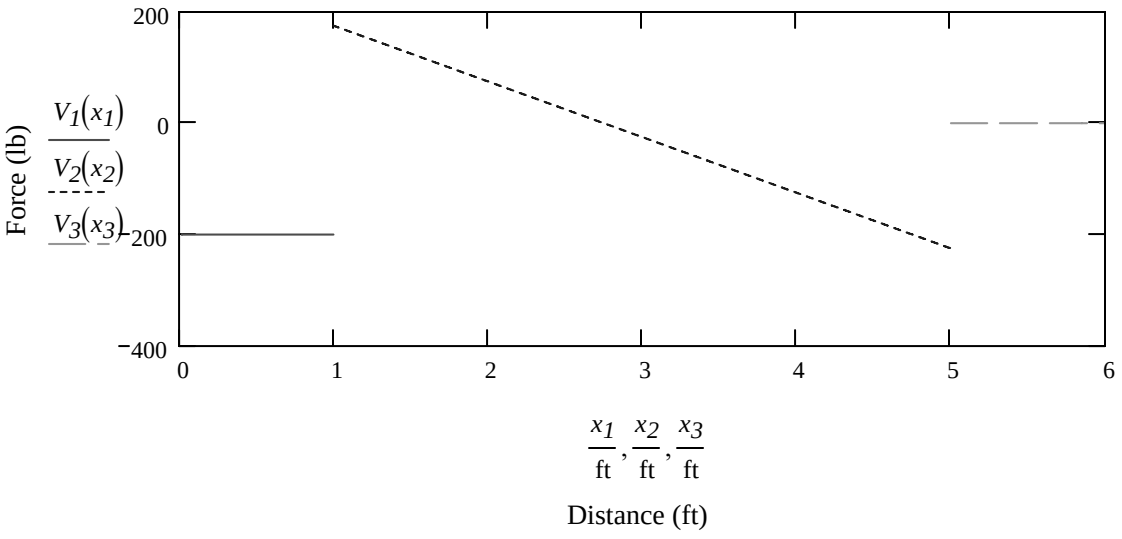
$$F(a+b) - A\,b + w\,b\left(\frac{b}{2}\right) - M = 0 \qquad A = \frac{F(a+b) + \left(\frac{w\,b^2}{2}\right) - M}{b} \qquad A = 375.00\,\text{lb}$$

$$x_1 = 0, 0.01a .. a \qquad V_1(x_1) = -F\frac{1}{\text{lb}} \qquad M_1(x_1) = -F\,x_1\frac{1}{\text{lb}\cdot\text{ft}}$$

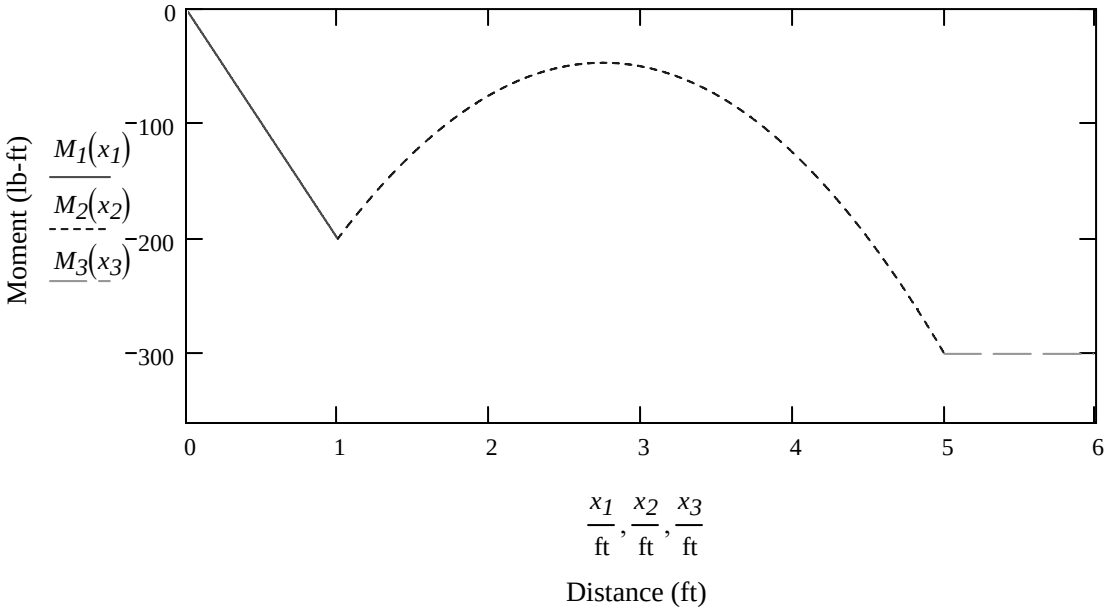
$$x_2 = a, 1.01a .. a + b \qquad V_2(x_2) = \left[-F + A - w(x_2 - a)\right]\frac{1}{\text{lb}}$$

$$M_2(x_2) = \left[-F\,x_2 + A(x_2 - a) - w\frac{(x_2 - a)^2}{2}\right]\frac{1}{\text{lb}\cdot\text{ft}}$$

$$x_3 = a + b, 1.01(a+b) .. a + b + c \qquad V_3(x_3) = 0\frac{1}{\text{lb}} \qquad M_3(x_3) = -M\frac{1}{\text{lb}\cdot\text{ft}}$$

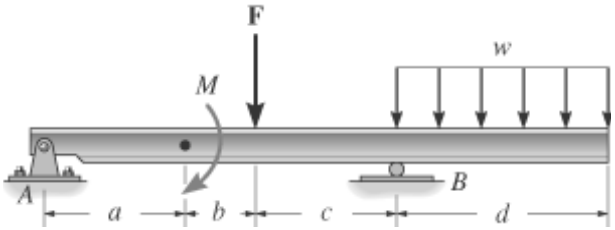


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Problem 7-75

Draw the shear and moment diagrams for the beam.



Units Used:

$kN = 10^3\ N$

Given: $F = 8\ kN$ $M = 20\ kN\cdot m$ $w = 15\ \frac{kN}{m}$ $a = 2\ m$ $b = 1\ m$

$c = 2\ m$ $d = 3\ m$

Solution:

$$-A(a+b+c) - M + F\,c - w\,d\left(\frac{d}{2}\right) = 0 \qquad A = \frac{F\,c - M - w\left(\frac{d^2}{2}\right)}{a+b+c} \qquad A = -14.30\ kN$$

$$x_1 = 0, 0.01a .. a \qquad V_1(x_1) = A\frac{1}{kN} \qquad M_1(x_1) = A\,x_1\frac{1}{kN\cdot m}$$

$$x_2 = a, 1.01a .. a+b \qquad V_2(x_2) = A\frac{1}{kN} \qquad M_2(x_2) = (A\,x_2 + M)\frac{1}{kN\cdot m}$$

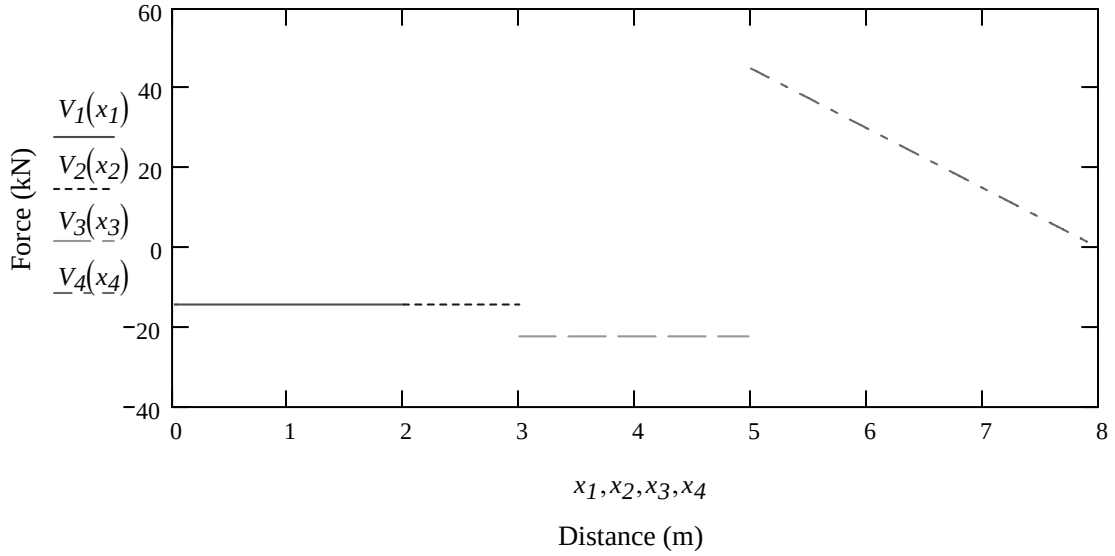
$$x_3 = a+b, 1.01(a+b) .. a+b+c \qquad V_3(x_3) = (A - F)\frac{1}{kN}$$

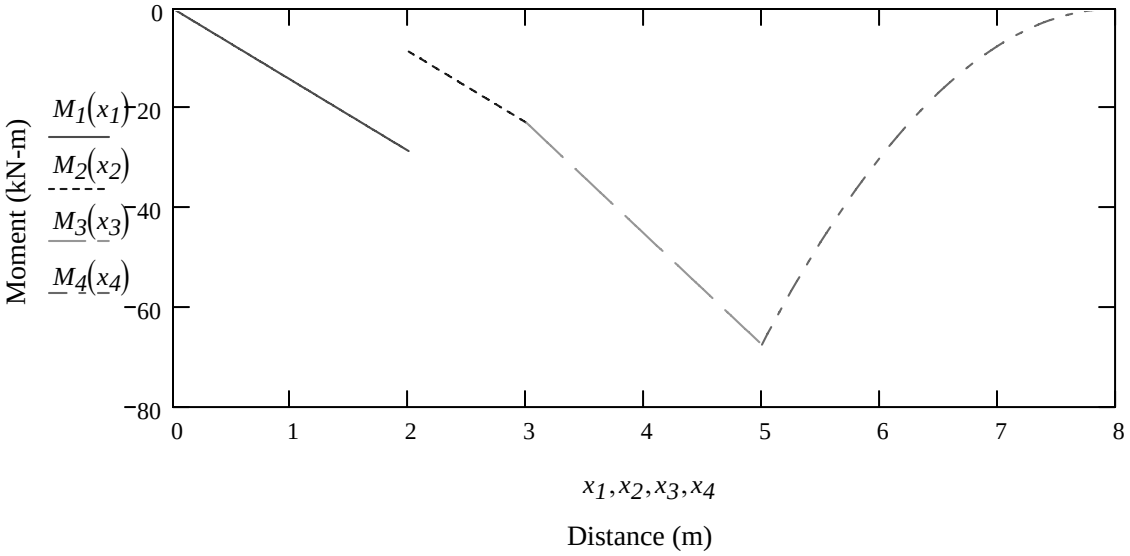
$$M_3(x_3) = \left[A\,x_3 + M - F(x_3 - a - b)\right]\frac{1}{kN\cdot m}$$

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$x_4 = a + b + c, 1.01(a + b + c) .. a + b + c + d$

$$V_4(x_4) = w(a + b + c + d - x_4)\frac{1}{\text{kN}} \qquad M_4(x_4) = -w\frac{(a + b + c + d - x_4)^2}{2}\frac{1}{\text{kN}\cdot\text{m}}$$

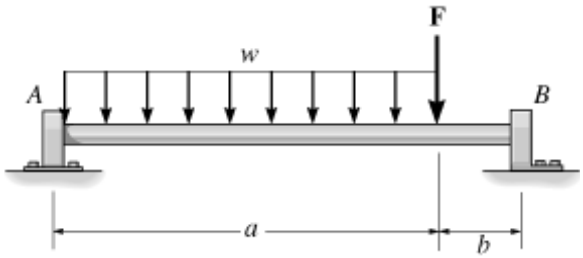




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Problem 7-76

Draw the shear and moment diagrams for the shaft. The support at *A* is a thrust bearing and at *B* it is a journal bearing.



Units Used:

$\text{kN} = 10^3\ \text{N}$

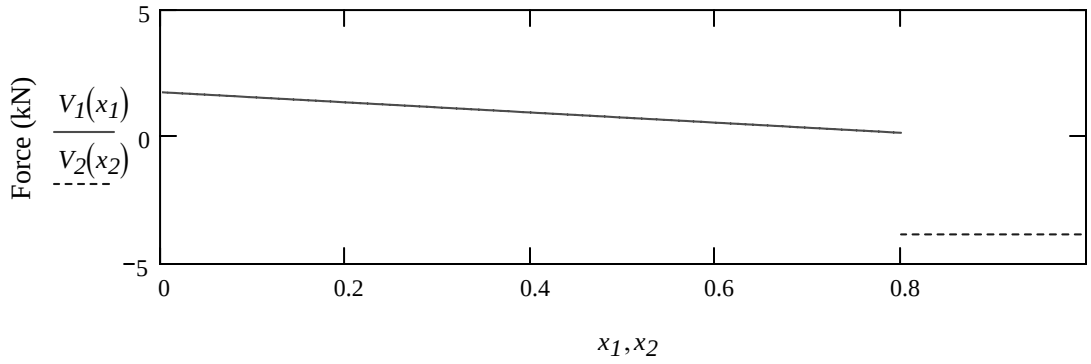
Given: $w = 2\ \frac{\text{kN}}{\text{m}}$ $F = 4\ \text{kN}$ $a = 0.8\ \text{m}$ $b = 0.2\ \text{m}$

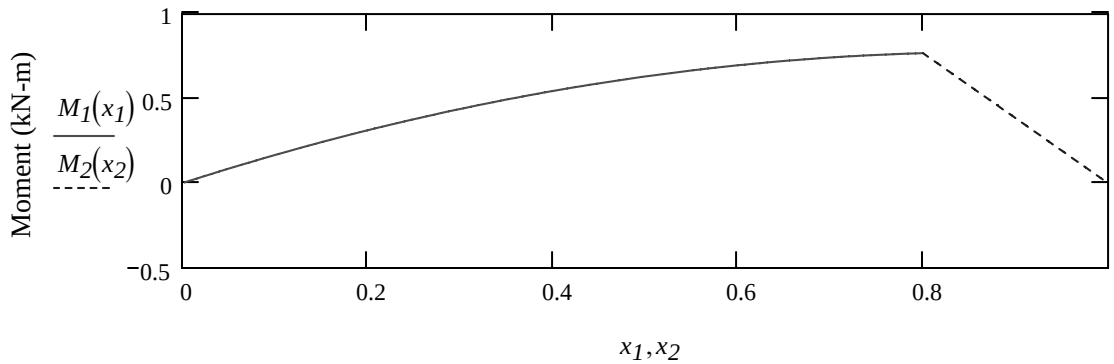
Solution: $B(a+b) - F\,a - w\,a\left(\frac{a}{2}\right) = 0$ $B = \frac{F\,a + w\left(\frac{a^2}{2}\right)}{a+b}$ $B = 3.84\,\text{kN}$

$A + B - w\,a - F = 0$ $A = w\,a + F - B$ $A = 1.76\,\text{kN}$

$x_1 = 0, 0.01a\,..\,a$ $V_1(x_1) = \left(A - w\,x_1\right)\frac{1}{\text{kN}}$ $M_1(x_1) = \left[A\,x_1 - w\left(\frac{x_1^2}{2}\right)\right]\frac{1}{\text{kN}\cdot\text{m}}$

$x_2 = a, 1.01a\,..\,a+b$ $V_2(x_2) = -B\,\frac{1}{\text{kN}}$ $M_2(x_2) = B\left(a+b-x_2\right)\frac{1}{\text{kN}\cdot\text{m}}$

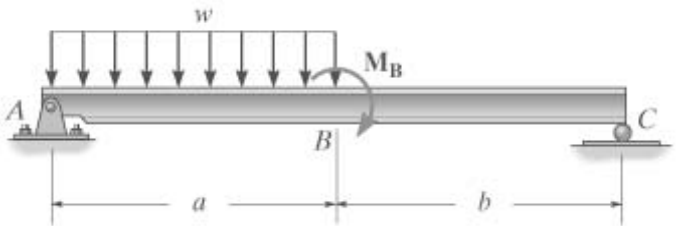




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Problem 7-77

Draw the shear and moment diagrams for the beam.



Given:

$w = 20\ \frac{\text{lb}}{\text{ft}}$ $M_B = 160\ \text{lb}\cdot\text{ft}$ $a = 20\ \text{ft}$ $b = 20\ \text{ft}$

Solution:

$-wa\left(\frac{a}{2}\right) - M_B + C_y(a+b) = 0$ $C_y = \left(w\frac{a^2}{2} + M_B\right)\left(\frac{1}{a+b}\right)$ $C_y = 104.00\,\text{lb}$

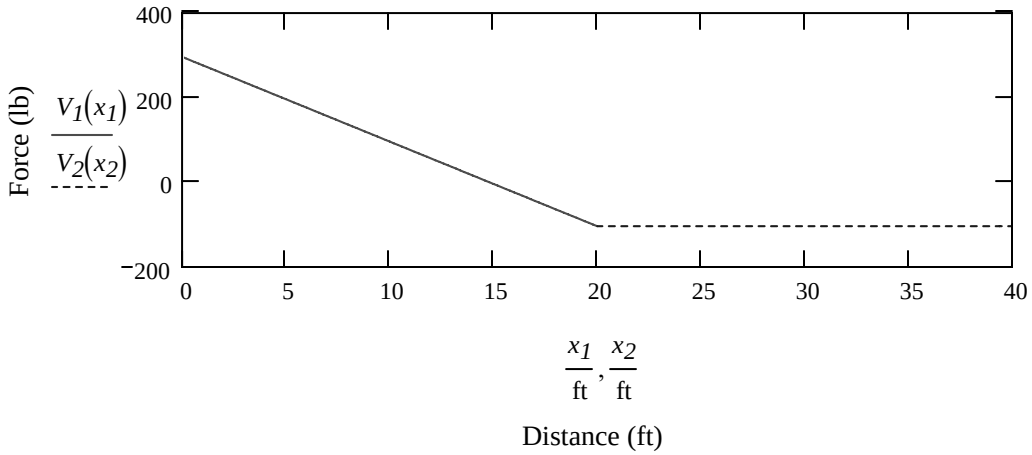
$A_y - wa + C_y = 0$ $A_y = wa - C_y$ $A_y = 296.00\,\text{lb}$

$x_1 = 0, 0.01a .. a$

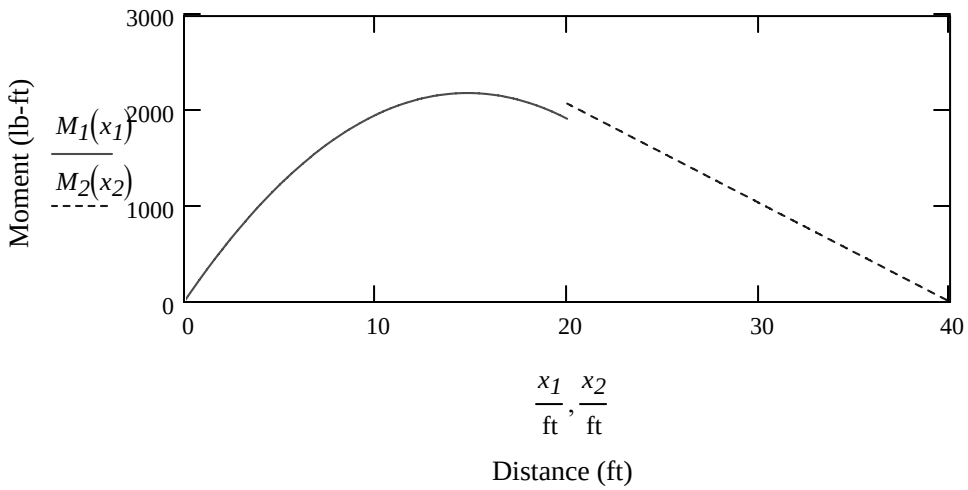
$V_1(x) = \left(A_y - wx\right)\frac{1}{\text{lb}}$ $M_1(x) = \left(A_y x - w\frac{x^2}{2}\right)\frac{1}{\text{lb}\cdot\text{ft}}$

$x_2 = a, 1.01a .. a + b$

$V_2(x) = -C_y\frac{1}{\text{lb}}$ $M_2(x) = C_y(a+b-x)\frac{1}{\text{lb}\cdot\text{ft}}$

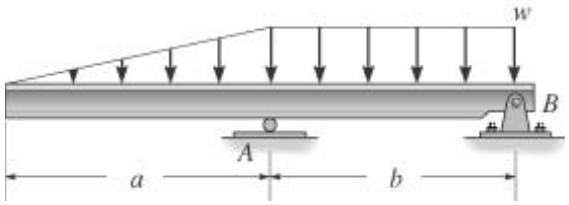


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Problem 7-78

The beam will fail when the maximum moment is M_{max} or the maximum shear is V_{max}. Determine the largest distributed load w the beam will support.



Units used: $\text{kip} = 10^3\ \text{lb}$

Given: $M_{max} = 30\ \text{kip}\cdot\text{ft}$ $V_{max} = 8\ \text{kip}$ $a = 6\ \text{ft}$ $b = 6\ \text{ft}$

Solution: Set $w = 1\ \frac{\text{kip}}{\text{ft}}$ and then scale the answer at the end

$$-A\,b + w\frac{a}{2}\left(\frac{a}{3} + b\right) + w\,b\frac{b}{2} = 0 \qquad A = \frac{w\frac{a}{2}\left(\frac{a}{3} + b\right) + w\frac{b^2}{2}}{b} \qquad A = 7.00\,\text{kip}$$

$$A + B - w\,b - w\frac{a}{2} = 0 \qquad B = w\left(b + \frac{a}{2}\right) - A \qquad B = 2.00\,\text{kip}$$

Shear limit - check critical points to the left and right of A and at B

$$V_{big} = \max\left(\left|B\right|, \left|w\frac{a}{2}\right|, \left|w\frac{a}{2} - A\right|\right) \qquad V_{big} = 4.00\,\text{kip}$$

$$w_{shear} = \left(\frac{V_{max}}{V_{big}}\right)w \qquad w_{shear} = 2.00\,\frac{\text{kip}}{\text{ft}}$$

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Moment limit - check critical points at A and betwen A and B

$$M_A = -w\left(\frac{a}{2}\right)\left(\frac{a}{3}\right) \quad x = \frac{B}{w} \qquad M_{AB} = Bx - w\left(\frac{x^2}{2}\right) \qquad \begin{pmatrix} M_A \\ M_{AB} \end{pmatrix} = \begin{pmatrix} -6.00 \\ 2.00 \end{pmatrix} \text{kip}\cdot\text{ft}$$

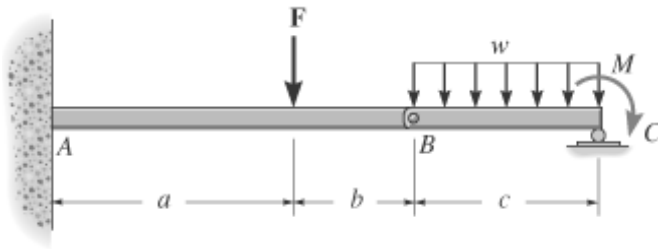
$$M_{big} = \max\left(\left|M_A\right|, \left|M_{AB}\right|\right) \qquad M_{big} = 6.00\,\text{kip}\cdot\text{ft}$$

$$w_{moment} = \left(\frac{M_{max}}{M_{big}}\right)w \qquad w_{moment} = 5.00\,\frac{\text{kip}}{\text{ft}}$$

Choose the critical case $\quad w_{ans} = \min\left(w_{shear}, w_{moment}\right) \qquad w_{ans} = 2.00\,\frac{\text{kip}}{\text{ft}}$

Problem 7-79

The beam consists of two segments pin connected at B. Draw the shear and moment diagrams for the beam.



Given:

$$F = 700\ \text{lb} \qquad w = 150\,\frac{\text{lb}}{\text{ft}} \qquad M = 800\ \text{lb}\,\text{ft} \qquad a = 8\ \text{ft} \qquad b = 4\ \text{ft} \qquad c = 6\ \text{ft}$$

Solution:

$$-wc\left(\frac{c}{2}\right) - M + Cc = 0 \qquad C = \frac{w\left(\frac{c^2}{2}\right) + M}{c} \qquad C = 583.33\,\text{lb}$$

$$B + C - wc = 0 \qquad B = wc - C \qquad B = 316.67\,\text{lb}$$

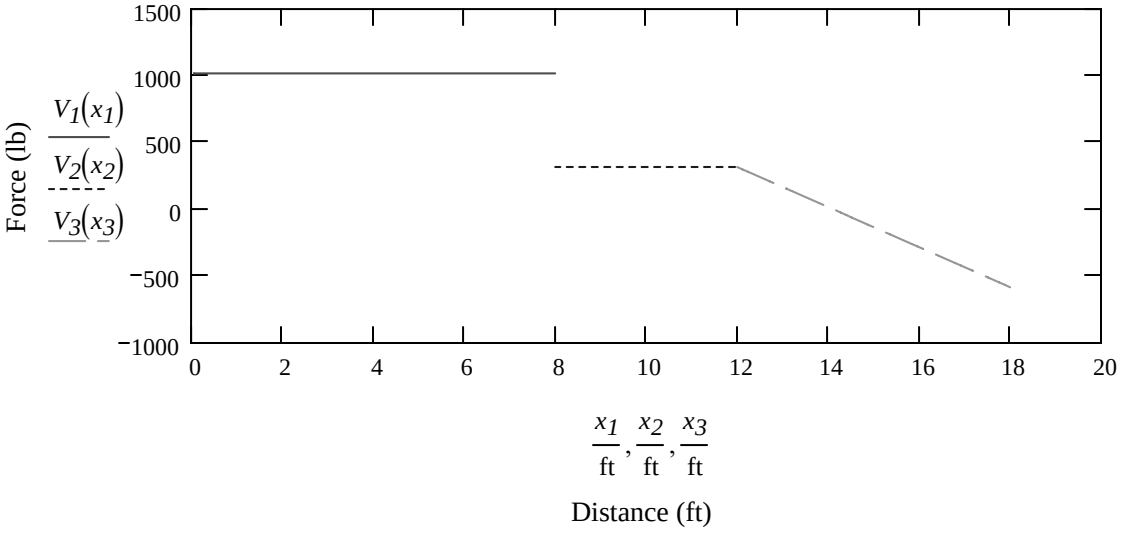
$$x_1 = 0, 0.01a .. a \qquad V_1(x_1) = (B + F)\frac{1}{\text{lb}} \qquad M_1(x_1) = \left[-F(a - x_1) - B(a + b - x_1)\right]\frac{1}{\text{lb}\cdot\text{ft}}$$

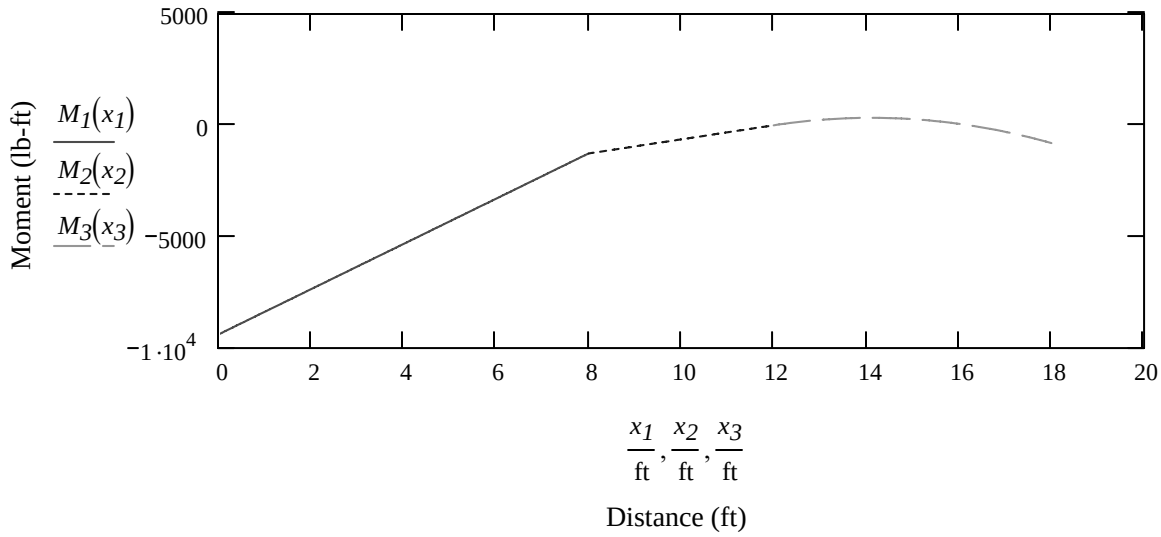
$$x_2 = a, 1.01a .. a + b \qquad V_2(x_2) = B\frac{1}{\text{lb}} \qquad M_2(x_2) = -B(a + b - x_2)\frac{1}{\text{lb}\cdot\text{ft}}$$

$$x_3 = a + b, 1.01(a + b) .. a + b + c \qquad V_3(x_3) = \left[-C + w(a + b + c - x_3)\right]\frac{1}{\text{lb}}$$

$$M_3(x_3) = \left[C(a + b + c - x_3) - w\frac{(a + b + c - x_3)^2}{2} - M\right]\frac{1}{\text{lb}\cdot\text{ft}}$$

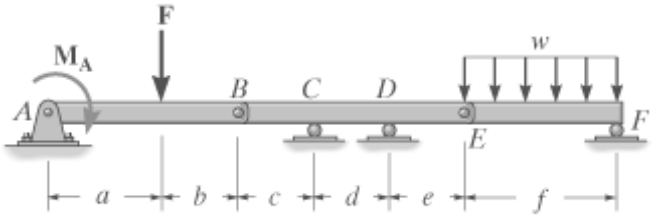
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Problem 7-80

The beam consists of three segments pin connected at *B* and *E*. Draw the shear and moment diagrams for the beam.



Units Used: $\text{kN} = 10^3\ \text{N}$

Given: $M_A = 8\ \text{kN}\cdot\text{m}$ $F = 15\ \text{kN}$ $w = 3\ \frac{\text{kN}}{\text{m}}$ $a = 3\ \text{m}$ $b = 2\ \text{m}$

$c = 2\ \text{m}$ $d = 2\ \text{m}$ $e = 2\ \text{m}$ $f = 4\ \text{m}$

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Guesses $A_y = 1\ \text{N}$ $B_y = 1\ \text{N}$ $C_y = 1\ \text{N}$

$D_y = 1\ \text{N}$ $E_y = 1\ \text{N}$ $F_y = 1\ \text{N}$

Given

$$A_y + C_y + D_y + F_y - F - w\,f = 0 \qquad F\,b - M_A - A_y(a+b) = 0$$

$$-w\,f\left(\frac{f}{2}\right) + F_y\,f = 0 \qquad M_A + F\,a + B_y(a+b) = 0$$

$$B_y + C_y + D_y + E_y = 0 \qquad -B_y\,c + D_y\,d + E_y(d+e) = 0$$

$$\begin{pmatrix} A_y \\ B_y \\ C_y \\ D_y \\ E_y \\ F_y \end{pmatrix} = \text{Find}\left(A_y, B_y, C_y, D_y, E_y, F_y\right) \qquad \begin{pmatrix} A_y \\ B_y \\ C_y \\ D_y \\ E_y \\ F_y \end{pmatrix} = \begin{pmatrix} 4.40 \\ -10.60 \\ 15.20 \\ 1.40 \\ -6.00 \\ 6.00 \end{pmatrix} \text{kN}$$

$x_1 = 0, 0.01a .. a$

$$V_1(x) = A_y \frac{1}{\text{kN}} \qquad M_1(x) = \left(A_y x + M_A\right)\frac{1}{\text{kN}\cdot\text{m}}$$

$x_2 = a, 1.01a .. a + b$

$$V_2(x) = \left(A_y - F\right)\frac{1}{\text{kN}} \qquad M_2(x) = \left[A_y x + M_A - F(x-a)\right]\frac{1}{\text{kN}\cdot\text{m}}$$

$x_3 = a + b, 1.01(a+b) .. a + b + c$

$$V_3(x) = B_y \frac{1}{\text{kN}} \qquad M_3(x) = B_y(x - a - b)\,\frac{1}{\text{kN}\cdot\text{m}}$$

$x_4 = a + b + c, 1.01(a+b+c) .. a + b + c + d$

$$V_4(x) = \left(B_y + C_y\right)\frac{1}{\text{kN}} \qquad M_4(x) = \left[B_y(x - a - b) + C_y(x - a - b - c)\right]\frac{1}{\text{kN}\cdot\text{m}}$$

$x_5 = a + b + c + d, 1.01(a+b+c+d) .. a + b + c + d + e$

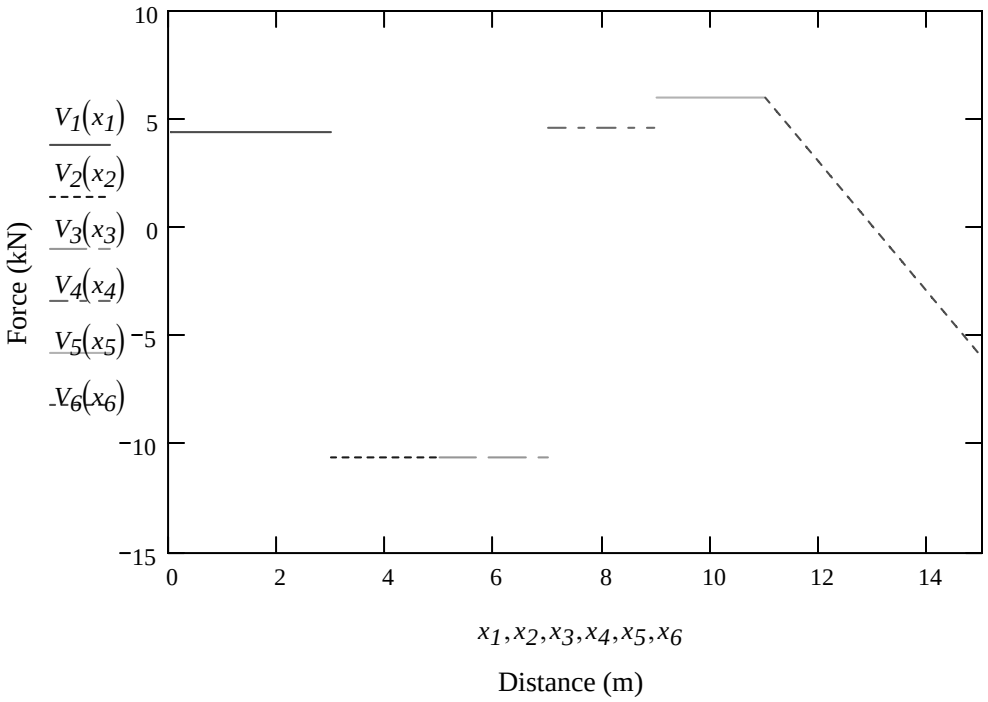
$$V_5(x) = -E_y \frac{1}{\text{kN}} \qquad M_5(x) = E_y(a + b + c + d + e - x)\,\frac{1}{\text{kN}\cdot\text{m}}$$

$x_6 = a + b + c + d + e, 1.01(a+b+c+d+e) .. a + b + c + d + e + f$

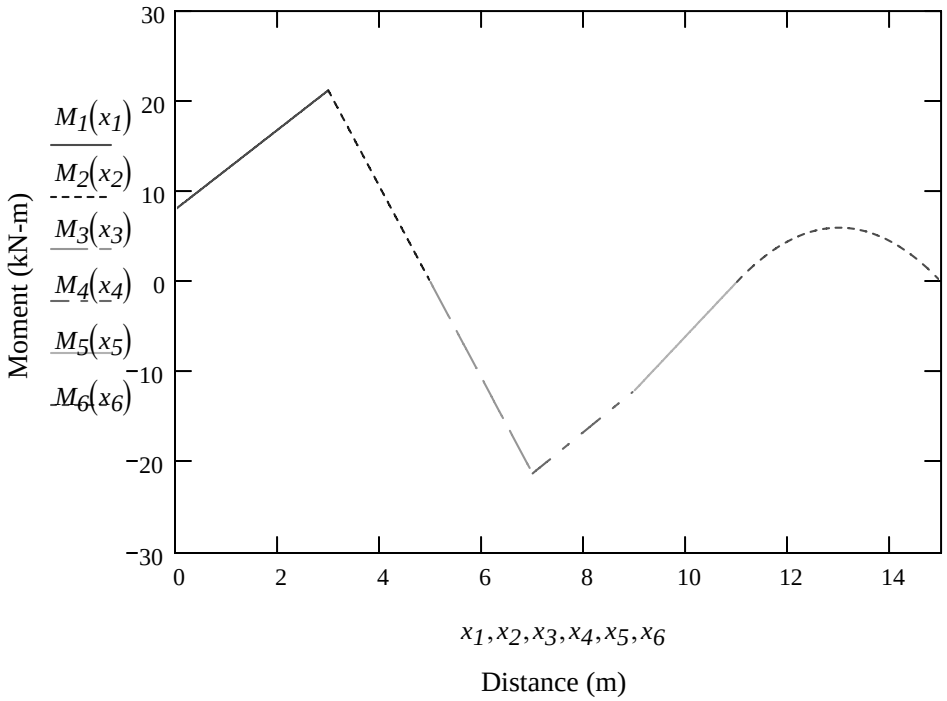
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$$V_6(x) = \left[-F_y + w(a + b + c + d + e + f - x)\right]\frac{1}{\text{kN}}$$

$$M_6(x) = \left[F_y(a + b + c + d + e + f - x) - w\frac{(a + b + c + d + e + f - x)^2}{2}\right]\frac{1}{\text{kN}\cdot\text{m}}$$

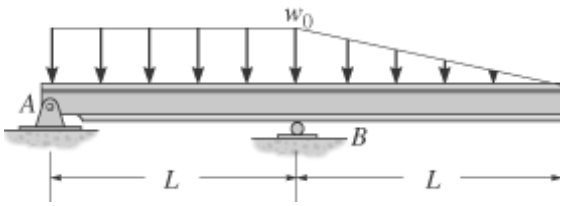


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Problem 7-81

Draw the shear and moment diagrams for the beam.



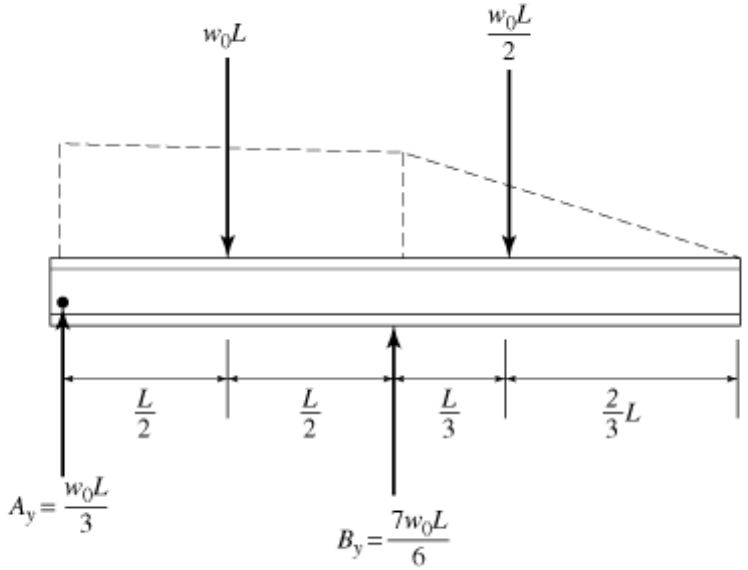
Solutions:

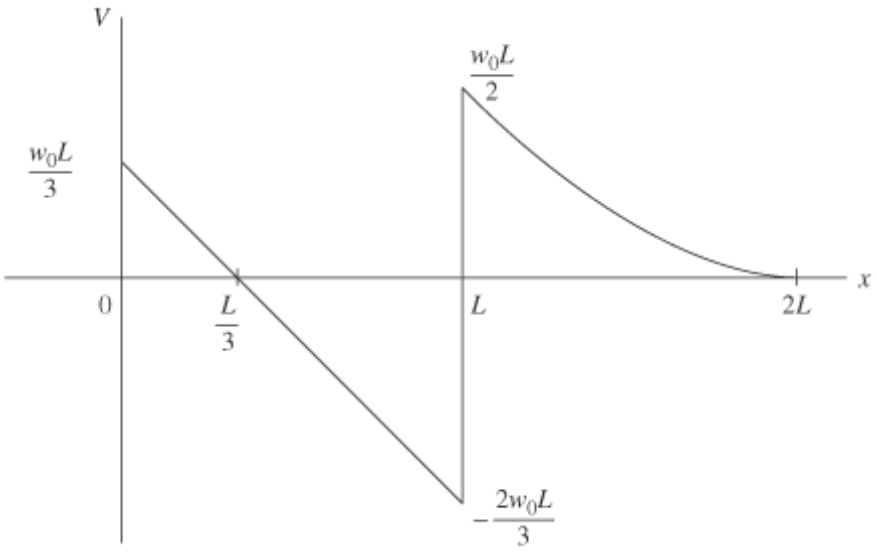
Support Reactions:

$$\curvearrowleft \quad \Sigma M_x = 0; \qquad B_y L - w_0 L\left(\frac{L}{2}\right) - \frac{w_0 L}{2}\left(\frac{4L}{3}\right) = 0 \qquad B_y = \frac{7w_0 L}{6}$$

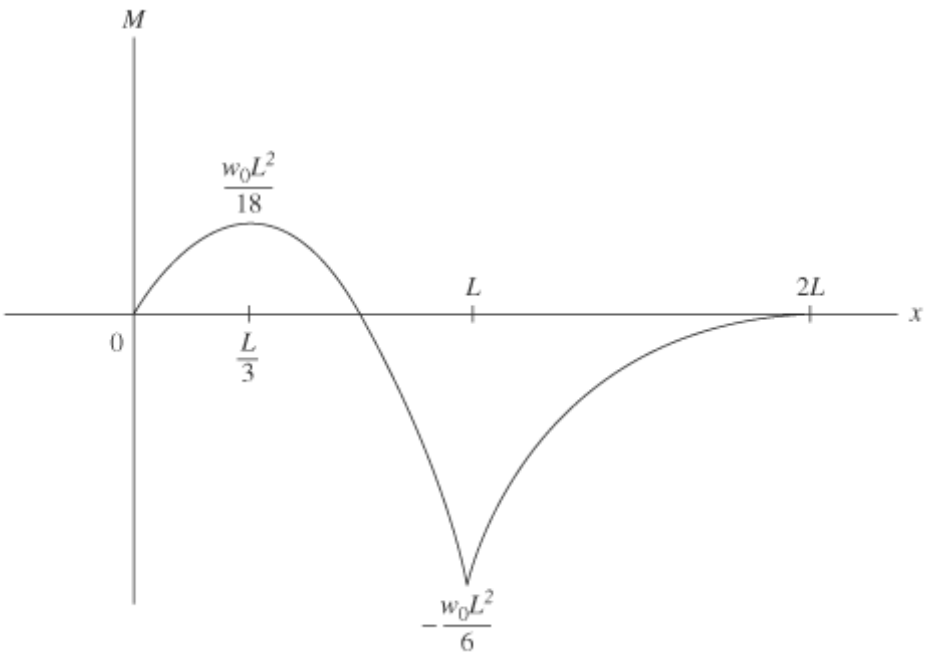
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$$+\uparrow \Sigma F_y = 0; \qquad A_y + \left(\frac{7w_0 L}{6}\right) - w_0 L - \left(\frac{w_0 L}{2}\right) = 0 \qquad A_y = \frac{w_0 L}{3}$$



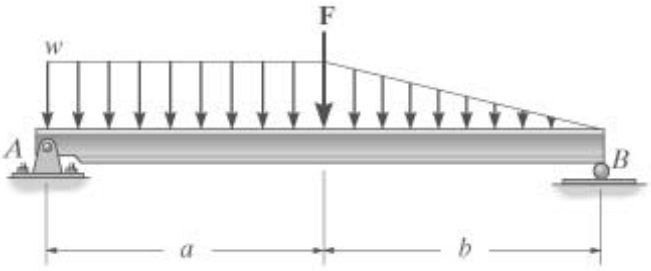


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Problem 7-82

Draw the shear and moment diagrams for the beam.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$F = 2000 \text{ lb}$ $a = 9 \text{ ft}$

$w = 500 \frac{\text{lb}}{\text{ft}}$ $b = 9 \text{ ft}$

Solution:

$\frac{1}{2}wb\left(\frac{2b}{3}\right) + Fb + wa\left(b + \frac{a}{2}\right) - A_y(a + b) = 0$ $A_y = \frac{\left(\frac{wb^2}{3}\right) + Fb + wa\left(b + \frac{a}{2}\right)}{a + b}$

$A_y + B_y - F - wa - \frac{1}{2}wb = 0$ $B_y = wa + \frac{1}{2}wb - A_y + F$

$A_y = 5.13 \text{ kip}$ $B_y = 3.63 \text{ kip}$

$x_1 = 0, 0.01a .. a$

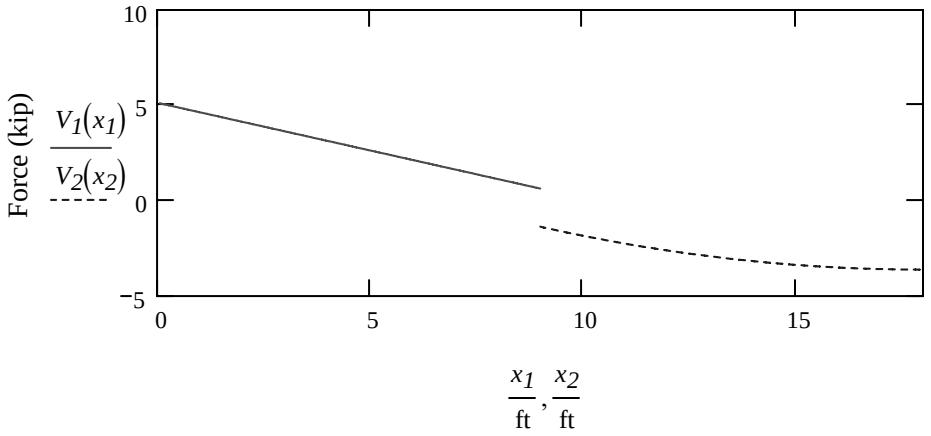
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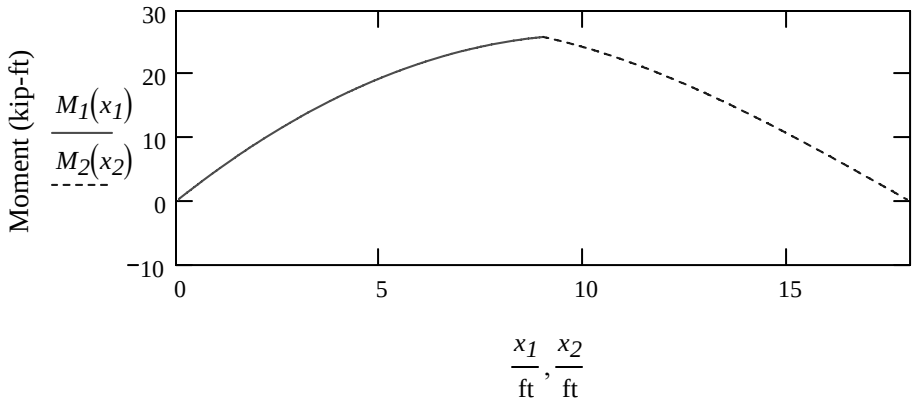
$$V_1(x) = \left(A_y - wx\right)\frac{1}{\text{kip}} \qquad M_1(x) = \left[A_y x - wx\left(\frac{x}{2}\right)\right]\frac{1}{\text{kip}\cdot\text{ft}}$$

$$x_2 = a, 1.01a .. a + b$$

$$V_2(x) = \left[-B_y + \frac{1}{2}w\left(\frac{a+b-x}{b}\right)(a+b-x)\right]\frac{1}{\text{kip}}$$

$$M_2(x) = \left[B_y(a+b-x) - \frac{1}{2}w\left(\frac{a+b-x}{b}\right)(a+b-x)\left(\frac{a+b-x}{3}\right)\right]\frac{1}{\text{kip}\cdot\text{ft}}$$

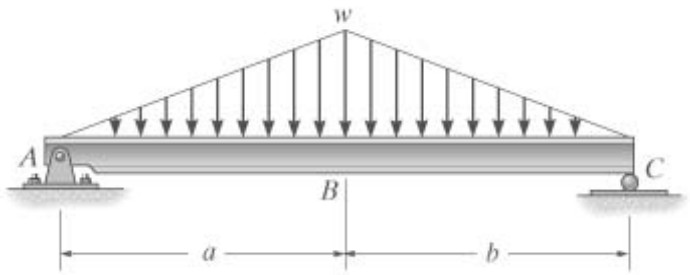




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Problem 7-83

Draw the shear and moment diagrams for the beam.



Units Used:

$$\text{kN} = 10^3\ \text{N}$$

Given:

$$w = 3\ \frac{\text{kN}}{\text{m}} \qquad a = 3\ \text{m} \qquad b = 3\ \text{m}$$

Solution:

$$-A_y(a+b) + \left(\frac{wb}{2}\right)\left(\frac{2b}{3}\right) + \left(\frac{wa}{2}\right)\left(b + \frac{a}{3}\right) = 0 \qquad A_y = \frac{\frac{wb^2}{3} + \left(\frac{wa}{2}\right)\left(b + \frac{a}{3}\right)}{a+b}$$

$$A_y + B_y - \frac{w}{2}(a+b) = 0 \qquad B_y = \frac{w}{2}(a+b) - A_y$$

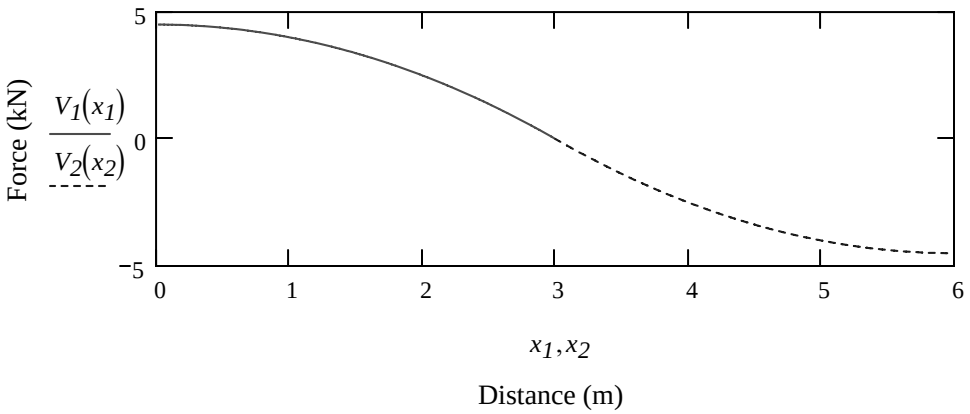
$$x_1 = 0, 0.01a .. a$$

$$V_1(x) = \left[A_y - \frac{1}{2}w\left(\frac{x}{a}\right)x\right]\frac{1}{\text{kN}} \qquad M_1(x) = \left[A_y x - \frac{1}{2}w\left(\frac{x}{a}\right)x\left(\frac{x}{3}\right)\right]\frac{1}{\text{kN}\cdot\text{m}}$$

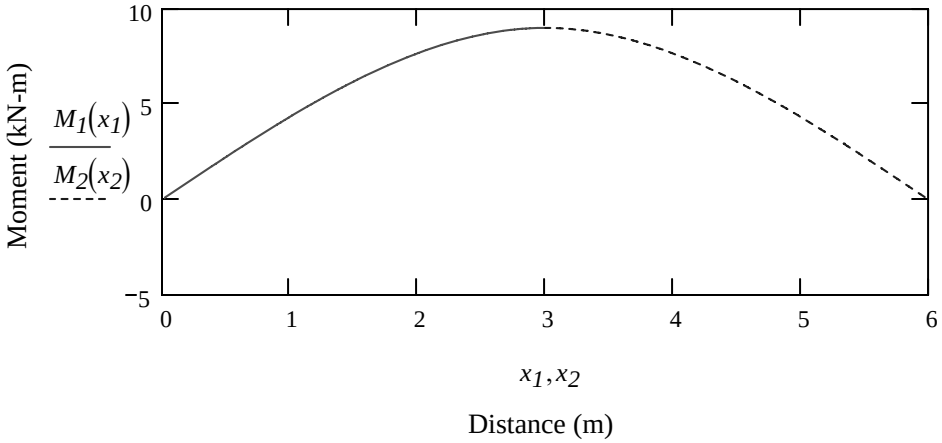
$$x_2 = a, 1.01a .. a + b$$

$$V_2(x) = \left[-B_y + \frac{1}{2}w\left(\frac{a+b-x}{b}\right)(a+b-x)\right]\frac{1}{\text{kN}}$$

$$M_2(x) = \left[B_y(a+b-x) - \frac{1}{2}w\left(\frac{a+b-x}{b}\right)(a+b-x)\left(\frac{a+b-x}{3}\right)\right]\frac{1}{\text{kN}\cdot\text{m}}$$

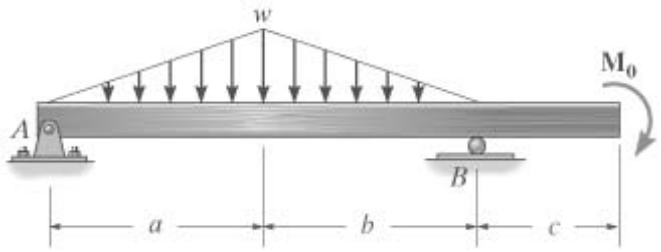


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Problem 7-84

Draw the shear and moment diagrams for the beam.



Units Used:

$\text{kip} = 10^3\ \text{lb}$

Given:

$w = 100\ \frac{\text{lb}}{\text{ft}} \qquad M_0 = 9\ \text{kip}\cdot\text{ft} \qquad a = 6\ \text{ft} \qquad b = 6\ \text{ft} \qquad c = 4\ \text{ft}$

Guesses: $A_y = 1\ \text{lb} \qquad B_y = 1\ \text{lb}$

Given $\qquad -M_0 + B_y(a+b) - \frac{1}{2}wa\left(\frac{2a}{3}\right) - \frac{1}{2}wb\left(a+\frac{b}{3}\right) = 0$

$$A_y + B_y - \frac{1}{2}w(a+b) = 0$$

$\begin{pmatrix} A_y \\ B_y \end{pmatrix} = \text{Find}(A_y, B_y) \qquad \begin{pmatrix} A_y \\ B_y \end{pmatrix} = \begin{pmatrix} -0.45 \\ 1.05 \end{pmatrix} \text{kip}$

$x_1 = 0, 0.01a .. a$

$V_1(x) = \left[A_y - \frac{1}{2}w\left(\frac{x}{a}\right)x\right]\frac{1}{\text{lb}} \qquad M_1(x) = \left[A_y x - \frac{1}{2}w\left(\frac{x}{a}\right)x\left(\frac{x}{3}\right)\right]\frac{1}{\text{kip}\cdot\text{ft}}$

$x_2 = a, 1.01a .. a + b$

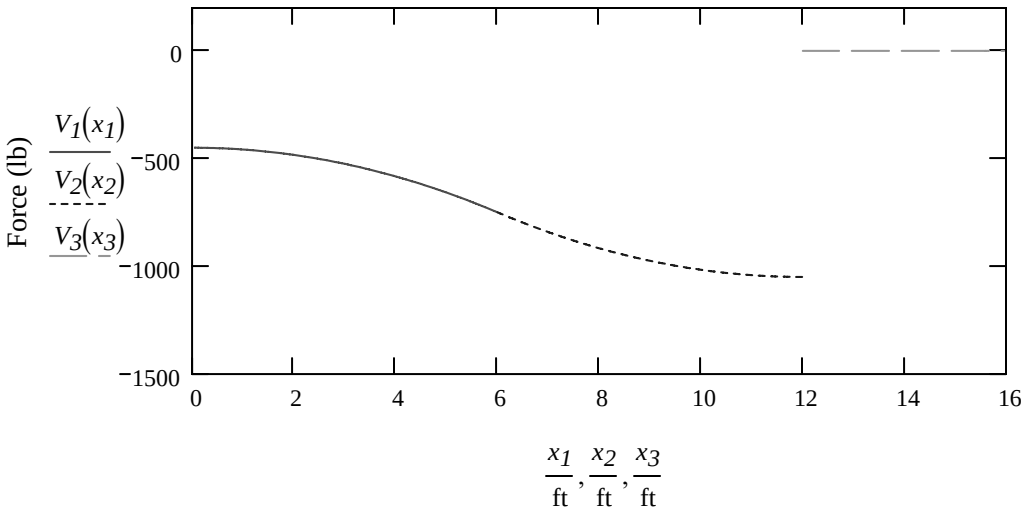
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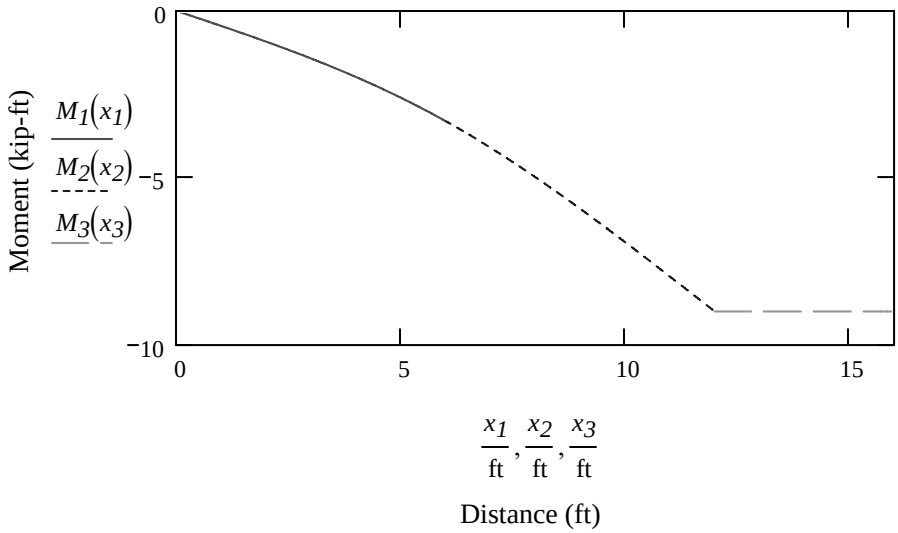
$$V_2(x) = \left[-B_y + \frac{1}{2}w\left(\frac{a+b-x}{b}\right)(a+b-x)\right]\frac{1}{\text{lb}}$$

$$M_2(x) = \left[B_y(a+b-x) - M_0 - \frac{1}{2}w\left(\frac{a+b-x}{b}\right)(a+b-x)\left(\frac{a+b-x}{3}\right)\right]\frac{1}{\text{kip}\cdot\text{ft}}$$

$$x_3 = a+b, 1.01(a+b) .. a+b+c$$

$$V_3(x) = 0 \qquad M_3(x) = -M_0\frac{1}{\text{kip}\cdot\text{ft}}$$

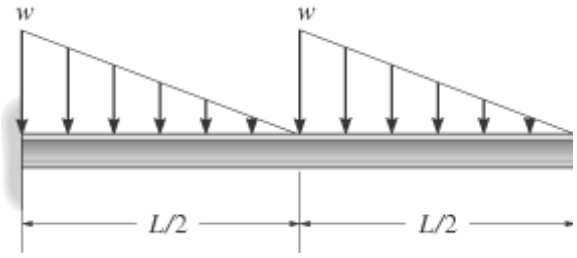




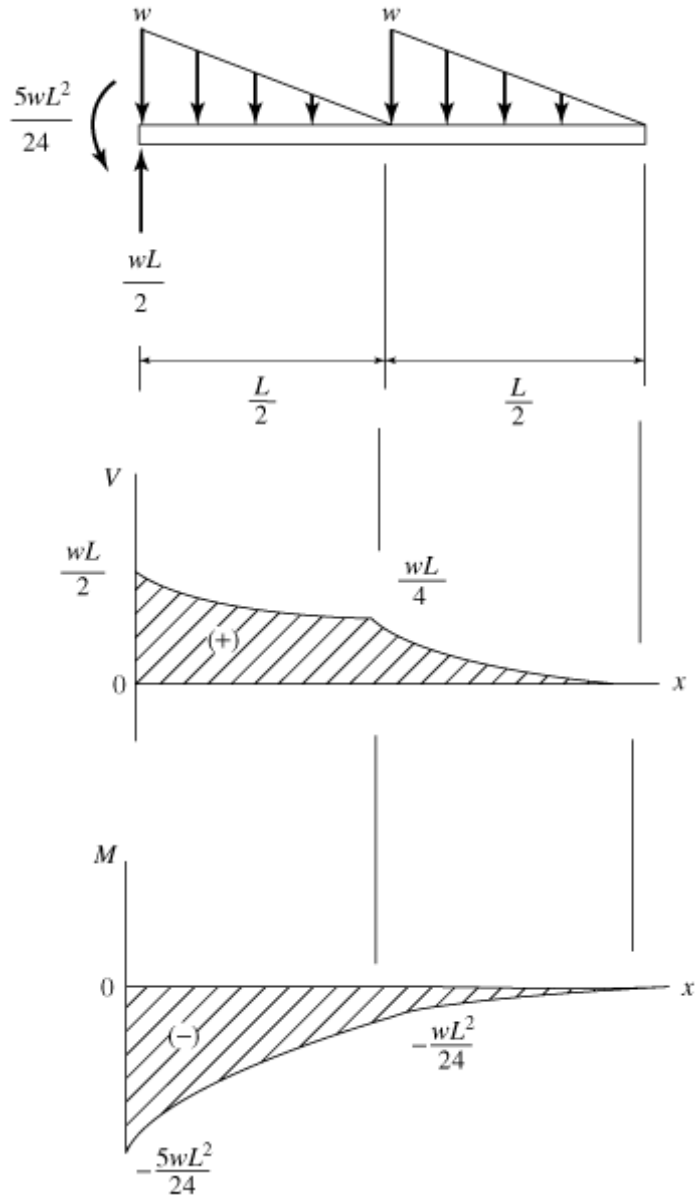
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Problem 7-85

Draw the shear and moment diagrams for the beam.



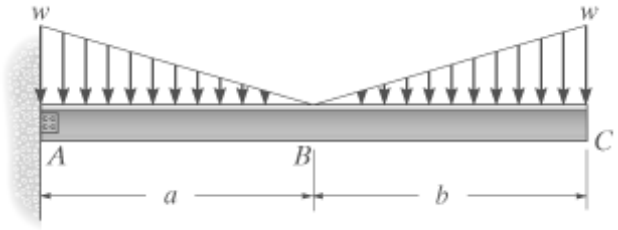
Solution:



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Problem 7-86

Draw the shear and moment diagrams for the beam.



Units Used:

$$\text{kN} = 10^3 \text{ N}$$

Given: $w = 2 \dfrac{\text{kN}}{\text{m}} \qquad a = 3 \text{ m} \qquad b = 3 \text{ m}$

Solution:

$x_1 = 0, 0.01a .. a$

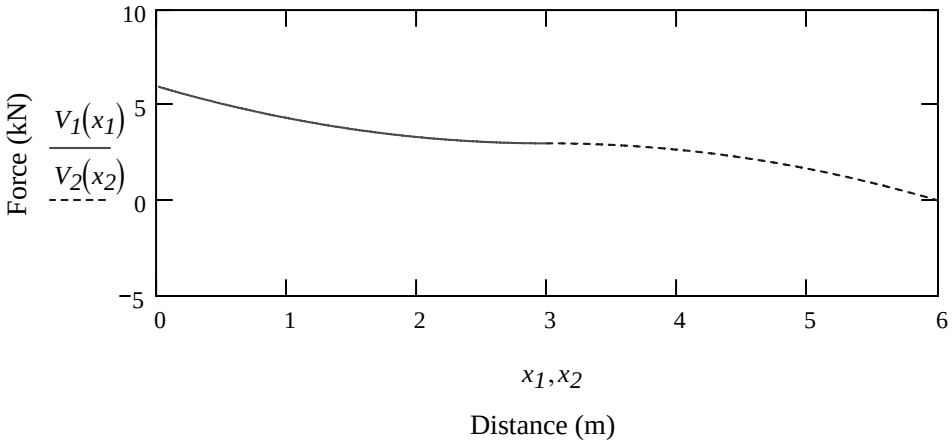
$$V_1(x) = \left[\frac{1}{2}wb + \frac{1}{2}w\left(\frac{a-x}{a}\right)(a-x)\right]\frac{1}{\text{kN}}$$

$$M_1(x) = \left[\frac{-1}{2}wb\left(\frac{2b}{3} + a - x\right) - \frac{1}{2}w\left(\frac{a-x}{a}\right)(a-x)\left(\frac{a-x}{3}\right)\right]\frac{1}{\text{kN}\cdot\text{m}}$$

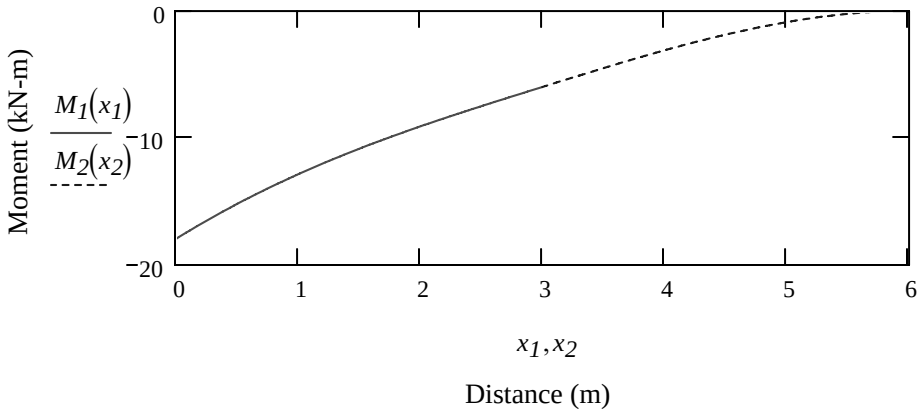
$x_2 = a, 1.01a .. a + b$

$$V_2(x) = \left[\frac{1}{2}wb - \frac{1}{2}w\left(\frac{x-a}{a}\right)(x-a)\right]\frac{1}{\text{kN}}$$

$$M_2(x) = \left[\frac{-1}{2}wb\left(a + \frac{2b}{3} - x\right) - \frac{1}{2}w\left(\frac{x-a}{b}\right)(x-a)\left(\frac{x-a}{3}\right)\right]\frac{1}{\text{kN}\cdot\text{m}}$$



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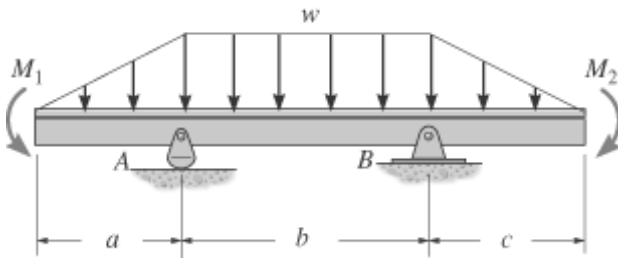


Problem 7-87

Draw the shear and moment diagrams for the beam.

Units Used:

$\text{kip} = 10^3 \text{ lb}$



Given:

$w = 5 \frac{\text{kip}}{\text{ft}}$ $M_1 = 15 \text{ kip·ft}$ $M_2 = 15 \text{ kip·ft}$ $a = 6 \text{ ft}$ $b = 10 \text{ ft}$ $c = 6 \text{ ft}$

Solution:

$$M_1 - A\,b - M_2 + w\left(\frac{a}{2}\right)\left(b + \frac{a}{3}\right) + w\,b\left(\frac{b}{2}\right) - w\left(\frac{c}{2}\right)\left(\frac{c}{3}\right) = 0 \qquad A + B - w\,b - w\left(\frac{a+c}{2}\right) = 0$$

$$A = \frac{M_1 - M_2 + w\left(\frac{a}{2}\right)\left(b + \frac{a}{3}\right) + w\left(\frac{b^2}{2}\right) - w\left(\frac{c^2}{6}\right)}{b}$$

$$B = w\left(b + \frac{a+c}{2}\right) - A \qquad \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 40.00 \\ 40.00 \end{pmatrix} \text{kip}$$

$x_1 = 0, 0.01a .. a$ $V_1(x) = \left(-w\frac{x}{a}\right)\frac{x}{2}\frac{1}{\text{kip}}$

$$M_{1p}(x) = \left[\left(-w\frac{x}{a}\right)\frac{x}{2}\frac{x}{3} - M_1\right]\frac{1}{\text{kip·ft}}$$

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$x_2 = a, 1.01a .. a + b$

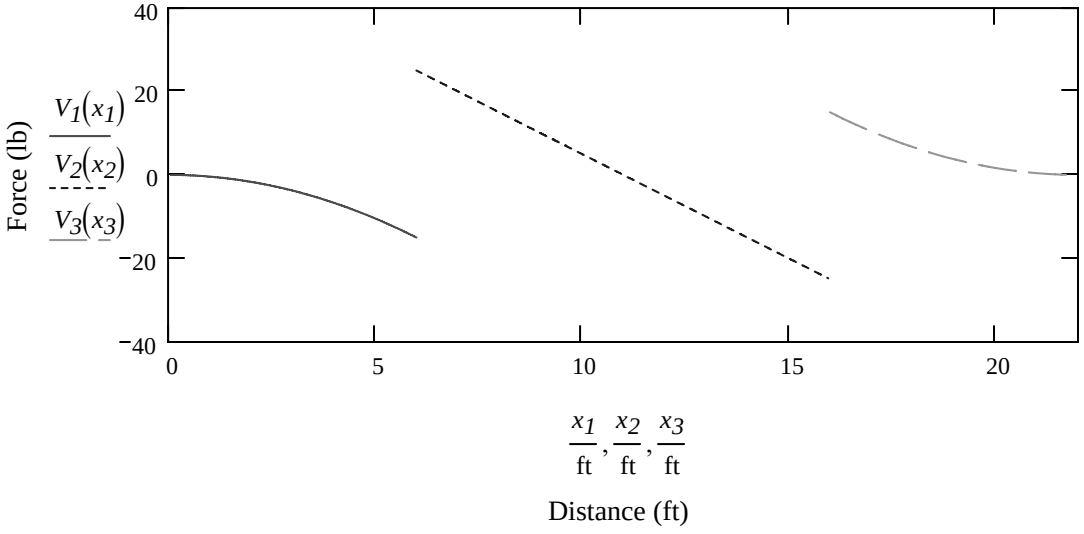
$$V_2(x) = \left[A - w\frac{a}{2} - w(x - a)\right]\frac{1}{\text{kip}}$$

$$M_{2p}(x) = \left[-M_1 - w\frac{a}{2}\left(x - \frac{2a}{3}\right) + A(x - a) - w(x - a)\left(\frac{x - a}{2}\right)\right]\frac{1}{\text{kip}\cdot\text{ft}}$$

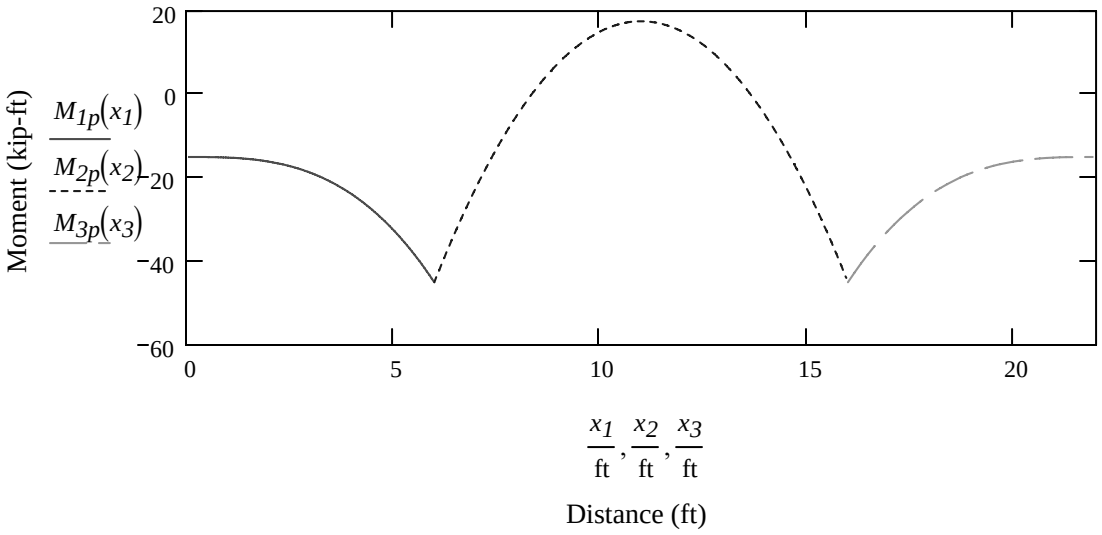
$x_3 = a + b, 1.01(a + b) .. a + b + c$

$$V_3(x) = w\left(\frac{a + b + c - x}{c}\right)\left(\frac{a + b + c - x}{2}\right)\frac{1}{\text{kip}}$$

$$M_{3p}(x) = \left[-w\left(\frac{a + b + c - x}{c}\right)\left(\frac{a + b + c - x}{2}\right)\left(\frac{a + b + c - x}{3}\right) - M_2\right]\frac{1}{\text{kip}\cdot\text{ft}}$$

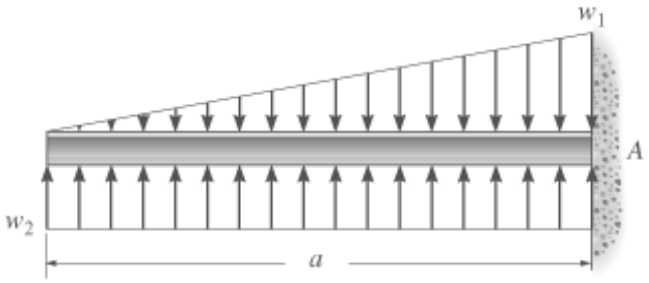


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Problem 7-88

Draw the shear and moment diagrams for the beam.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$w_1 = 2 \frac{\text{kip}}{\text{ft}}$ $w_2 = 1 \frac{\text{kip}}{\text{ft}}$ $a = 15 \text{ ft}$

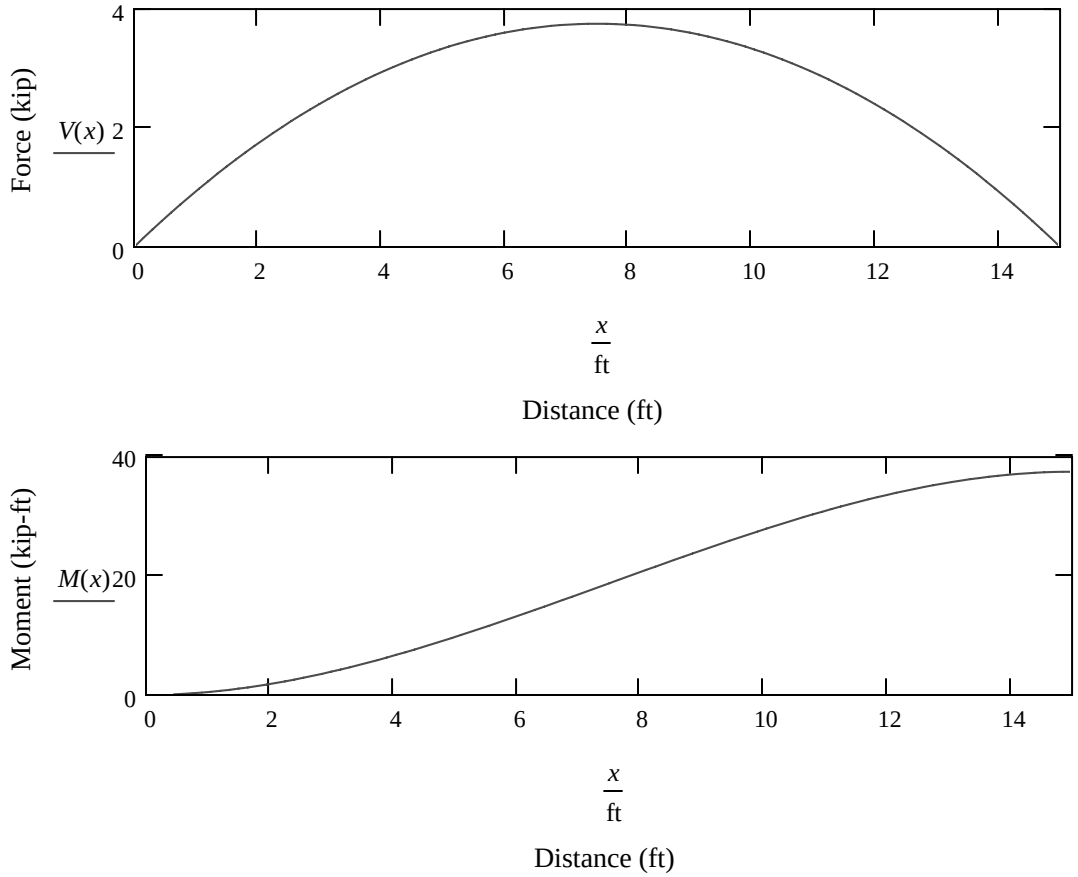
Solution:

$x = 0, 0.01a .. a$

$V(x) = \left[w_2 x - \left(w_1 \frac{x}{a} \right) \frac{x}{2} \right] \frac{1}{\text{kip}}$

$M(x) = \left[w_2 x \frac{x}{2} - \left(w_1 \frac{x}{a} \right) \frac{x}{2} \frac{x}{3} \right] \frac{1}{\text{kip} \cdot \text{ft}}$

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Problem 7-89

Determine the force P needed to hold the cable in the position shown, i.e., so segment BC remains horizontal. Also, compute the sag y_B and the maximum tension in the cable.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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Given:

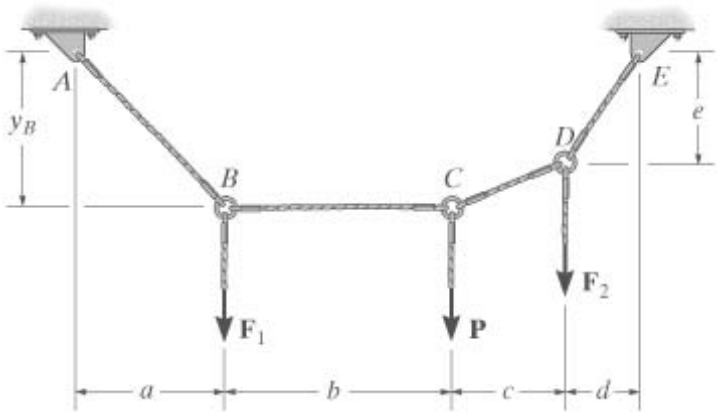
$a = 4$ m $\quad F_1 = 4$ kN

$b = 6$ m $\quad F_2 = 6$ kN

$c = 3$ m

$d = 2$ m

$e = 3$ m



Solution:

Initial guesses:

$y_B = 1$ m $\quad P = 1$ kN $\quad T_{AB} = 1$ kN $\quad T_{BC} = 1$ kN $\quad T_{CD} = 1$ kN $\quad T_{DE} = 1$ kN

Given

$$\left(\frac{-a}{\sqrt{a^2 + y_B^2}}\right) T_{AB} + T_{BC} = 0$$

$$\left(\frac{y_B}{\sqrt{a^2 + y_B^2}}\right) T_{AB} - F_1 = 0$$

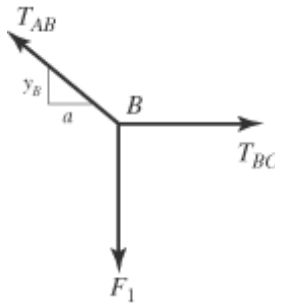
$$-T_{BC} + \left[\frac{c}{\sqrt{c^2 + (y_B - e)^2}}\right] T_{CD} = 0$$

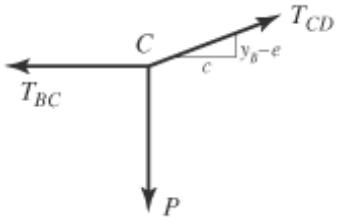
$$\left[\frac{y_B - e}{\sqrt{c^2 + (y_B - e)^2}}\right] T_{CD} - P = 0$$

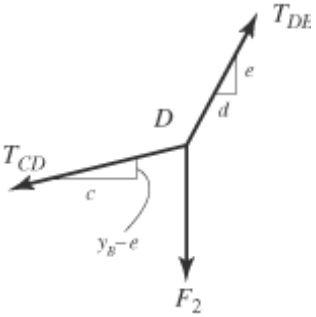
$$\left[\frac{-c}{\sqrt{c^2 + (y_B - e)^2}}\right] T_{CD} + \left(\frac{d}{\sqrt{d^2 + e^2}}\right) T_{DE} = 0$$

$$\left[\frac{-(y_B - e)}{\sqrt{c^2 + (y_B - e)^2}}\right] T_{CD} + \left(\frac{e}{\sqrt{e^2 + d^2}}\right) T_{DE} - F_2 = 0$$

$$\begin{pmatrix} y_B \\ P \\ T_{AB} \\ T_{BC} \\ T_{CD} \\ T_{DE} \end{pmatrix} = \text{Find}(y_B, P, T_{AB}, T_{BC}, T_{CD}, T_{DE})$$







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$T_{max} = \max\left(T_{AB}, T_{BC}, T_{CD}, T_{DE}\right)$ $y_B = 3.53$ m

$P = 800.00$ N

$T_{max} = 8.17$ kN

Problem 7-90

Cable *ABCD* supports the lamp of mass M_1 and the lamp of mass M_2. Determine the maximum tension in the cable and the sag of point *B*.

Given:

$M_1 = 10$ kg

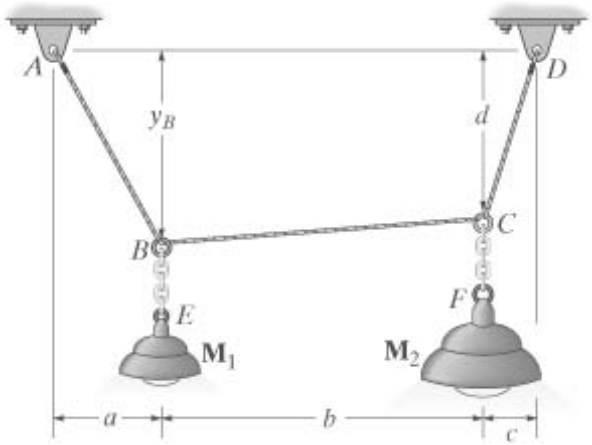
$M_2 = 15$ kg

$a = 1$ m

$b = 3$ m

$c = 0.5$ m

$d = 2$ m



Solution:

Guesses $y_B = 1$ m $T_{AB} = 1$ N $T_{BC} = 1$ N $T_{CD} = 1$ N

Given

$$\left(\frac{-a}{\sqrt{a^2 + y_B{}^2}}\right) T_{AB} + \left[\frac{b}{\sqrt{b^2 + \left(y_B - d\right)^2}}\right] T_{BC} = 0$$

$$\left(\frac{y_B}{\sqrt{a^2 + y_B{}^2}}\right) T_{AB} + \left[\frac{y_B - d}{\sqrt{b^2 + \left(y_B - d\right)^2}}\right] T_{BC} - M_1 g = 0$$

$$\left[\frac{-b}{\sqrt{b^2 + \left(y_B - d\right)^2}}\right] T_{BC} + \left(\frac{c}{\sqrt{c^2 + d^2}}\right) T_{CD} = 0$$

$$\left[\frac{-\left(y_B - d\right)}{\sqrt{b^2 + \left(y_B - d\right)^2}}\right] T_{BC} + \left(\frac{d}{\sqrt{c^2 + d^2}}\right) T_{CD} - M_2 g = 0$$

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$$\begin{pmatrix} y_B \\ T_{AB} \\ T_{BC} \\ T_{CD} \end{pmatrix} = \text{Find}\left(y_B, T_{AB}, T_{BC}, T_{CD}\right) \qquad \begin{pmatrix} T_{AB} \\ T_{BC} \\ T_{CD} \end{pmatrix} = \begin{pmatrix} 100.163 \\ 38.524 \\ 157.243 \end{pmatrix} \text{N}$$

$$T_{max} = \max\left(T_{AB}, T_{BC}, T_{CD}\right) \qquad T_{max} = 157.2 \text{ N} \qquad y_B = 2.43 \text{ m}$$

Problem 7-91

The cable supports the three loads shown. Determine the sags y_B and y_D of points B and D.

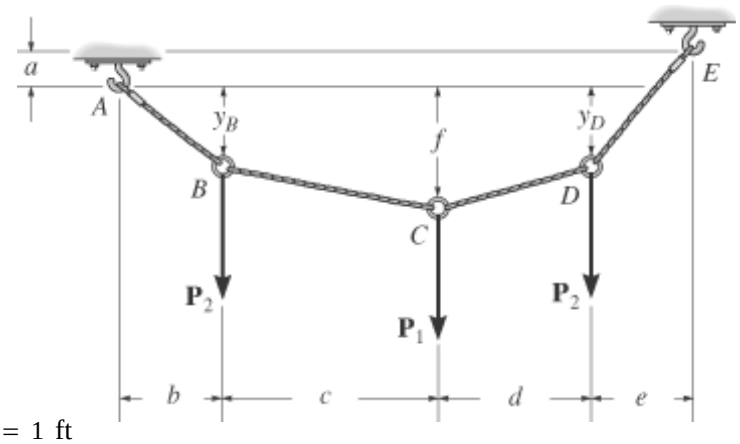
Given:

$a = 4$ ft $\quad e = 12$ ft

$b = 12$ ft $\quad f = 14$ ft

$c = 20$ ft $\quad P_1 = 400$ lb

$d = 15$ ft $\quad P_2 = 250$ lb



Solution:

Guesses $y_B = 1$ ft $\quad y_D = 1$ ft

$T_{AB} = 1$ lb $\quad T_{BC} = 1$ lb

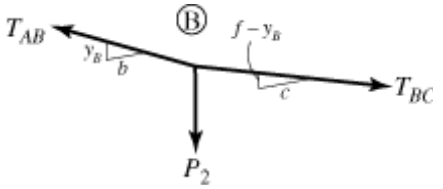
$T_{CD} = 1$ lb $\quad T_{DE} = 1$ lb

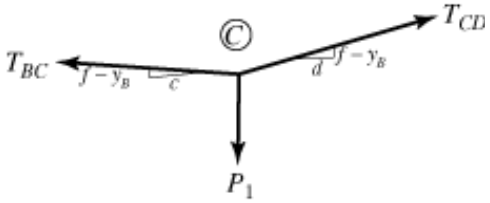
Given

$$\left(\frac{-b}{\sqrt{b^2 + y_B^2}}\right) T_{AB} + \left[\frac{c}{\sqrt{c^2 + \left(f - y_B\right)^2}}\right] T_{BC} = 0$$

$$\left(\frac{y_B}{\sqrt{b^2 + y_B^2}}\right) T_{AB} - \left[\frac{f - y_B}{\sqrt{c^2 + \left(f - y_B\right)^2}}\right] T_{BC} - P_2 = 0$$

$$\left[\frac{-c}{\sqrt{c^2 + \left(f - y_B\right)^2}}\right] T_{BC} + \left[\frac{d}{\sqrt{d^2 + \left(f - y_D\right)^2}}\right] T_{CD} = 0$$



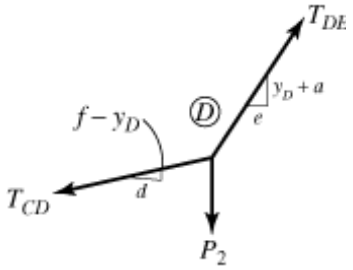


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$$\left[\frac{f-y_B}{\sqrt{c^2+(f-y_B)^2}}\right]T_{BC}+\left[\frac{f-y_D}{\sqrt{d^2+(f-y_D)^2}}\right]T_{CD}-P_1=0$$

$$\left[\frac{-d}{\sqrt{d^2+(f-y_D)^2}}\right]T_{CD}+\left[\frac{e}{\sqrt{e^2+(a+y_D)^2}}\right]T_{DE}=0$$

$$\left[\frac{-(f-y_D)}{\sqrt{d^2+(f-y_D)^2}}\right]T_{CD}+\left[\frac{a+y_D}{\sqrt{e^2+(a+y_D)^2}}\right]T_{DE}-P_2=0$$



$$\begin{pmatrix} T_{AB}\\ T_{BC}\\ T_{CD}\\ T_{DE}\\ y_B\\ y_D \end{pmatrix}=\text{Find}\left(T_{AB},T_{BC},T_{CD},T_{DE},y_B,y_D\right)$$

$$\begin{pmatrix} T_{AB}\\ T_{BC}\\ T_{CD}\\ T_{DE} \end{pmatrix}=\begin{pmatrix} 675.89\\ 566.90\\ 603.86\\ 744.44 \end{pmatrix}\text{ lb}$$

$$\begin{pmatrix} y_B\\ y_D \end{pmatrix}=\begin{pmatrix} 8.67\\ 7.04 \end{pmatrix}\text{ft}$$

Problem 7-92

The cable supports the three loads shown. Determine the magnitude of $\mathbf{P_1}$ and find the sag y_D for the given data.

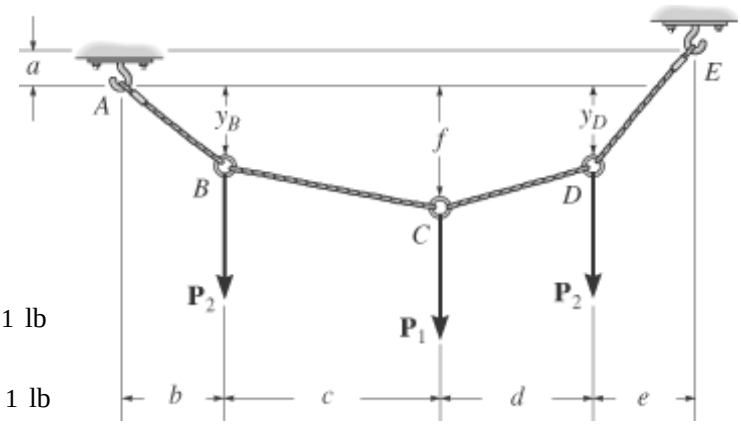
Given:

$P_2 = 300$ lb $\quad c = 20$ ft

$y_B = 8$ ft $\quad d = 15$ ft

$a = 4$ ft $\quad e = 12$ ft

$b = 12$ ft $\quad f = 14$ ft



Solution:

Guesses $\quad P_1 = 1$ lb $\quad T_{AB} = 1$ lb

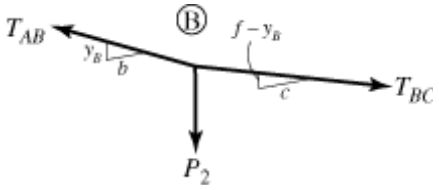
$\quad T_{BC} = 1$ lb $\quad T_{CD} = 1$ lb

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$T_{DE} = 1\ \text{lb} \qquad y_D = 1\ \text{ft}$

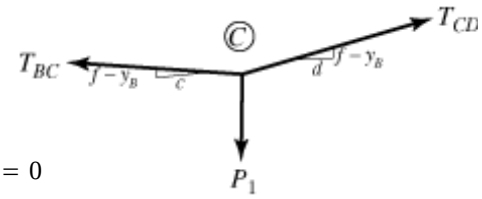
Given

$$\left(\frac{-b}{\sqrt{b^2 + y_B^2}}\right)T_{AB} + \left[\frac{c}{\sqrt{c^2 + (f - y_B)^2}}\right]T_{BC} = 0$$



$$\left(\frac{y_B}{\sqrt{b^2 + y_B^2}}\right)T_{AB} - \left[\frac{f - y_B}{\sqrt{c^2 + (f - y_B)^2}}\right]T_{BC} - P_2 = 0$$

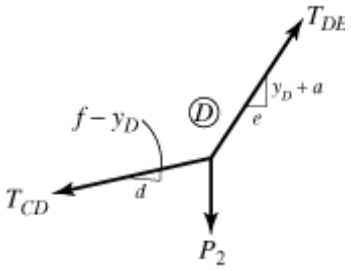
$$\left[\frac{-c}{\sqrt{c^2 + (f - y_B)^2}}\right]T_{BC} + \left[\frac{d}{\sqrt{d^2 + (f - y_D)^2}}\right]T_{CD} = 0$$



$$\left[\frac{f - y_B}{\sqrt{c^2 + (f - y_B)^2}}\right]T_{BC} + \left[\frac{f - y_D}{\sqrt{d^2 + (f - y_D)^2}}\right]T_{CD} - P_1 = 0$$

$$\left[\frac{-d}{\sqrt{d^2 + (f - y_D)^2}}\right]T_{CD} + \left[\frac{e}{\sqrt{e^2 + (a + y_D)^2}}\right]T_{DE} = 0$$

$$\left[\frac{-(f - y_D)}{\sqrt{d^2 + (f - y_D)^2}}\right]T_{CD} + \left[\frac{a + y_D}{\sqrt{e^2 + (a + y_D)^2}}\right]T_{DE} - P_2 = 0$$



$$\begin{pmatrix} T_{AB} \\ T_{BC} \\ T_{CD} \\ T_{DE} \\ P_1 \\ y_D \end{pmatrix} = \text{Find}(T_{AB}, T_{BC}, T_{CD}, T_{DE}, P_1, y_D) \qquad \begin{pmatrix} T_{AB} \\ T_{BC} \\ T_{CD} \\ T_{DE} \end{pmatrix} = \begin{pmatrix} 983.33 \\ 854.21 \\ 916.11 \\ 1084.68 \end{pmatrix} \text{lb}$$

$P_1 = 658\,\text{lb}$

$y_D = 6.44\,\text{ft}$

Problem 7-93

The cable supports the loading shown. Determine the distance x_B the force at point B acts from A.

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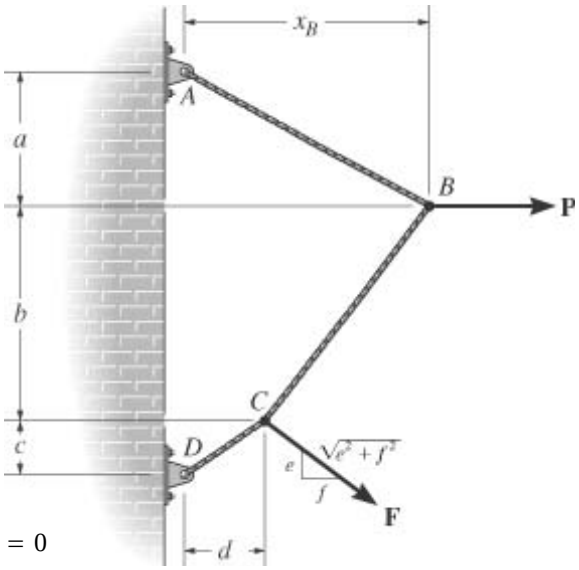
Given:

$P = 40$ lb $\quad c = 2$ ft

$F = 30$ lb $\quad d = 3$ ft

$a = 5$ ft $\quad e = 3$

$b = 8$ ft $\quad f = 4$



Solution:

The initial guesses:

$T_{AB} = 10$ lb $\quad T_{CD} = 30$ lb

$T_{BC} = 20$ lb $\quad x_B = 5$ ft

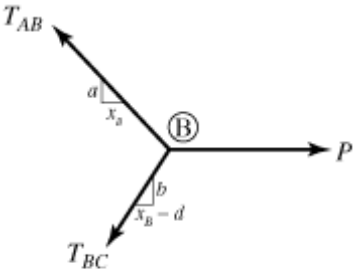
Given

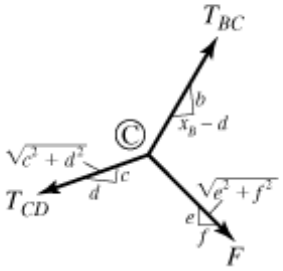
$$\left(\frac{-x_B}{\sqrt{x_B^2 + a^2}}\right)T_{AB} - \left[\frac{x_B - d}{\sqrt{(x_B - d)^2 + b^2}}\right]T_{BC} + P = 0$$

$$\left(\frac{a}{\sqrt{x_B^2 + a^2}}\right)T_{AB} - \left[\frac{b}{\sqrt{(x_B - d)^2 + b^2}}\right]T_{BC} = 0$$

$$\left[\frac{x_B - d}{\sqrt{(x_B - d)^2 + b^2}}\right]T_{BC} - \left(\frac{d}{\sqrt{c^2 + d^2}}\right)T_{CD} + \left(\frac{f}{\sqrt{e^2 + f^2}}\right)F = 0$$

$$\left[\frac{b}{\sqrt{(x_B - d)^2 + b^2}}\right]T_{BC} - \left(\frac{c}{\sqrt{c^2 + d^2}}\right)T_{CD} - \left(\frac{e}{\sqrt{e^2 + f^2}}\right)F = 0$$





$$\begin{pmatrix} T_{AB} \\ T_{CD} \\ T_{BC} \\ x_B \end{pmatrix} = \text{Find}(T_{AB}, T_{CD}, T_{BC}, x_B) \qquad \begin{pmatrix} T_{AB} \\ T_{CD} \\ T_{BC} \end{pmatrix} = \begin{pmatrix} 50.90 \\ 36.70 \\ 38.91 \end{pmatrix} \text{lb} \qquad x_B = 4.36\,\text{ft}$$

Problem 7-94

The cable supports the loading shown. Determine the magnitude of the horizontal force **P**.

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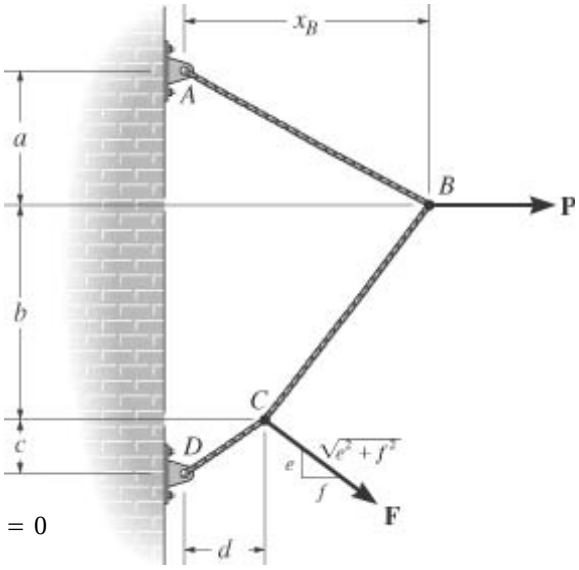
Given:

$F = 30$ lb $\quad c = 2$ ft

$x_B = 6$ ft $\quad d = 3$ ft

$a = 5$ ft $\quad e = 3$

$b = 8$ ft $\quad f = 4$



Solution:

The initial guesses:

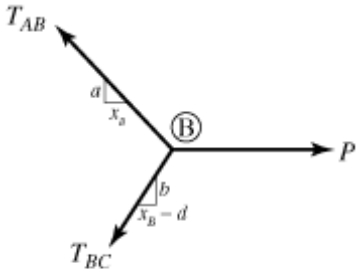
$T_{AB} = 10$ lb $\quad T_{CD} = 30$ lb

$T_{BC} = 20$ lb $\quad P = 10$ lb

Given

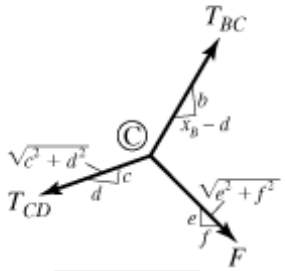
$$\left(\frac{-x_B}{\sqrt{x_B^2 + a^2}}\right)T_{AB} - \left[\frac{x_B - d}{\sqrt{(x_B - d)^2 + b^2}}\right]T_{BC} + P = 0$$

$$\left(\frac{a}{\sqrt{a^2 + x_B^2}}\right)T_{AB} - \left[\frac{b}{\sqrt{b^2 + (x_B - d)^2}}\right]T_{BC} = 0$$



$$\left(\frac{-d}{\sqrt{c^2 + d^2}}\right)T_{CD} + \left[\frac{x_B - d}{\sqrt{b^2 + (x_B - d)^2}}\right]T_{BC} + \left(\frac{f}{\sqrt{e^2 + f^2}}\right)F = 0$$

$$\left(\frac{-c}{\sqrt{c^2 + d^2}}\right)T_{CD} + \left[\frac{b}{\sqrt{b^2 + (x_B - d)^2}}\right]T_{BC} - \left(\frac{e}{\sqrt{e^2 + f^2}}\right)F = 0$$

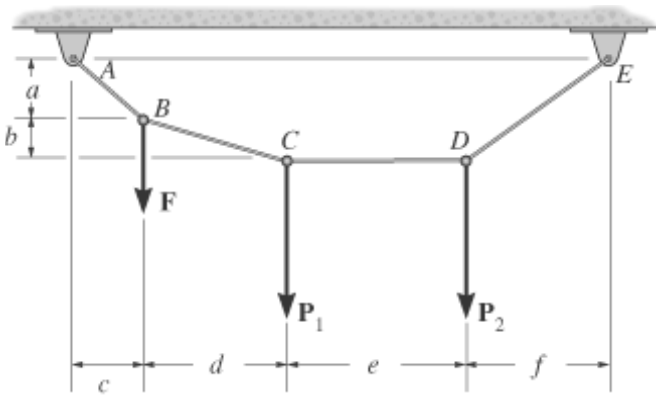


$$\begin{pmatrix} T_{AB} \\ T_{BC} \\ T_{CD} \\ P \end{pmatrix} = \text{Find}(T_{AB}, T_{BC}, T_{CD}, P) \qquad \begin{pmatrix} T_{AB} \\ T_{BC} \\ T_{CD} \end{pmatrix} = \begin{pmatrix} 70.81 \\ 48.42 \\ 49.28 \end{pmatrix} \text{lb} \qquad P = 71.40 \text{ lb}$$

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Problem 7-95

Determine the forces P_1 and P_2 needed to hold the cable in the position shown, i.e., so segment *CD* remains horizontal. Also, compute the maximum tension in the cable.



Given: $\text{kN} = 10^3\ \text{N}$

$F = 5\ \text{kN}$ $d = 4\ \text{m}$

$a = 1.5\ \text{m}$ $e = 5\ \text{m}$

$b = 1\ \text{m}$ $f = 4\ \text{m}$

$c = 2\ \text{m}$

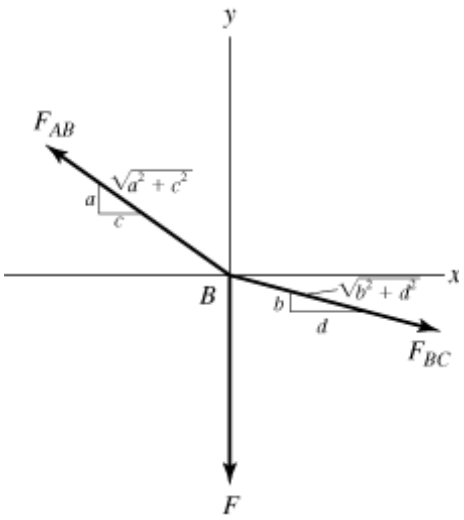
Solution:

Guesses

$F_{AB} = 1\ \text{kN}$ $F_{BC} = 1\ \text{kN}$

$F_{CD} = 1\ \text{kN}$ $F_{DE} = 1\ \text{kN}$

$P_1 = 1\ \text{kN}$ $P_2 = 1\ \text{kN}$



Given

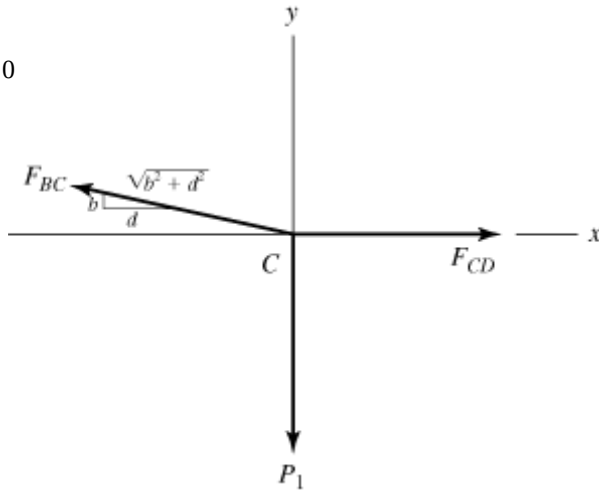
$$\left(\frac{-c}{\sqrt{a^2+c^2}}\right)F_{AB}+\left(\frac{d}{\sqrt{b^2+d^2}}\right)F_{BC}=0$$

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$$\left(\frac{a}{\sqrt{a^2+c^2}}\right)F_{AB}-\left(\frac{b}{\sqrt{b^2+d^2}}\right)F_{BC}-F=0$$

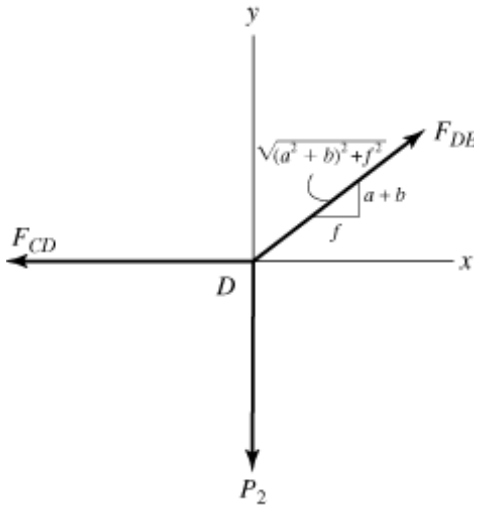
$$\left(\frac{-d}{\sqrt{b^2+d^2}}\right)F_{BC}+F_{CD}=0$$

$$\left(\frac{b}{\sqrt{b^2+d^2}}\right)F_{BC}-P_1=0$$



$$-F_{CD}+\left[\frac{f}{\sqrt{f^2+(a+b)^2}}\right]F_{DE}=0$$

$$\left[\frac{a+b}{\sqrt{f^2+(a+b)^2}}\right]F_{DE}-P_2=0$$



$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CD} \\ F_{DE} \\ P_1 \\ P_2 \end{pmatrix} = \text{Find}\left(F_{AB},F_{BC},F_{CD},F_{DE},P_1,P_2\right)$$

$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CD} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} 12.50 \\ 10.31 \\ 10.00 \\ 11.79 \end{pmatrix}\text{kN} \qquad \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} 2.50 \\ 6.25 \end{pmatrix}\text{kN}$$

$$T_{max} = \max\left(F_{AB},F_{BC},F_{CD},F_{DE}\right) \qquad T_{max} = 12.50\,\text{kN}$$

$$F_{max} = \max\left(F_{AB},F_{BC},F_{CD},F_{DE}\right) \qquad F_{max} = 12.50\,\text{kN}$$

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Problem 7-96

The cable supports the loading shown. Determine the distance x_B from the wall to point B.

Given:

$W_1 = 8$ lb

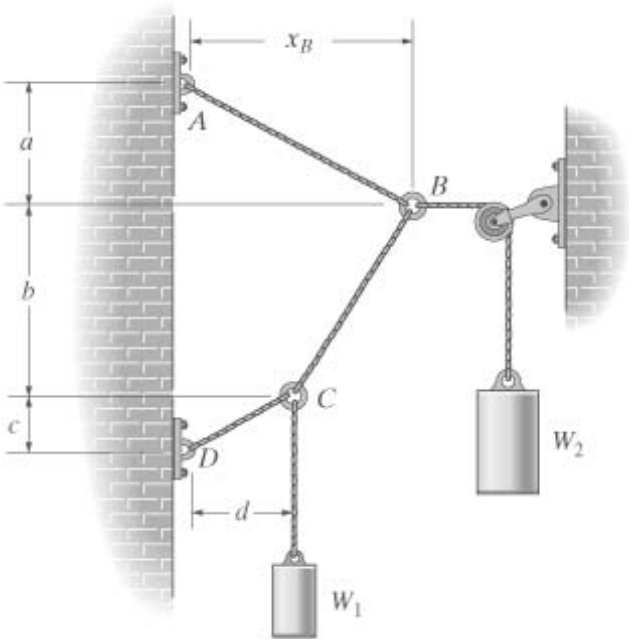
$W_2 = 15$ lb

$a = 5$ ft

$b = 8$ ft

$c = 2$ ft

$d = 3$ ft



Solution:

Guesses

$T_{AB} = 1$ lb $\quad T_{BC} = 1$ lb

$T_{CD} = 1$ lb $\quad x_B = 1$ ft

Given

$$\left(\frac{-x_B}{\sqrt{a^2 + x_B{}^2}}\right) T_{AB} - \left[\frac{x_B - d}{\sqrt{b^2 + (x_B - d)^2}}\right] T_{BC} + W_2 = 0$$

$$\left(\frac{a}{\sqrt{a^2 + x_B{}^2}}\right) T_{AB} - \left[\frac{b}{\sqrt{b^2 + (x_B - d)^2}}\right] T_{BC} = 0$$

$$\left[\frac{x_B - d}{\sqrt{b^2 + (x_B - d)^2}}\right] T_{BC} - \left(\frac{d}{\sqrt{c^2 + d^2}}\right) T_{CD} = 0$$

$$\left[\frac{b}{\sqrt{b^2 + (x_B - d)^2}}\right] T_{BC} - \left(\frac{c}{\sqrt{c^2 + d^2}}\right) T_{CD} - W_1 = 0$$

$$\begin{pmatrix} T_{AB} \\ T_{BC} \\ T_{CD} \\ x_B \end{pmatrix} = \text{Find}(T_{AB}, T_{BC}, T_{CD}, x_B) \qquad \begin{pmatrix} T_{AB} \\ T_{BC} \\ T_{CD} \end{pmatrix} = \begin{pmatrix} 15.49 \\ 10.82 \\ 4.09 \end{pmatrix} \text{lb} \qquad x_B = 5.65\,\text{ft}$$

Problem 7-97

Determine the maximum uniform loading w, measured in lb/ft, that the cable can support if it is

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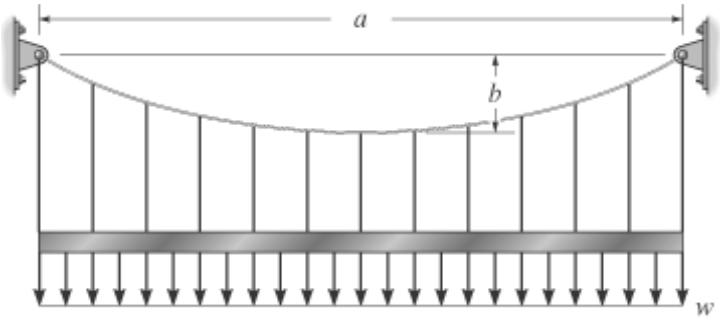
capable of sustaining a maximum tension T_{max} before it will break.

Given:

$T_{max} = 3000 \text{ lb}$

$a = 50 \text{ ft}$

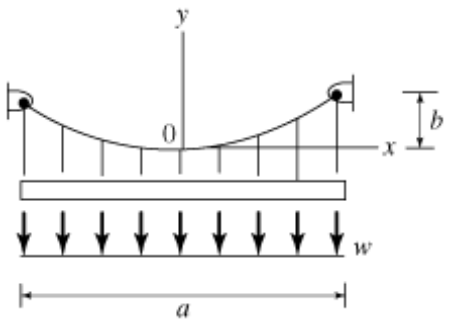
$b = 6 \text{ ft}$



Solution:

$$y = \frac{1}{F_H}\int\int w\,dx\,dx = \frac{wx^2}{2F_H}$$

$$y = \left(\frac{w}{2F_H}\right)x^2$$



$$x = \frac{a}{2} \qquad y = b \qquad F_H = \frac{wa^2}{8b}$$

$$\left(\frac{dy}{dx}\right) = \tan\left(\theta_{max}\right) = \frac{w}{F_H}\left(\frac{a}{2}\right) = \frac{4b}{a} \qquad \theta_{max} = \operatorname{atan}\left(\frac{4b}{a}\right) \qquad \theta_{max} = 25.64 \text{ deg}$$

$$T_{max} = \frac{F_H}{\cos\left(\theta_{max}\right)} = \frac{wa^2}{8b\cos\left(\theta_{max}\right)} \qquad w = \frac{T_{max}8b\cos\left(\theta_{max}\right)}{a^2} \qquad w = 51.93\,\frac{\text{lb}}{\text{ft}}$$

Problem 7-98

The cable is subjected to a uniform loading w. Determine the maximum and minimum tension in the cable.

Units Used:

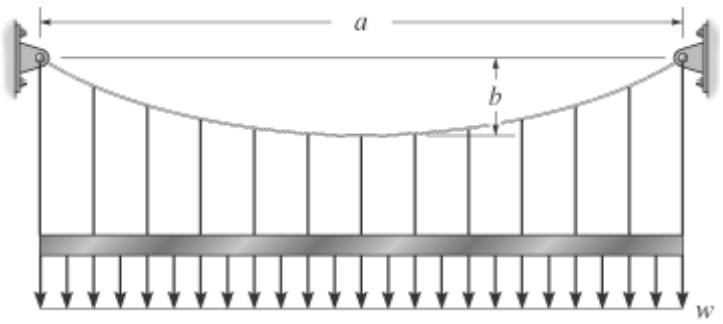
$\text{kip} = 10^3 \text{ lb}$

Given:

$w = 250\,\frac{\text{lb}}{\text{ft}}$

$a = 50 \text{ ft}$

$b = 6 \text{ ft}$



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Solution:

$$y = \frac{w x^2}{2F_H} \qquad b = \frac{w}{2F_H}\left(\frac{a}{2}\right)^2 \qquad F_H = \frac{w a^2}{8b} \qquad F_H = 13021\,\text{lb}$$

$$\tan\left(\theta_{max}\right) = \frac{\mathrm{d}}{\mathrm{d}x} y\left(x = \frac{a}{2}\right) = \frac{w}{F_H}\left(\frac{a}{2}\right) \qquad \theta_{max} = \operatorname{atan}\left(\frac{w a}{2F_H}\right) \qquad \theta_{max} = 25.64\,\text{deg}$$

$$T_{max} = \frac{F_H}{\cos\left(\theta_{max}\right)} \qquad T_{max} = 14.44\,\text{kip}$$

The minimum tension occurs at $\theta = 0$ deg

$$T_{min} = F_H \qquad T_{min} = 13.0\,\text{kip}$$

Problem 7-99

The cable is subjected to the triangular loading. If the slope of the cable at A is zero, determine the equation of the curve $y = f(x)$ which defines the cable shape AB, and the maximum tension developed in the cable.

Units Used:

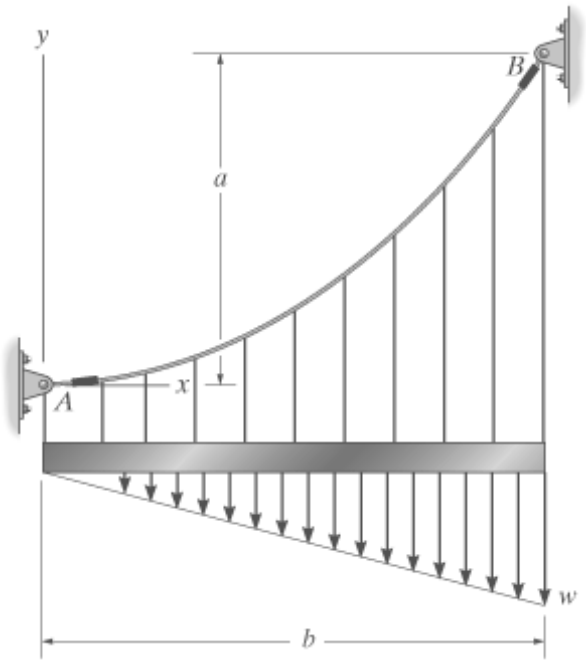
$\text{kip} = 10^3$ lb

Given:

$w = 250\ \frac{\text{lb}}{\text{ft}}$

$a = 20$ ft

$b = 30$ ft



Solution:

$$y = \frac{1}{F_H}\int\int \frac{w x}{b}\,\mathrm{d}x\,\mathrm{d}x$$

$$y = \frac{1}{F_H}\left(\frac{w x^3}{6b} + c_1 x + c_2\right)$$

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Apply boundary conditions $y = x = 0$ and $\frac{d}{dx}y = 0, x = 0$ Thus $C_1 = C_2 = 0$

$y = \frac{w x^3}{6 F_H b}$ set $y = a$ $x = b$ $a = \frac{w b^3}{6 F_H b}$

$F_H = \frac{w b^2}{6 a}$ $F_H = 1.875\,\text{kip}$ $\theta_{max} = \text{atan}\left(\frac{w b^2}{2 F_H b}\right)$ $\theta_{max} = 63.43\,\text{deg}$

$T_{max} = \frac{F_H}{\cos(\theta_{max})}$ $T_{max} = 4.19\,\text{kip}$

Problem 7-100

The cable supports a girder which has weight density γ. Determine the tension in the cable at points *A*, *B*, and *C*.

Units used:

$\text{kip} = 10^3\ \text{lb}$

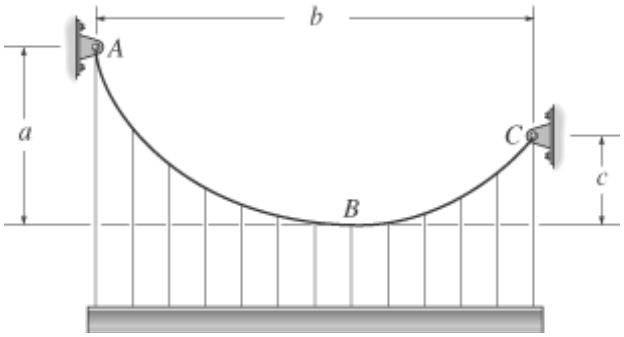
Given:

$\gamma = 850\ \frac{\text{lb}}{\text{ft}}$

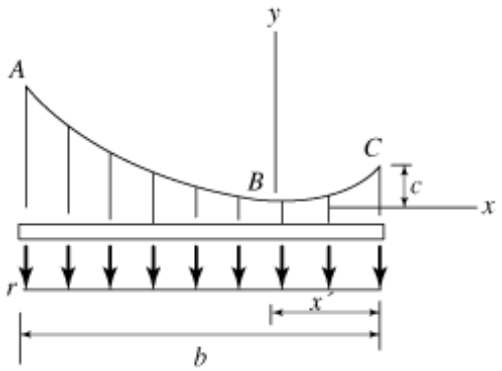
$a = 40\ \text{ft}$

$b = 100\ \text{ft}$

$c = 20\ \text{ft}$



Solution:



$y = \frac{1}{F_H}\int\int \gamma\, dx\, dx$

$y = \frac{\gamma x^2}{2 F_H}$ $\frac{d}{dx}y = \frac{\gamma x}{F_H}$ Guesses $x_1 = 1\ \text{ft}$ $F_H = 1\ \text{lb}$

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Given $c = \frac{\gamma x_1^2}{2F_H}$ $a = \frac{\gamma}{2F_H}(b - x_1)^2$ $\begin{pmatrix} x_1 \\ F_H \end{pmatrix} = \text{Find}(x_1, F_H)$ $F_H = 36.46\,\text{kip}$

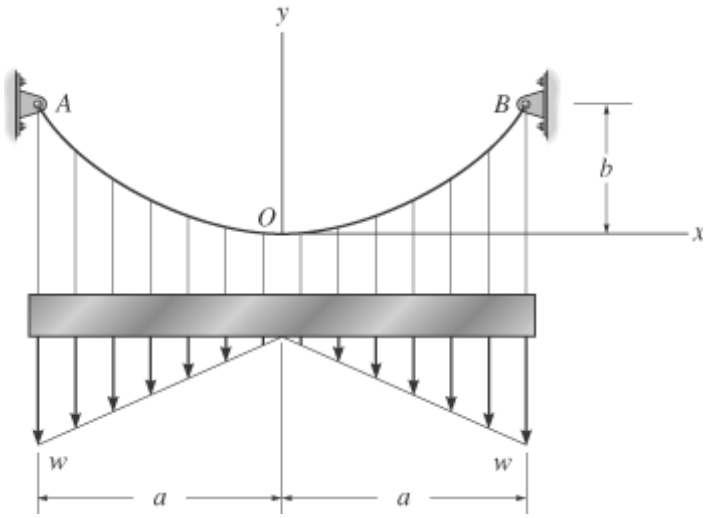
$\tan(\theta_A) = \frac{\gamma}{F_H}(x_1 - b)$ $\theta_A = \text{atan}\left[\frac{\gamma}{F_H}(x_1 - b)\right]$ $\theta_A = -53.79\,\text{deg}$

$\tan(\theta_C) = \frac{\gamma}{F_H}x_1$ $\theta_C = \text{atan}\left(\frac{\gamma}{F_H}x_1\right)$ $\theta_C = 44.00\,\text{deg}$

$T_A = \frac{F_H}{\cos(\theta_A)}$ $T_B = F_H$ $T_C = \frac{F_H}{\cos(\theta_C)}$ $\begin{pmatrix} T_A \\ T_B \\ T_C \end{pmatrix} = \begin{pmatrix} 61.71 \\ 36.46 \\ 50.68 \end{pmatrix}\,\text{kip}$

Problem 7-101

The cable is subjected to the triangular loading. If the slope of the cable at point *O* is zero, determine the equation of the curve $y = f(x)$ which defines the cable shape *OB*, and the maximum tension developed in the cable.



Units used:

$\text{kip} = 10^3\,\text{lb}$

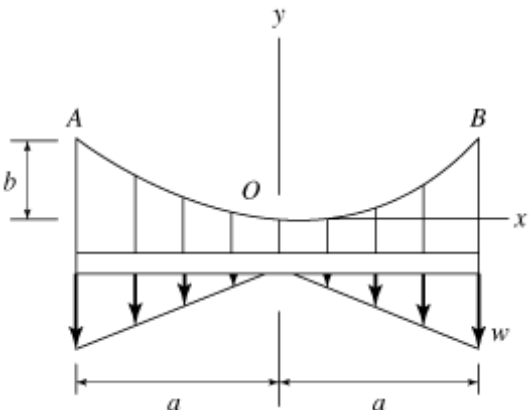
Given:

$w = 500\,\frac{\text{lb}}{\text{ft}}$ $b = 8\,\text{ft}$

$a = 15\,\text{ft}$

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Solution:



$$y = \frac{1}{F_H}\left(\int \int \frac{wx}{a}\,dx\,dx\right)$$

$$y = \frac{1}{F_H}\left[\frac{w}{a}\left(\frac{x^3}{6}\right) + C_1 x + C_2\right]$$

$$\frac{d}{dx}y = \frac{1}{F_H}\left(\frac{wx^2}{2a}\right) + \left(\frac{C_1}{F_H}\right)$$

At $x = 0$, $\frac{d}{dx}y = 0$, $C_1 = 0$

At $x = 0$, $y = 0$, $C_2 = 0$

$$y = \frac{wx^3}{6aF_H} \qquad \frac{d}{dx}y = \frac{wx^2}{2aF_H}$$

At $x = a$, $y = b$ $\quad b = \frac{wa^3}{6aF_H}$ $\quad F_H = \frac{1}{6}w\left(\frac{a^2}{b}\right)$ $\quad F_H = 2343.75\ \text{lb}$

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$$\frac{d}{dx}y = \tan(\theta_{max}) = \frac{w a^2}{2 a F_H} \qquad (\theta_{max}) = \operatorname{atan}\left(\frac{w a^2}{2 a F_H}\right) \qquad (\theta_{max}) = 57.99\ \text{deg}$$

$$T_{max} = \frac{F_H}{\cos(\theta_{max})} \qquad T_{max} = 4.42\ \text{kip}$$

Problem 7-102

The cable is subjected to the parabolic loading $w = w_0(1 - (2x/a)^2)$. Determine the equation $y = f(x)$ which defines the cable shape AB and the maximum tension in the cable.

Units Used:

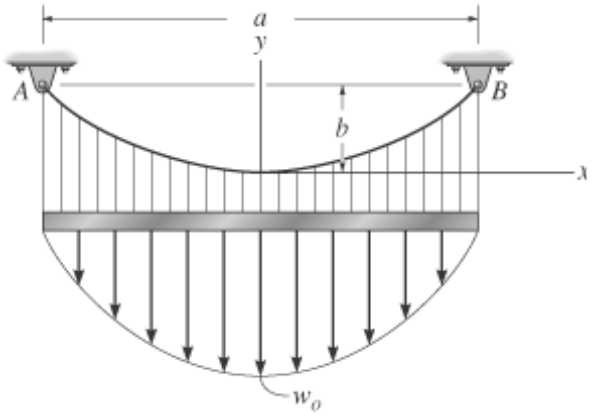
$$\text{kip} = 10^3\ \text{lb}$$

Given:

$$w = w_0\left[1 - \left(\frac{2x}{a}\right)^2\right]$$

$$a = 100\ \text{ft} \qquad w_0 = 150\ \frac{\text{lb}}{\text{ft}}$$

$$b = 20\ \text{ft}$$



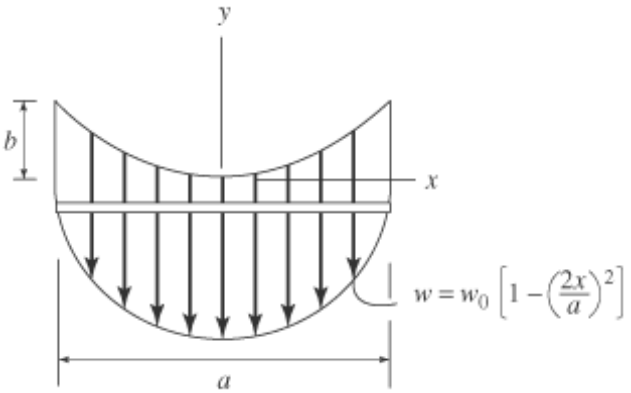
Solution:

$$y = \frac{1}{F_H}\int\int w(x)\,dx\,dx$$

$$y = \frac{1}{F_H}\int w_0\left(x - \frac{4x^3}{3a^2}\right) + C_1\,dx$$

$$y = \frac{1}{F_H}\left(\frac{w_0 x^2}{2} - \frac{x^4 w_0}{3a^2} + C_1 x + C_2\right)$$

$$\frac{dy}{dx} = \frac{1}{F_H}\left(w_0 x - \frac{4 w_0 x^3}{3a^2} + C_1\right)$$



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At $x = 0$ $\frac{dy}{dx} = 0$ $C_1 = 0$

At $x = 0$ $y = 0$ $C_2 = 0$

Thus

$$y = \frac{1}{F_H}\left(\frac{w_0 x^2}{2} - \frac{x^4 w_0}{3a^2}\right) \qquad \frac{dy}{dx} = \frac{1}{F_H}\left(w_0 x - \frac{4 w_0 x^3}{3a^2}\right)$$

At $x = \frac{a}{2}$ we have

$$b = \frac{1}{F_H}\left[\frac{w_0}{2}\left(\frac{a}{2}\right)^2 - \frac{w_0}{3a^2}\left(\frac{a}{2}\right)^4\right] = \frac{5}{48}\left(\frac{w_0 a^2}{F_H}\right) \qquad F_H = \frac{5 w_0 a^2}{48 b} \qquad F_H = 7812.50 \text{ lb}$$

$$\tan(\theta_{max}) = \frac{d}{dx} y\left(\frac{a}{2}\right) = \frac{1}{F_H}\left[w_0\left(\frac{a}{2}\right) - \frac{4 w_0}{3a^2}\left(\frac{a}{2}\right)^3\right] = \frac{w_0 a}{3 F_H}$$

$$\theta_{max} = \operatorname{atan}\left(\frac{w_0 a}{3 F_H}\right) \qquad \theta_{max} = 32.62 \text{ deg}$$

$$T_{max} = \frac{F_H}{\cos(\theta_{max})} \qquad T_{max} = 9.28 \text{ kip}$$

Problem 7-103

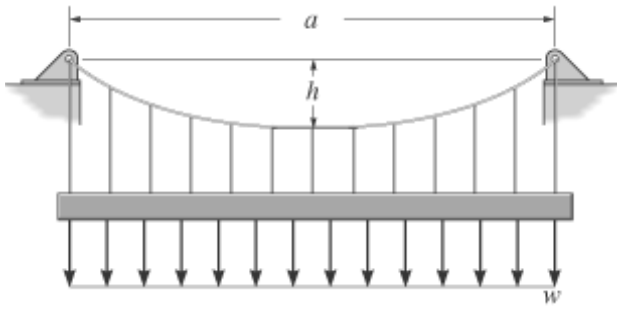
The cable will break when the maximum tension reaches T_{max}. Determine the minimum sag h if it supports the uniform distributed load w.

Given: $\text{kN} = 10^3 \text{ N}$

$T_{max} = 10 \text{ kN}$

$w = 600 \frac{\text{N}}{\text{m}}$

$a = 25 \text{ m}$



Solution:

The equation of the cable:

$$y = \frac{1}{F_H}\int\int w \, dx \, dx \qquad y = \frac{1}{F_H}\left(\frac{w x^2}{2} + C_1 x + C_2\right) \qquad \frac{dy}{dx} = \frac{1}{F_H}(w x + C_1)$$

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Boundary Conditions:

$y = 0$ at $x = 0$, then from Eq.[1] $\quad 0 = \frac{1}{F_H}(C_2) \quad C_2 = 0$

$\frac{d}{dx}y = 0$ at $x = 0$, then from Eq.[2] $\quad 0 = \frac{1}{F_H}(C_1) \quad C_1 = 0$

Thus, $\quad y = \left(\frac{w}{2F_H}\right)x^2 \quad \frac{dy}{dx} = \frac{w}{F_H}x \quad h = \frac{w}{2F_H}\left(\frac{a}{2}\right)^2 \quad F_H = \frac{w}{2h}\left(\frac{a}{2}\right)^2$

$$\tan(\theta_{max}) = \frac{wa}{2F_H} \qquad \cos(\theta_{max}) = \frac{2F_H}{\sqrt{4F_H^2 + (wa)^2}}$$

$$T_{max} = \frac{F_H}{\cos(\theta_{max})} = \sqrt{F_H^2 + \frac{(wa)^2}{4}} = \frac{wa}{2}\sqrt{\frac{a^2}{16h^2} + 1}$$

Guess $\quad h = 1$ m

Given $\quad T_{max} = \frac{wa}{2}\sqrt{\frac{a^2}{16h^2} + 1} \quad h = \text{Find}(h) \quad h = 7.09$ m

---

Problem 7-104

A fiber optic cable is suspended over the poles so that the angle at the supports is θ. Determine the minimum tension in the cable and the sag. The cable has a mass density ρ and the supports are at the same elevation.

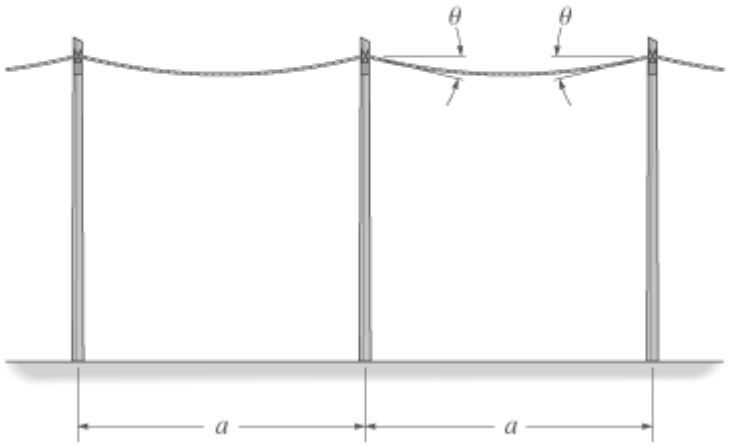
Given:

$\theta = 22$ deg

$\rho = 0.9 \frac{\text{kg}}{\text{m}}$

$a = 30$ m

$g = 9.81 \frac{\text{m}}{\text{s}^2}$



Solution:

$\theta_{max} = \theta$

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$w_0 = \rho g$

$$\frac{d}{dx}y = \tan(\theta) = \sinh\left(w_0 \frac{\frac{a}{2}}{F_H}\right)$$

$$F_H = \frac{w_0\left(\frac{a}{2}\right)}{\operatorname{asinh}(\tan(\theta))} \qquad F_H = 336 \text{ N}$$

$$T_{max} = \frac{F_H}{\cos(\theta)} \qquad T_{max} = 363 \text{ N}$$

$$h = \frac{F_H}{w_0}\left(\cosh\left(\frac{w_0 \frac{a}{2}}{F_H}\right) - 1\right) \qquad h = 2.99 \text{ m}$$

Problem 7-105

A cable has a weight density γ and is supported at points that are a distance d apart and at the same elevation. If it has a length L, determine the sag.

Given:

$$\gamma = 3 \frac{\text{lb}}{\text{ft}} \qquad d = 500 \text{ ft} \qquad L = 600 \text{ ft}$$

Solution:

Guess $F_H = 100$ lb

Given $\quad \frac{L}{2} - \left[\frac{F_H}{\gamma}\sinh\left[\frac{\gamma}{F_H}\left(\frac{d}{2}\right)\right]\right] = 0 \qquad F_H = \text{Find}(F_H) \qquad F_H = 704.3 \text{ lb}$

$$h = \frac{F_H}{\gamma}\left(\cosh\left(\frac{1}{2}\frac{\gamma}{F_H}d\right) - 1\right) \qquad h = 146 \text{ ft}$$

Problem 7-106

Show that the deflection curve of the cable discussed in Example 7.15 reduces to Eq. (4) in Example 7.14 when the *hyperbolic cosine function* is expanded in terms of a series and only the

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first two terms are retained. (The answer indicates that the *catenary* may be replaced by a *parabola* in the analysis of problems in which the sag is small. In this case, the cable weight is assumed to be uniformly distributed along the horizontal.)

Solution:

$$\cosh(x) = 1 + \frac{x^2}{2!} + ..$$

Substituting into

$$y = \frac{F_H}{w_0}\left(\cosh\left(\frac{w_0}{F_H}x\right) - 1\right) = \frac{F_H}{w_0}\left(1 + \frac{{w_0}^2 x^2}{2{F_H}^2} + .. - 1\right) = \frac{w_0 x^2}{2F_H}$$

Using the boundary conditions $y = h$ at $x = \frac{L}{2}$

$$h = \frac{w_0}{2F_H}\left(\frac{L}{2}\right)^2 \qquad F_H = \frac{w_0 L^2}{8h} \qquad \text{We get} \qquad y = \frac{4h}{L^2}x^2$$

Problem 7-107

A uniform cord is suspended between two points having the same elevation. Determine the sag-to-span ratio so that the maximum tension in the cord equals the cord's total weight.

Solution:

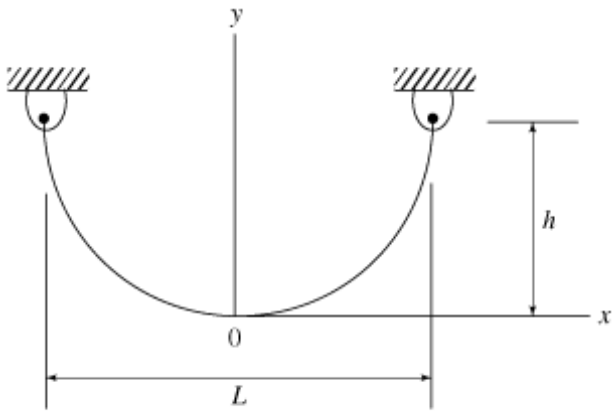
$$s = \frac{F_H}{w_0}\sinh\left(\frac{w_0}{F_H}x\right)$$

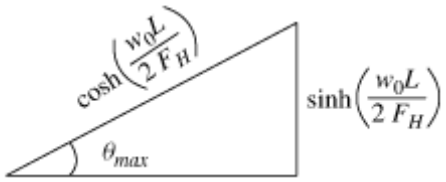
$$y = \frac{F_H}{w_0}\left(\cosh\left(\frac{w_0}{F_H}x\right) - 1\right)$$

At $x = \frac{L}{2}$

$$\frac{\mathrm{d}}{\mathrm{d}x}y_{max} = \tan\left(\theta_{max}\right) = \sinh\left(\frac{w_0 L}{2F_H}\right)$$

$$\cos\left(\theta_{max}\right) = \frac{1}{\cosh\left(\frac{w_0 L}{2F_H}\right)}$$





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$$T_{max} = \frac{F_H}{\cos(\theta_{max})} \qquad w_0 2s = F_H \cosh\left(\frac{w_0 L}{2F_H}\right)$$

$$2F_H \sinh\left(\frac{w_0 L}{2F_H}\right) = F_H \cosh\left(\frac{w_0 L}{2F_H}\right) \qquad \tanh\left(\frac{w_0 L}{2F_H}\right) = \frac{1}{2}$$

$$k_1 = \operatorname{atanh}(0.5) \qquad k_1 = 0.55 \qquad \frac{w_0 L}{2F_H} = k_1$$

when $x = \frac{L}{2}$ $y = h$ $h = \frac{F_H}{W_0}\left(\cosh(k_1) - 1\right)$ $k_2 = \cosh(k_1) - 1$ $k_2 = 0.15$

$$h = k_2\left(\frac{F_H}{w_0}\right) \qquad L = k_1\left(\frac{2F_h}{w_0}\right) \qquad ratio = \frac{h}{L} = \frac{k_2}{2k_1} \qquad ratio = \frac{k_2}{2k_1} \qquad ratio = 0.14$$

Problem 7-108

A cable has a weight denisty γ. If it can span a distance L and has a sag h determine the length of the cable. The ends of the cable are supported from the same elevation.

Given:

$$\gamma = 2\ \frac{\text{lb}}{\text{ft}} \qquad L = 100\ \text{ft} \qquad h = 12\ \text{ft}$$

Solution:

From Eq. (5) of Example 7-15 :

$$h = \frac{F_H}{\gamma}\left[\frac{\left(\frac{\gamma}{2}\frac{L}{F_H}\right)^2}{2}\right] \qquad F_H = \frac{1}{8}\gamma\left(\frac{L^2}{h}\right) \qquad F_H = 208.33\,\text{lb}$$

From Eq. (3) of Example 7-15:

$$\frac{l}{2} = \left(\frac{F_H}{\gamma}\right)\sinh\left[\frac{\gamma}{F_H}\left(\frac{L}{2}\right)\right] \qquad l = 2\left(\frac{F_H}{\gamma}\right)\sinh\left(\frac{1}{2}\gamma\frac{L}{F_H}\right) \qquad l = 104\,\text{ft}$$

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Problem 7-109

The transmission cable having a weight density γ is strung across the river as shown. Determine the required force that must be applied to the cable at its points of attachment to the towers at *B* and *C*.

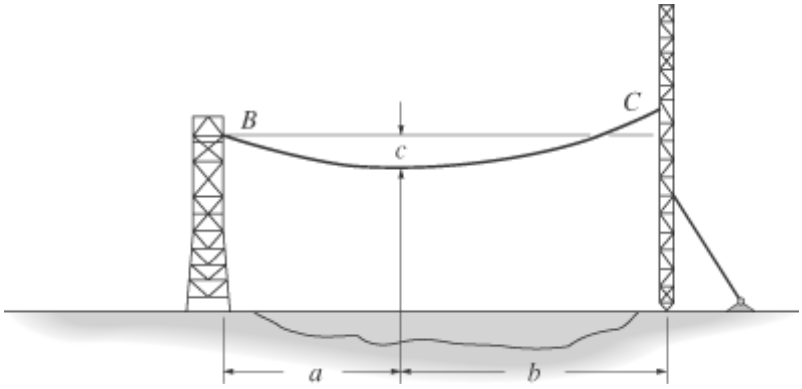
Units Used:

$\text{kip} = 10^3\ \text{lb}$

Given:

$\gamma = 20\ \frac{\text{lb}}{\text{ft}} \qquad b = 75\ \text{ft}$

$a = 50\ \text{ft} \qquad c = 10\ \text{ft}$



Solution:

From Example 7-15,

$$y = \frac{F_H}{\gamma}\left[\cosh\left(\frac{\gamma}{F_H}x\right) - 1\right) \qquad \frac{dy}{dx} = \sinh\left(\frac{\gamma x}{F_H}\right)$$

Guess $F_H = 1000\ \text{lb}$

Given At *B*: $c = \frac{F_H}{\gamma}\left(\cosh\left(-\frac{\gamma a}{F_H}\right) - 1\right) \qquad F_H = \text{Find}(F_H) \qquad F_H = 2.53\,\text{kip}$

$$\tan(\theta_B) = \sinh\left(-\frac{\gamma a}{F_H}\right) \qquad \theta_B = \text{atan}\left(\sinh\left(\frac{-\gamma a}{F_H}\right)\right) \qquad \theta_B = -22.06\,\text{deg}$$

$$\tan(\theta_C) = \sinh\left(\frac{\gamma b}{F_H}\right) \qquad \theta_C = \text{atan}\left(\sinh\left(\frac{\gamma b}{F_H}\right)\right) \qquad \theta_C = 32.11\,\text{deg}$$

$$T_B = \frac{F_H}{\cos(\theta_B)} \qquad T_C = \frac{F_H}{\cos(\theta_C)} \qquad \begin{pmatrix} T_B \\ T_C \end{pmatrix} = \begin{pmatrix} 2.73 \\ 2.99 \end{pmatrix}\text{kip}$$

Problem 7-110

Determine the maximum tension developed in the cable if it is subjected to a uniform load *w*.

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Units Used:

$MN = 10^6\ N$

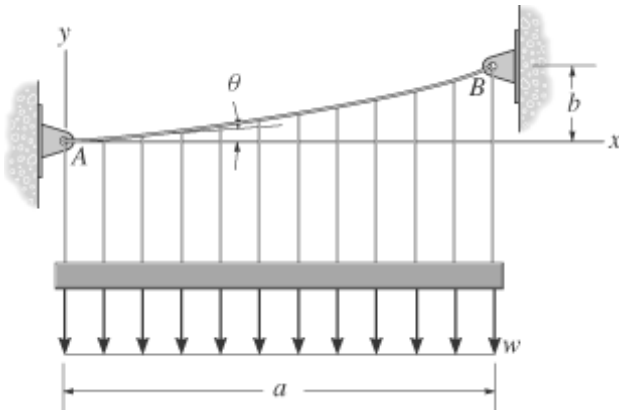
Given:

$w = 600\ \frac{N}{m}$

$a = 100\ m$

$b = 20\ m$

$\theta = 10\ deg$



Solution:

The Equation of the Cable:

$$y = \frac{1}{F_H}\int\int w(x)\,dx\,dx = \frac{1}{F_H}\left(\frac{w x^2}{2} + C_1 x + C_2\right)$$

$$\frac{dy}{dx} = \frac{1}{F_H}(w x + C_1)$$

Initial Guesses: $C_1 = 1\ N$ $C_2 = 1\ N\cdot m$ $F_H = 1\ N$

Given *Boundary Conditions:*

$x = 0$ $0 = \frac{1}{F_H}C_2$ $\tan(\theta) = \frac{1}{F_H}(C_1)$

$y = b$ at $x = a$ $b = \left(\frac{w}{2F_H}\right)a^2 + \left(\frac{C_1}{F_H}\right)a$

$$\begin{pmatrix} C_1 \\ C_2 \\ F_H \end{pmatrix} = \text{Find}(C_1, C_2, F_H)$$ $C_1 = 0.22\,MN$ $C_2 = 0.00\,N\cdot m$ $F_H = 1.27\,MN$

$\tan(\theta_{max}) = \frac{1}{F_H}(w a + C_1)$ $\theta_{max} = \text{atan}\left(\frac{w a + C_1}{F_H}\right)$ $\theta_{max} = 12.61\,deg$

$$T_{max} = \frac{F_H}{\cos(\theta_{max})}$$ $T_{max} = 1.30\,MN$

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Problem 7-111

A chain of length L has a total mass M and is suspended between two points a distance d apart. Determine the maximum tension and the sag in the chain.

Given:

$$L = 40\ \text{m} \qquad M = 100\ \text{kg} \qquad d = 10\ \text{m} \qquad g = 9.81\ \frac{\text{m}}{\text{s}^2}$$

Solution:

$$w = M\frac{g}{L}$$

$$s = \left(\frac{F_H}{w}\right)\sinh\left(\frac{w}{F_H}x\right) \qquad y = \left(\frac{F_H}{w}\right)\left(\cosh\left(\frac{w}{F_H}x\right) - 1\right) \qquad \frac{\mathrm{d}}{\mathrm{d}x}y = \sinh\left(\frac{w}{F_H}x\right)$$

Guesses $F_H = 10\ \text{N}$ $h = 10\ \text{m}$

Given $\frac{L}{2} = \left(\frac{F_H}{w}\right)\sinh\left(\frac{w}{F_H}\frac{d}{2}\right)$ $h = \frac{F_H}{w}\left(\cosh\left(\frac{w}{F_H}\frac{d}{2}\right) - 1\right)$ $\begin{pmatrix} F_H \\ h \end{pmatrix} = \text{Find}\left(F_H, h\right)$

$$\theta_{max} = \text{atan}\left(\sinh\left(\frac{w}{F_H}\frac{d}{2}\right)\right) \qquad T_{max} = \frac{F_H}{\cos\left(\theta_{max}\right)}$$

$F_H = 37.57\ \text{N}$ $h = 18.53\ \text{m}$ $T_{max} = 492\ \text{N}$

Problem 7-112

The cable has a mass density ρ and has length L. Determine the vertical and horizontal components of force it exerts on the top of the tower.

Given:

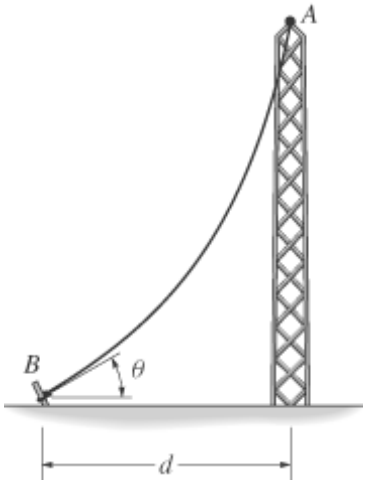
$\rho = 0.5\ \frac{\text{kg}}{\text{m}}$

$L = 25\ \text{m}$

$\theta = 30\ \text{deg}$

$d = 15\ \text{m}$

$\text{g} = 9.81\ \frac{\text{m}}{\text{s}^2}$



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Solution:

$$x = \int \frac{1}{\sqrt{1 + \frac{1}{F_H{}^2}\left(\int \rho g \, \mathrm{ds}\right)^2}} \, \mathrm{ds}$$

Performing the integration yields:

$$x = \frac{F_H}{\rho g}\left(\operatorname{asinh}\left(\frac{\rho g s + C_1}{F_H}\right) + C_2\right)$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{F_H}\int \rho g \, \mathrm{ds} = \frac{1}{F_H}\left(\rho g s + C_1\right)$$

At $s = 0$; $\frac{\mathrm{d}y}{\mathrm{d}x} = \tan(\theta)$ Hence $C_1 = F_H \tan(\theta)$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\rho g s}{F_H} + \tan(\theta)$$

Applying boundary conditions at $x = 0$; $s = 0$ to Eq.[1] and using the result $C_1 = F_H \tan(\theta)$ yields $C_2 = -\operatorname{asinh}(\tan(\theta))$. Hence

Guess $F_H = 1$ N

Given

$$d = \left(\frac{F_H}{\rho g}\right)\left[\operatorname{asinh}\left[\left(\frac{1}{F_H}\right)\left(\rho g L + F_H \tan(\theta)\right)\right] - \left(\operatorname{asinh}(\tan(\theta))\right)\right]$$

$F_H = \operatorname{Find}(F_H)$ $F_H = 73.94$ N

At A $\tan(\theta_A) = \frac{\rho g L}{F_H} + \tan(\theta)$ $\theta_A = \operatorname{atan}\left(\frac{\rho g L}{F_H} + \tan(\theta)\right)$ $\theta_A = 65.90 \deg$

$F_A = \frac{F_H}{\cos(\theta_A)}$ $F_{Ax} = F_A \cos(\theta_A)$ $F_{Ay} = F_A \sin(\theta_A)$ $\begin{pmatrix} F_{Ax} \\ F_{Ay} \end{pmatrix} = \begin{pmatrix} 73.94 \\ 165.31 \end{pmatrix}$ N

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Problem 7-113

A cable of length L is suspended between two points a distance d apart and at the same elevation. If the minimum tension in the cable is T_{min}, determine the total weight of the cable and the maximum tension developed in the cable.

Units Used: $\text{kip} = 10^3\ \text{lb}$

Given: $L = 50\ \text{ft}$ $d = 15\ \text{ft}$ $T_{min} = 200\ \text{lb}$

Solution: $T_{min} = F_H$ $F_H = T_{min}$ $F_H = 200\,\text{lb}$

From Example 7-15: $s = \left(\frac{F_H}{w_0}\right)\sinh\left(\frac{w_0 x}{F_H}\right)$

Guess $w_0 = 1\ \frac{\text{lb}}{\text{ft}}$

Given $\frac{L}{2} = \left(\frac{F_H}{w_0}\right)\sinh\left(\frac{w_0}{F_H}\frac{d}{2}\right)$ $w_0 = \text{Find}(w_0)$ $w_0 = 79.93\,\frac{\text{lb}}{\text{ft}}$

$Totalweight = w_0 L$ $Totalweight = 4.00\,\text{kip}$

$\tan(\theta_{max}) = \frac{w_0}{F_H}\frac{L}{2}$ $\theta_{max} = \text{atan}\left[\frac{w_0\left(\frac{L}{2}\right)}{F_H}\right]$ $\theta_{max} = 84.28\,\text{deg}$

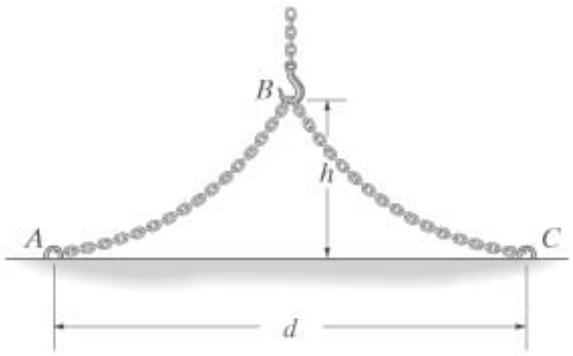
Then,

$T_{max} = \frac{F_H}{\cos(\theta_{max})}$ $T_{max} = 2.01\,\text{kip}$

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Problem 7-114

The chain of length L is fixed at its ends and hoisted at its midpoint B using a crane. If the chain has a weight density w, determine the minimum height h of the hook in order to lift the chain *completely* off the ground. What is the horizontal force at pin A or C when the chain is in this position? *Hint:* When h is a minimum, the slope at A and C is zero.



Given:

$L = 80\ \text{ft}$

$d = 60\ \text{ft}$

$w = 0.5\ \frac{\text{lb}}{\text{ft}}$

Solution:

Guesses $F_H = 10\ \text{lb}$ $h = 1\ \text{ft}$

Given $h = \frac{F_H}{w}\left(\cosh\left(\frac{w}{F_H}\frac{d}{2}\right) - 1\right)$ $\frac{L}{2} = \left(\frac{F_H}{w}\right)\sinh\left(\frac{w}{F_H}\frac{d}{2}\right)$

$\begin{pmatrix} h \\ F_H \end{pmatrix} = \text{Find}(h, F_H)$ $F_A = F_H$ $F_C = F_H$ $\begin{pmatrix} F_A \\ F_C \end{pmatrix} = \begin{pmatrix} 11.1 \\ 11.1 \end{pmatrix}\text{lb}$

$h = 23.5\,\text{ft}$

Problem 7-115

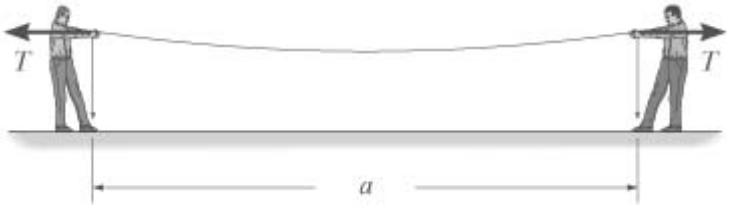
A steel tape used for measurement in surveying has a length L and a total weight W. How much horizontal tension must be applied to the tape so that the distance marked on the ground is a? In practice the calculation should also include the effects of elastic stretching and temperature changes on the tape's length.

Given:

$L = 100\ \text{ft}$

$W = 2\ \text{lb}$

$a = 99.90\ \text{ft}$



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Solution:

$$w_0 = \frac{W}{L} \qquad w_0 = 0.02\,\frac{\text{lb}}{\text{ft}}$$

Guess $F_H = 10\ \text{lb}$

Given $\dfrac{L}{2} - \left(\dfrac{F_H}{w_0}\sinh\left(\dfrac{w_0}{F_H}\dfrac{a}{2}\right)\right) = 0 \qquad F_H = \text{Find}(F_H) \qquad F_H = 12.9\,\text{lb}$

Problem 7-116

A cable of weight W is attached between two points that are a distance d apart, having equal elevations. If the maximum tension developed in the cable is T_{max} determine the length L of the cable and the sag h.

Given: $W = 100\ \text{lb} \qquad d = 50\ \text{ft} \qquad T_{max} = 75\ \text{lb}$

Solution:

Guesses $F_H = 20\ \text{lb} \quad L = 20\ \text{ft} \quad \theta_{max} = 20\ \text{deg} \quad h = 2\ \text{ft}$

Given

$$h = \frac{F_H L}{W}\left(\cosh\left(\frac{W}{F_H L}\frac{d}{2}\right) - 1\right) \qquad \tan(\theta_{max}) = \sinh\left(\frac{W}{F_H L}\frac{d}{2}\right)$$

$$T_{max} = \frac{F_H}{\cos(\theta_{max})} \qquad \frac{L}{2} = \left(\frac{F_H L}{W}\right)\sinh\left(\frac{W}{F_H L}\frac{d}{2}\right)$$

$$\begin{pmatrix} F_H \\ L \\ \theta_{max} \\ h \end{pmatrix} = \text{Find}(F_H, L, \theta_{max}, h) \qquad F_H = 55.90\,\text{lb} \qquad \begin{pmatrix} L \\ h \end{pmatrix} = \begin{pmatrix} 55.57 \\ 10.61 \end{pmatrix}\text{ft}$$

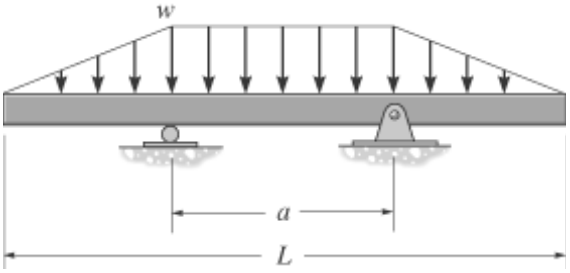
$$\theta_{max} = 41.81\,\text{deg}$$

Problem 7-117

Determine the distance a between the supports in terms of the beam's length L so that the moment in the *symmetric* beam is zero at the beam's center.

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Solution:



Support Reactions:

$\curvearrowleft + \Sigma M_D = 0;$

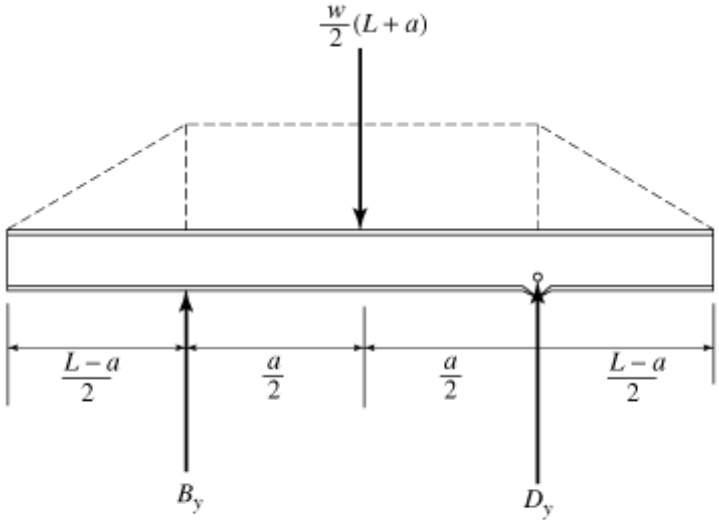
$$\frac{w}{2}(L + a)\left(\frac{a}{2}\right) - B_y(a) = 0$$

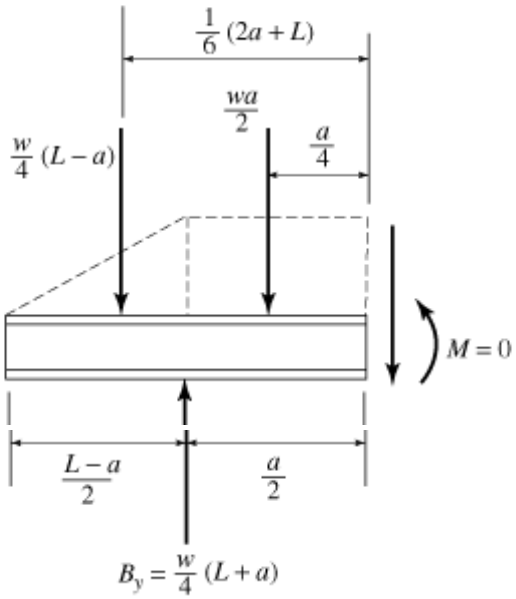
$$B_y = \frac{w}{4}(L + a)$$

Internal Forces:

$\curvearrowleft + \Sigma M_C = 0;$ $\quad w\left(\frac{a}{2}\right)\left(\frac{a}{4}\right) + \frac{1}{2}w\left(\frac{L-a}{2}\right)\left(\frac{2a+L}{6}\right) - \frac{w}{4}(L + a)\left(\frac{a}{2}\right) = 0$

$2a^2 + 2aL - L^2 = 0$ $\quad b = \frac{-2 + \sqrt{12}}{4}$ $\quad b = 0.366$ $\quad a = bL$





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Problem 7-118

Determine the internal normal force, shear force, and moment at point *D*.

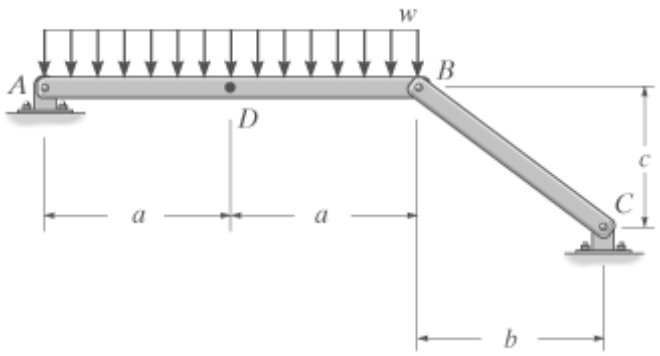
Given:

$w = 150\ \frac{\text{N}}{\text{m}}$

$a = 4\ \text{m}$

$b = 4\ \text{m}$

$c = 3\ \text{m}$



Solution:

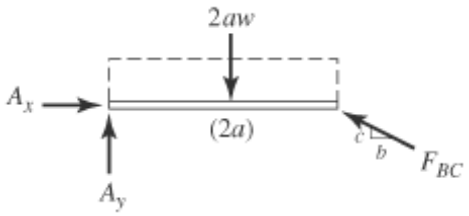
Guesses $A_x = 1\ \text{N}$ $A_y = 1\ \text{N}$ $F_{BC} = 1\ \text{N}$

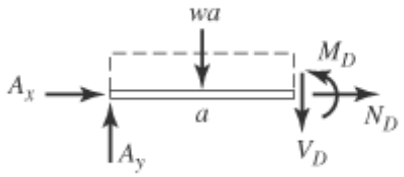
Given

$$A_x - \left(\frac{b}{\sqrt{b^2 + c^2}}\right) F_{BC} = 0$$

$$A_y - w(2a) + \left(\frac{c}{\sqrt{b^2 + c^2}}\right) F_{BC} = 0$$

$$w2a\,a - A_y(2a) = 0$$





$$\begin{pmatrix} A_x \\ A_y \\ F_{BC} \end{pmatrix} = \text{Find}\left(A_x, A_y, F_{BC}\right) \qquad \begin{pmatrix} A_x \\ A_y \\ F_{BC} \end{pmatrix} = \begin{pmatrix} 800 \\ 600 \\ 1000 \end{pmatrix} \text{N}$$

Guesses $N_D = 1\ \text{N}$ $V_D = 1\ \text{N}$ $M_D = 1\ \text{N}\cdot\text{m}$

Given $A_x + N_D = 0$ $A_y - wa - V_D = 0$ $-A_y a + wa\begin{pmatrix} a \\ 2 \end{pmatrix} + M_D = 0$

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$$\begin{pmatrix} N_D \\ V_D \\ M_D \end{pmatrix} = \text{Find}(N_D, V_D, M_D) \qquad \begin{pmatrix} N_D \\ V_D \end{pmatrix} = \begin{pmatrix} -800.00 \\ 0.00 \end{pmatrix} \text{N} \qquad M_D = 1200\,\text{N}\cdot\text{m}$$

Problem 7-119

The beam is supported by a pin at *C* and a rod *AB.* Determine the internal normal force, shear force, and moment at point *D.*

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F = 4\ \text{kN}$

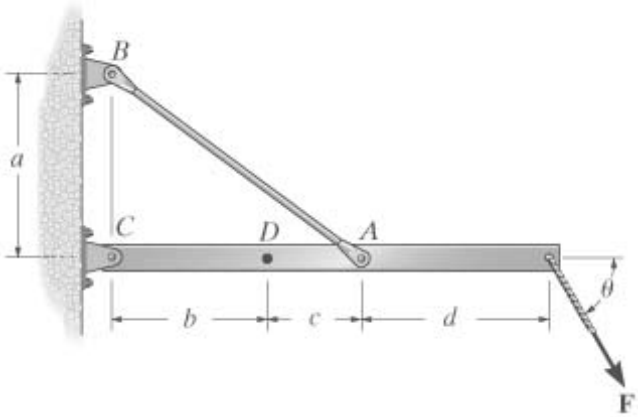
$a = 6\ \text{m}$

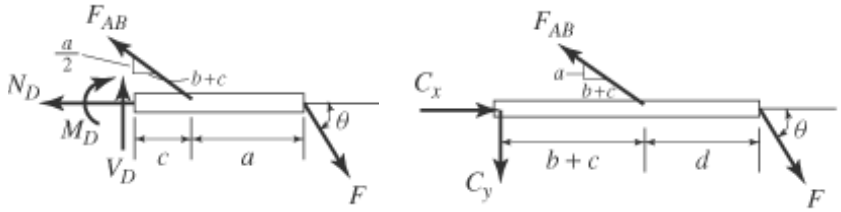
$b = 5\ \text{m}$

$c = 3\ \text{m}$

$d = 6\ \text{m}$

$\theta = 60\ \text{deg}$





Solution:

Guesses

$F_{AB} = 1\ \text{N} \quad N_D = 1\ \text{N} \quad V_D = 1\ \text{N} \quad M_D = 1\ \text{N}\cdot\text{m}$

Given $\quad -F\sin(\theta)(b+c+d) + \left[\frac{a}{\sqrt{a^2 + (b+c)^2}}\right] F_{AB}(b+c) = 0$

$$-N_D - \left[\frac{b+c}{\sqrt{a^2 + (b+c)^2}}\right] F_{AB} + F\cos(\theta) = 0$$

$$V_D + \left[\frac{a}{\sqrt{a^2 + (b+c)^2}}\right] F_{AB} - F\sin(\theta) = 0$$

$$\left[\frac{a}{\sqrt{a^2 + (b+c)^2}}\right] F_{AB}\,c - F\sin(\theta)(c+d) - M_D = 0$$

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$$\begin{pmatrix} F_{AB} \\ N_D \\ V_D \\ M_D \end{pmatrix} = \text{Find}(F_{AB}, N_D, V_D, M_D) \qquad \begin{pmatrix} N_D \\ V_D \end{pmatrix} = \begin{pmatrix} -6.08 \\ -2.6 \end{pmatrix} \text{kN} \qquad M_D = -12.99\ \text{kN}\cdot\text{m}$$

Problem 7-120

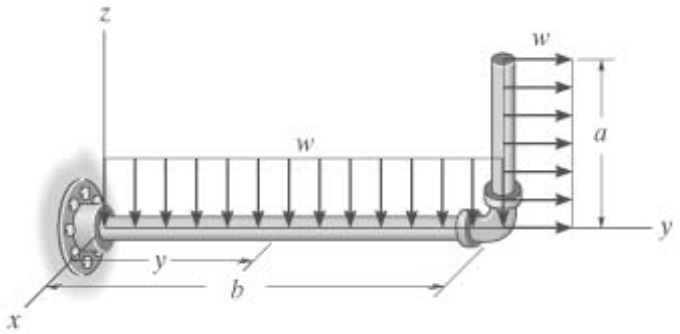
Express the shear and moment acting in the pipe as a function of y, where $0 \le y \le b$ ft.

Given:

$w = 4\ \frac{\text{lb}}{\text{ft}}$

$a = 2$ ft

$b = 4$ ft

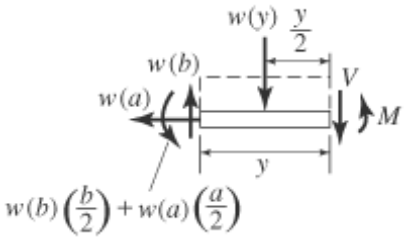


Solution:

$\Sigma F_y = 0;$ $\quad w\,b - w\,y - V = 0$

$$V(y) = w(b - y)$$

$$V(y) = 4\ \frac{\text{lb}}{\text{ft}}\ (4\ \text{ft} - y)$$



$\Sigma M = 0;$ $\quad M + w y\left(\frac{y}{2}\right) + w b\left(\frac{b}{2}\right) + w a\left(\frac{a}{2}\right) - w b\, y = 0$

$$M(y) = w\ b\ y - \frac{1}{2}\ w\ y^2 - \frac{1}{2}\ w\ b^2 - \frac{1}{2}\ w\ a^2$$

$$M(y) = 16\ \text{lb}\ y - 2\ \frac{\text{lb}}{\text{ft}}\ y^2 - 40\ \text{lb}\cdot\text{ft}$$

Problem 7-121

Determine the normal force, shear force, and moment at points B and C of the beam.

Given: $\quad \text{kN} = 10^3\ \text{N}$

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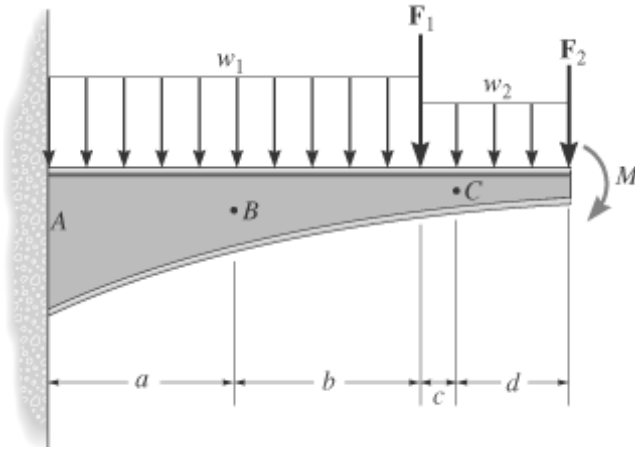
$a = 5\ \text{m}$ $F_2 = 6\ \text{kN}$

$b = 5\ \text{m}$ $w_1 = 2\ \frac{\text{kN}}{\text{m}}$

$c = 1\ \text{m}$

$d = 3\ \text{m}$ $w_2 = 1\ \frac{\text{kN}}{\text{m}}$

$F_1 = 7.5\ \text{kN}$ $M = 40\ \text{kN}\cdot\text{m}$



Solution:

Guesses

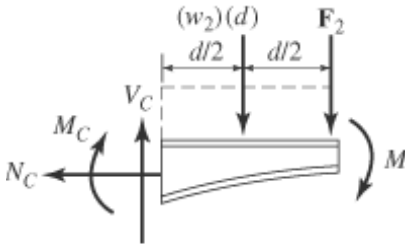
$N_C = 1\ \text{N}$ $V_C = 1\ \text{N}$ $M_C = 1\ \text{N}\cdot\text{m}$

Given

$-N_C = 0$ $V_C - w_2 d - F_2 = 0$

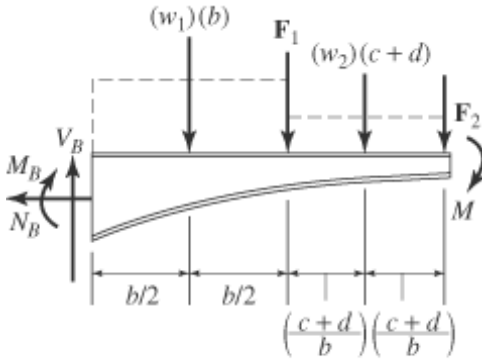
$-M_C - w_2 d\left(\frac{d}{2}\right) - F_2 d - M = 0$

$\begin{pmatrix} N_C \\ V_C \\ M_C \end{pmatrix} = \text{Find}(N_C, V_C, M_C)$



(a)

$\begin{pmatrix} N_C \\ V_C \end{pmatrix} = \begin{pmatrix} 0.00 \\ 9.00 \end{pmatrix}\text{kN}$ $M_C = -62.50\ \text{kN}\cdot\text{m}$



(b)

Guesses $N_B = 1\ \text{N}$ $V_B = 1\ \text{N}$ $M_B = 1\ \text{N}\cdot\text{m}$

Given

$-N_B = 0$ $V_B - w_1 b - w_2(c + d) - F_1 - F_2 = 0$

$-M_B - w_1 b\left(\frac{b}{2}\right) - F_1 b - w_2(c + d)\left(b + \frac{c + d}{2}\right) - F_2(b + c + d) - M = 0$

$\begin{pmatrix} N_B \\ V_B \\ M_B \end{pmatrix} = \text{Find}(N_B, V_B, M_B)$ $\begin{pmatrix} N_B \\ V_B \end{pmatrix} = \begin{pmatrix} 0.00 \\ 27.50 \end{pmatrix}\text{kN}$ $M_B = -184.50\ \text{kN}\cdot\text{m}$

Problem 7-122

The chain is suspended between points *A* and *B.* If it has a weight weight density *w* and the

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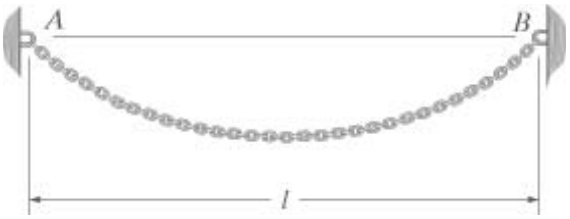
sag is *h*, determine the maximum tension in the chain.

Given:

$w = 0.5 \frac{\text{lb}}{\text{ft}}$

$L = 60 \text{ ft}$

$h = 3 \text{ ft}$

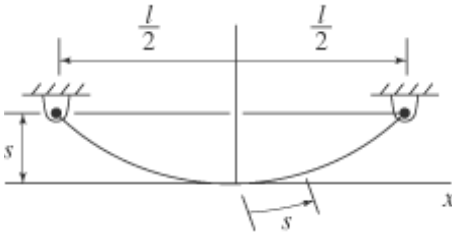


Solution:

Form Example 7-15

$$y = \frac{F_H}{w}\left(\cosh\left(\frac{wx}{F_H}\right) - 1\right)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}y = \sinh\left(\frac{wx}{F_H}\right)$$



Guess $F_H = 1 \text{ lb}$

Given $h = \frac{F_H}{w}\left(\cosh\left(\frac{w}{F_H}\frac{L}{2}\right) - 1\right)$ $F_H = \text{Find}(F_H)$ $F_H = 75.2 \text{ lb}$

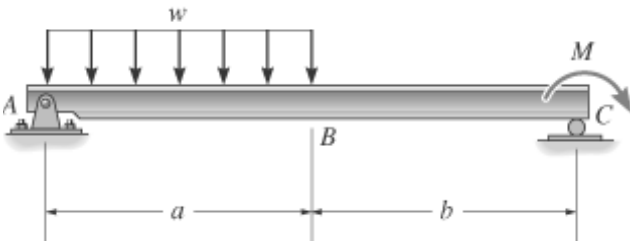
$\theta_{max} = \text{atan}\left(\sinh\left(\frac{w}{F_H}\frac{L}{2}\right)\right)$ $T_{max} = \frac{F_H}{\cos(\theta_{max})}$ $T_{max} = 76.7 \text{ lb}$

Problem 7-123

Draw the shear and moment diagrams for the beam.

Units Used:

$\text{kN} = 10^3 \text{ N}$



Given:

$w = 2 \frac{\text{kN}}{\text{m}}$ $a = 5 \text{ m}$ $b = 5 \text{ m}$ $M = 50 \text{ kN}\cdot\text{m}$

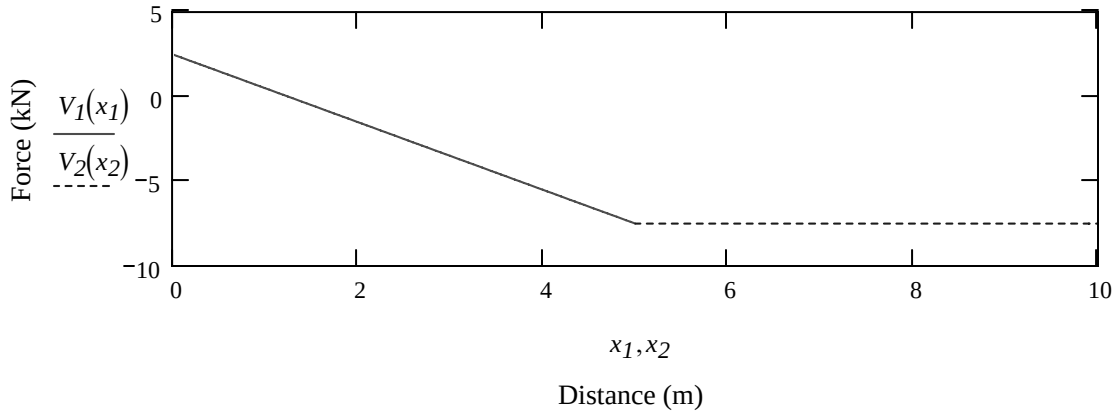
Solution: Guesses $A = 1 \text{ N}$ $C = 1 \text{ N}$

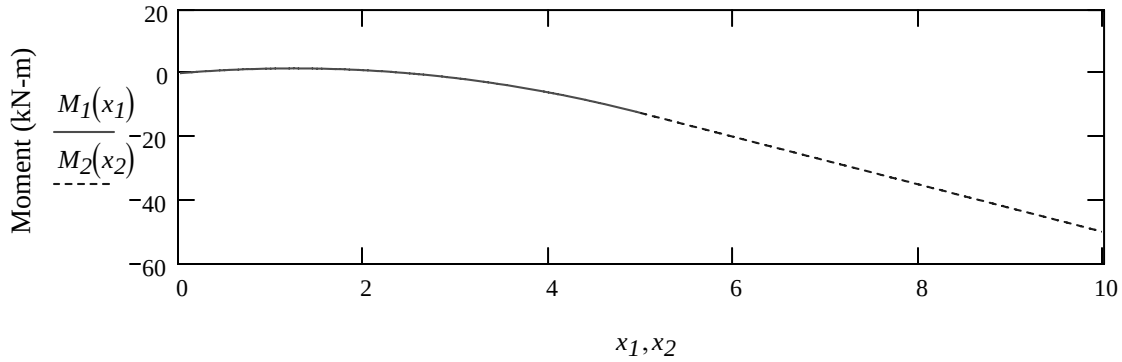
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Given $\quad wa\left(b+\frac{a}{2}\right)-A(a+b)-M=0 \qquad A+C-wa=0 \qquad \begin{pmatrix} A \\ C \end{pmatrix} = \text{Find}(A, C)$

$x_1 := 0, 0.01a .. a \qquad V_1(x_1) := \left(A - wx_1\right)\frac{1}{\text{kN}} \qquad M_1(x_1) := \left(Ax_1 - wx_1\frac{x_1}{2}\right)\frac{1}{\text{kN}\cdot\text{m}}$

$x_2 := a, 1.01a .. a+b \qquad V_2(x_2) := -C\frac{1}{\text{kN}} \qquad M_2(x_2) := \left[-M + C\left(a+b-x_2\right)\right]\frac{1}{\text{kN}\cdot\text{m}}$





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